

3.1 Find I_1 in the circuit in Fig. P3.1.

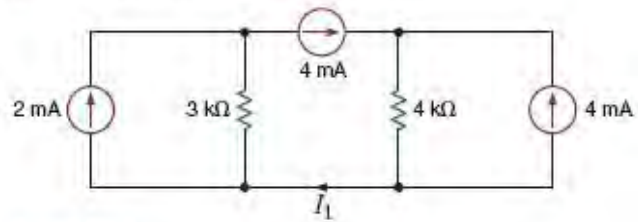


Figure P3.1

SOLUTION:

$$I_1 = 4 \text{ mA}$$

3.2 Find I_1 in the network in Fig. P3.2.

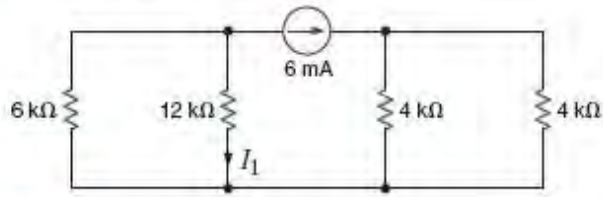


Figure P3.2

SOLUTION:

$$I_1 = \frac{-\frac{6}{k}(6k)}{18k} = -2mA$$

3.3 Find I_o in the circuit in Fig. P3.3.

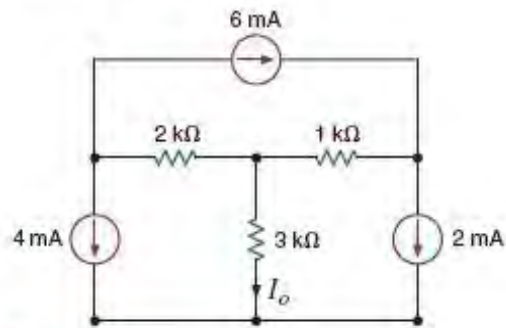


Figure P3.3

SOLUTION:

$$I_o = -6 \text{ mA}$$

3.4 Use nodal analysis to find V_1 in the circuit in Fig P3.4.

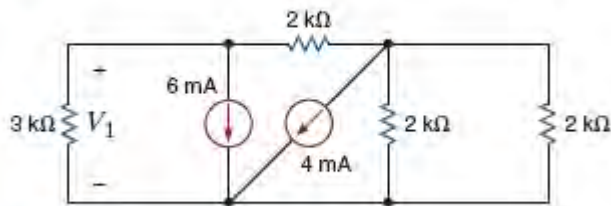
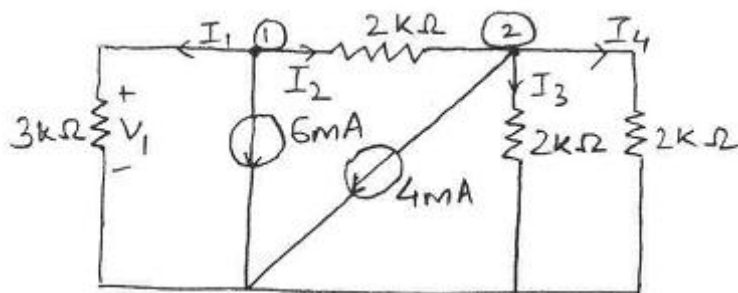


Figure P3.4

SOLUTION:



$$\text{KCL at } \textcircled{1}: I_1 + 6\text{m} + I_2 = 0$$

$$\frac{V_1}{3\text{k}} + \frac{V_1 - V_2}{2\text{k}} = -6\text{m}$$

$$2V_1 + 3V_1 - 3V_2 = -36$$

$$\boxed{5V_1 - 3V_2 = -36}$$

$$\text{KCL at } \textcircled{2}: I_2 = 4\text{m} + I_3 + I_4$$

$$\frac{V_1 - V_2}{2\text{k}} = 4\text{m} + \frac{V_2}{2\text{k}} + \frac{V_2}{2\text{k}}$$

$$V_1 - V_2 = 8 + V_2 + V_2$$

$$\boxed{V_1 - 3V_2 = 8}$$

$$5V_1 - 3V_2 = -36$$

$$V_1 - 3V_2 = 8$$

$$V_1 = -11\text{ V}$$

$$V_2 = -6.33\text{ V}$$

3.5 Find V_1 and V_2 in the circuit in Fig. P3.5 using nodal analysis.

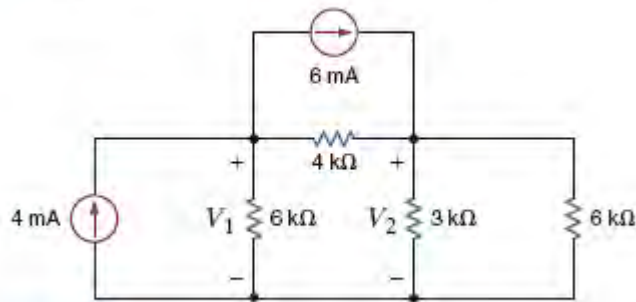
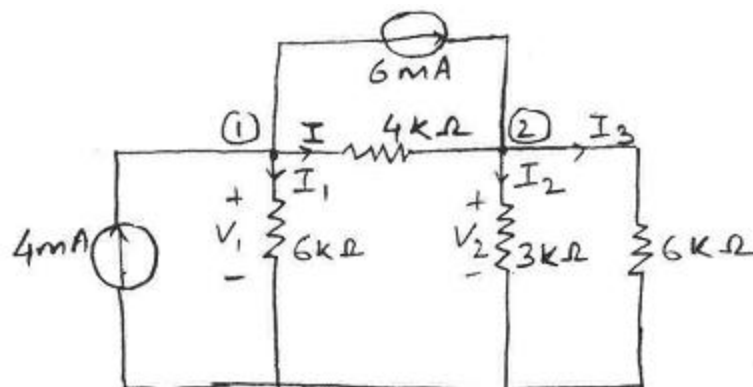


Figure P3.5

SOLUTION:



$$\text{KCL at } ①: 4\text{m} = 6\text{m} + I + I_1$$

$$\frac{V_1 - V_2}{4\text{k}} + \frac{V_1}{6\text{k}} = -2\text{m}$$

$$3V_1 - 3V_2 + 2V_1 = -24$$

$$\boxed{5V_1 - 3V_2 = -24}$$

$$\text{KCL at } ②: 6\text{m} + I = I_2 + I_3$$

$$\frac{V_2}{3\text{k}} + \frac{V_2}{6\text{k}} = 6\text{m} + \frac{V_1 - V_2}{4\text{k}}$$

$$4V_2 + 2V_2 = 72 + 3V_1 - 3V_2$$

$$\boxed{-3V_1 + 9V_2 = 72}$$

$$\begin{aligned}5V_1 - 3V_2 &= -24 \\ -3V_1 + 9V_2 &= 72\end{aligned}$$

$$V_1 = 0 \text{ V}$$

$$V_2 = 8 \text{ V}$$

% MATLAB Code and solution for Problem 3.4

$$G = [5, -3; -3, 9];$$

$$I_{matrix} = [-24; 72];$$

$$V = \text{inv}(G) * I_{matrix}$$

>>

V =

0.00
8.00

3.6 Use nodal analysis to find both V_1 and V_o in the circuit in Fig 3.6.

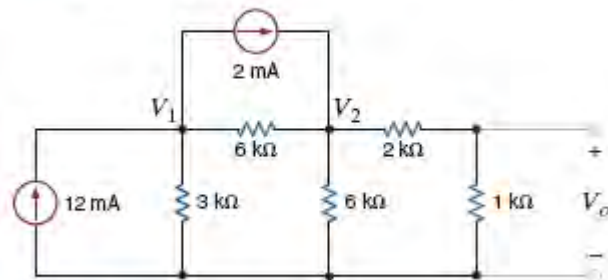
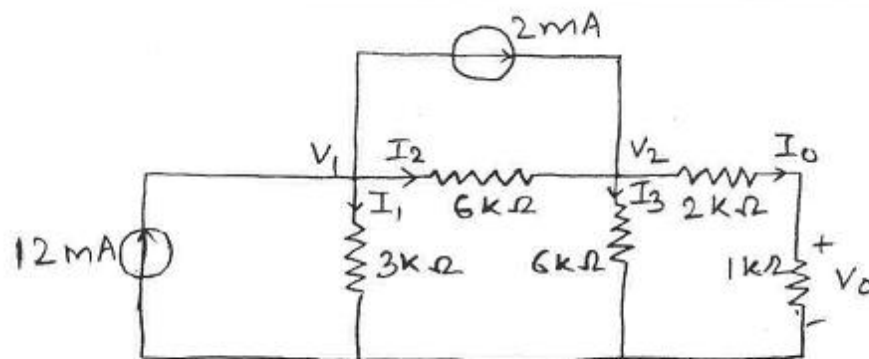


Figure P3.6

SOLUTION:



$$\text{KCL at } \textcircled{1} : 12\text{ m} = I_1 + I_2 + 2\text{ m}$$

$$\frac{V_1}{3\text{ k}} + \frac{V_1 - V_2}{6\text{ k}} = 10\text{ m}$$

$$2V_1 + V_1 - V_2 = 60$$

$$\boxed{3V_1 - V_2 = 60}$$

$$\text{KCL at } \textcircled{2} : 2\text{ m} + I_2 = I_3 + I_o$$

$$2\text{ m} + \frac{V_1 - V_2}{6\text{ k}} = \frac{V_2}{6\text{ k}} + \frac{V_2}{2\text{ k} + 1\text{ k}}$$

$$\frac{V_2}{6\text{ k}} + \frac{V_2}{3\text{ k}} - \frac{V_1}{6\text{ k}} + \frac{V_2}{6\text{ k}} = 2\text{ m}$$

$$V_2 + 2V_2 - V_1 + V_2 = 12$$

$$\boxed{-V_1 + 4V_2 = 12}$$

$$\begin{aligned} 3V_1 - V_2 &= 60 \\ -V_1 + 4V_2 &= 12 \end{aligned}$$

$$\begin{aligned} V_1 &= 252/11 \text{ V} \\ V_2 &= 96/11 \text{ V} \end{aligned}$$

$$\begin{aligned} I_o &= \frac{V_2}{3 \text{ k}} \\ &= \frac{32}{11} \text{ mA} \end{aligned}$$

$$I_o = 2.91 \text{ mA}$$

$$\begin{aligned} V &= \frac{V_2}{3} \\ &= 2.91 \text{ Volts} \end{aligned}$$

3.7 Find I_o in the circuit in Fig. P3.7 using nodal analysis.

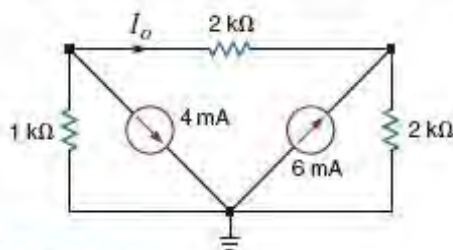


Figure P3.7

SOLUTION:

$$\begin{bmatrix} \frac{1}{k} + \frac{1}{2k} & -\frac{1}{2k} \\ -\frac{1}{2k} & \frac{1}{2k} + \frac{1}{2k} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -\frac{4}{k} \\ \frac{6}{k} \end{bmatrix}$$

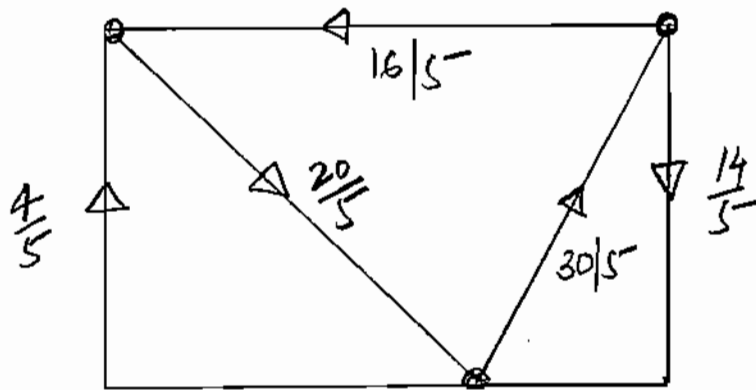
$$\Delta = \left(\frac{3}{2k} \right) \left(\frac{1}{k} \right) - \left(\frac{1}{4k^2} \right) = \frac{3}{2k^2} - \frac{1}{4k^2}$$

$$= \frac{6}{4k^2} - \frac{1}{4k^2} = \frac{5}{4k^2}$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \frac{4k^2}{5} \begin{bmatrix} \frac{1}{k} & \frac{1}{2k} \\ \frac{1}{2k} & \frac{3}{2k} \end{bmatrix} \begin{bmatrix} -4/k \\ 6/k \end{bmatrix}$$

$$= \frac{4k^2}{5} \begin{bmatrix} -\frac{4}{k^2} + \frac{3}{k^2} \\ -\frac{2}{k^2} + \frac{9}{k^2} \end{bmatrix} = \frac{4}{5} \begin{bmatrix} -1 \\ 7 \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} \\ \frac{28}{5} \end{bmatrix}$$

$$I_0 = \frac{1}{2k} \left(-\frac{4}{5} - \frac{28}{5} \right) = \frac{1}{2k} \left(\frac{-32}{5} \right) = -\frac{16}{5} \text{ mA}$$



3.8 Find I_o in the network in Fig. P3.8 using nodal analysis.

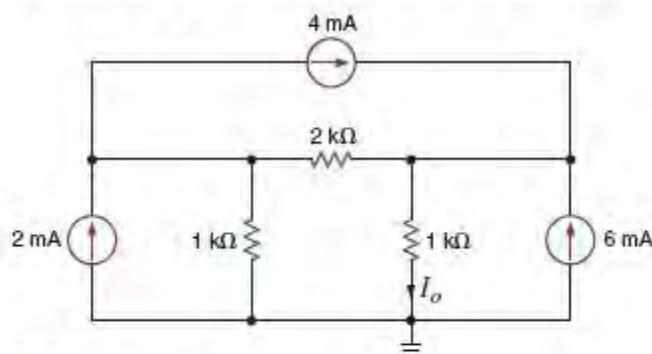


Figure P3.8

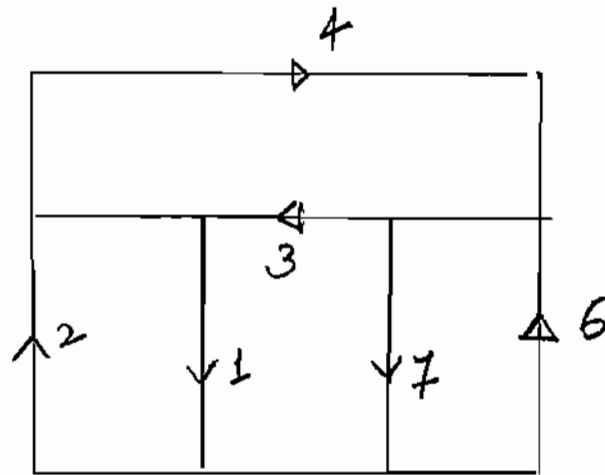
SOLUTION:

$$\begin{bmatrix} \frac{1}{k} + \frac{1}{2k} & -\frac{1}{2k} \\ -\frac{1}{2k} & \frac{1}{k} + \frac{1}{2k} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -\frac{2}{k} \\ \frac{10}{k} \end{bmatrix}$$

$$\Delta = \left(\frac{3}{2k}\right) \left(\frac{3}{2k}\right) - \frac{1}{4k^2} = \frac{9}{4k^2} - \frac{1}{4k^2} = \frac{2}{k^2}$$

$$\begin{aligned} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} &= \frac{k^2}{2} \begin{pmatrix} \frac{3}{2k} & \frac{1}{2k} \\ \frac{1}{2k} & \frac{3}{2k} \end{pmatrix} \begin{bmatrix} -\frac{2}{k} \\ \frac{10}{k} \end{bmatrix} \\ &= \frac{k^2}{2} \begin{bmatrix} -3/k^2 & +5/k^2 \\ -1/k^2 & 15/k^2 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} \end{aligned}$$

$$I_o = 7 \text{ mA}$$



3.9 Find I_o in the circuit in Fig. P3.9.

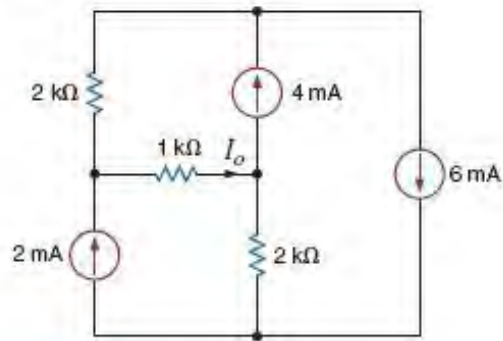
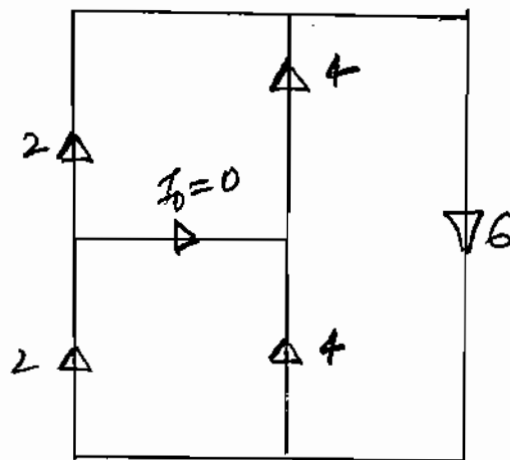


Figure P3.9

SOLUTION:



3.10 Find I_o in the circuit in Fig. P3.10 using nodal analysis.

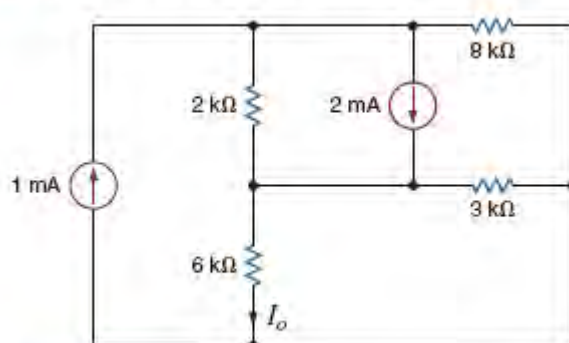
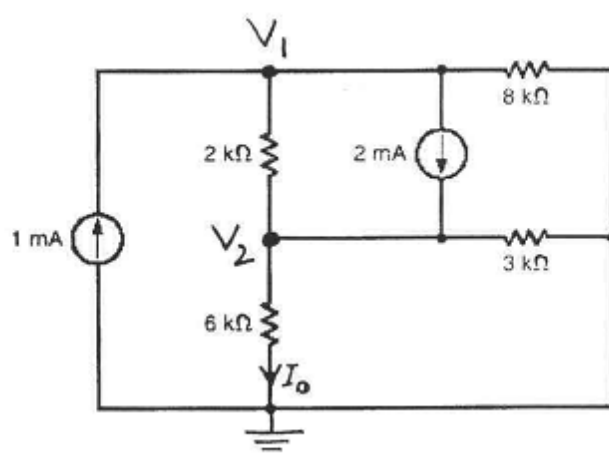


Figure P3.10

SOLUTION:



Let the two node voltages be V_1 & V_2

Applying KCL at node 1:

$$1 = \frac{V_1 - V_2}{2} + 2 + \frac{V_1}{8}$$

$$\Rightarrow 5V_1 - 4V_2 = -8 \quad \text{--- (1)}$$

Applying KCL at node 2:

$$\frac{V_1 - V_2}{2} + 2 = \frac{V_2}{3} + \frac{V_2}{6}$$

$$\Rightarrow V_1 - 2V_2 = -4 \quad \text{--- (2)}$$

From equation (1) and (2) we get

$$V_2 = 2 \text{ Volts} \quad \& \quad V_1 = 0 \text{ Volts}$$

$$\text{Current } I_o = \frac{V_2}{6k} = \frac{2}{6} \text{ mA}$$

$$I_o = 0.333 \text{ mA}$$

3.11 Use nodal analysis to find I_o in the network in Fig. P3.11.

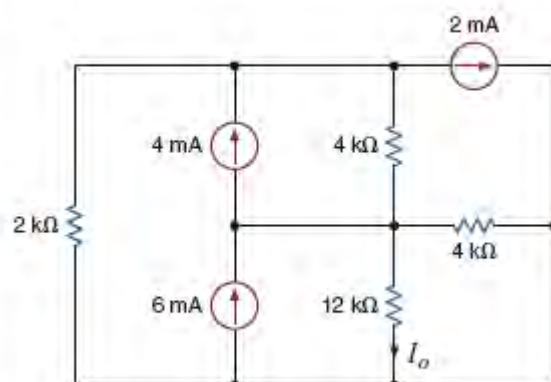
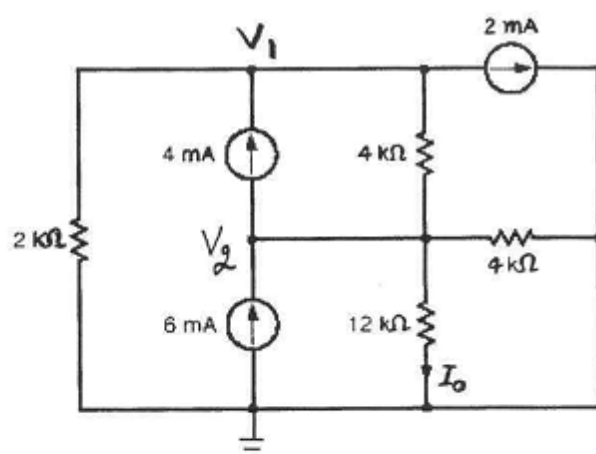


Figure P3.11

SOLUTION:



Let the two node
voltages be
 V_1 & V_2

KCL at node 1 :

$$\frac{V_1}{2} + \frac{V_1 - V_2}{4} + 2 = 4$$

$$\Rightarrow 3V_1 - V_2 = 8 \quad \text{--- (1)}$$

KCL at node 2 :

$$6 = 4 + \frac{V_2 - V_1}{4} + \frac{V_2}{4} + \frac{V_2}{12}$$

$$\Rightarrow -3V_1 + 7V_2 = 24 \quad - (2)$$

From equation ① & ②, we get

$$V_2 = \frac{32}{6} = \frac{16}{3} \text{ Volts}$$

$$I_0 = \frac{V_2}{12K} = \frac{16/3}{12} \text{ mA}$$

$$\Rightarrow I_0 = \frac{4}{9} \text{ mA} = 0.444 \text{ mA}$$

3.12 Find V_o in the network in Fig. P3.12 using nodal analysis.

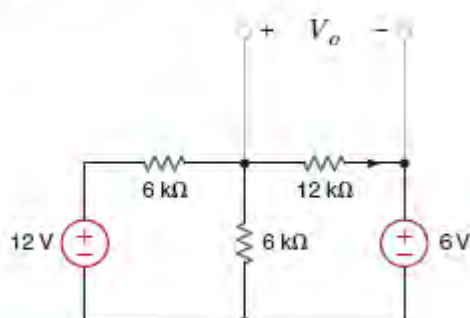
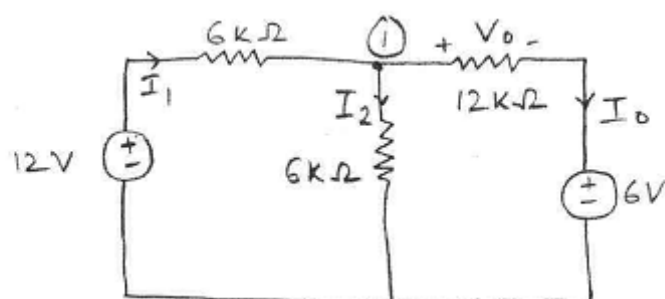


Figure P3.12

SOLUTION:



KCL at (1): $I_1 = I_2 + I_o$

$$\frac{12 - V_1}{6\text{K}} = \frac{V_1}{6\text{K}} + \frac{V_1 - 6}{12\text{K}}$$

$$24 - 2V_1 = 2V_1 + V_1 - 6$$

$$5V_1 = 30$$

$$V_1 = 6\text{V}$$

$$V_o = V_1 - 6$$

$$= 6 - 6$$

$$V_o = 0\text{V}$$

3.13 Find V_o in the circuit in Fig. P3.13 using nodal analysis.

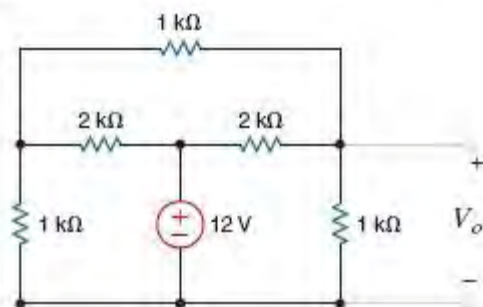


Figure P3.13

SOLUTION:

$$\frac{V_1}{k} + \frac{V_1 - V_3}{1k} + \frac{V_1 - 12}{2k} = 0$$

$$\frac{V_o}{k} + \frac{V_o - V_1}{1k} + \frac{V_o - 12}{2k} = 0$$

$$\begin{bmatrix} \left(\frac{1}{1k} + \frac{1}{k} + \frac{1}{2k}\right) & -\left(\frac{1}{k}\right) \\ -\frac{1}{k} & \frac{1}{k} + \frac{1}{k} + \frac{1}{2k} \end{bmatrix} \begin{bmatrix} V_1 \\ V_o \end{bmatrix} = \begin{bmatrix} 12/2k \\ 12/2k \end{bmatrix}$$

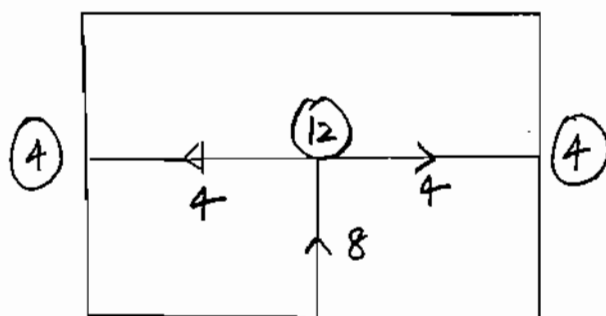
$$\begin{bmatrix} 5/2k & -1/k \\ -1/k & 5/2k \end{bmatrix} \begin{bmatrix} V_1 \\ V_o \end{bmatrix} = \begin{bmatrix} 6/k \\ 6/k \end{bmatrix}$$

$$\Delta = \frac{25}{4k^2} - \frac{1}{k^2} = \frac{21}{4k^2}$$

$$\begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = \frac{4k^2}{21} \begin{bmatrix} 5/2k & 1/k \\ 1/k & 5/2k \end{bmatrix} \begin{bmatrix} 6/k \\ 6/k \end{bmatrix}$$

$$= \frac{4}{21} \begin{bmatrix} 15 + 6 \\ 6 + 15 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

$$V_0 = 4V$$



3.14 Use nodal analysis to find V_o in the circuit in Fig. P3.14.

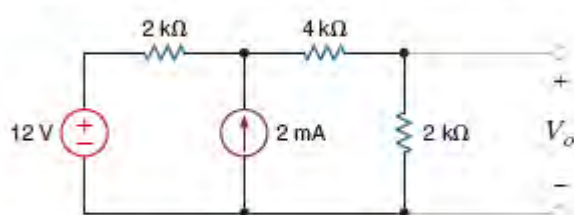
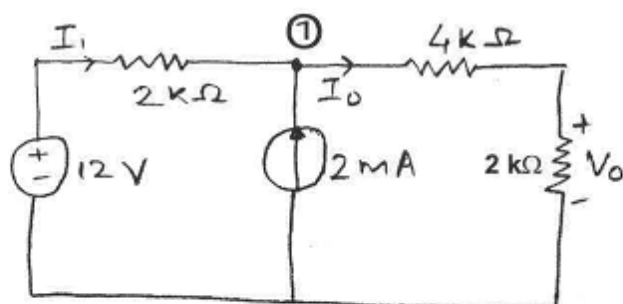


Figure P3.14

SOLUTION:



$$\text{KCL at } ① : I_1 + 2\text{ m} = I_o$$

$$\frac{12 - V_1}{2\text{ k}} + 2\text{ m} = \frac{V_1}{4\text{ k} + 2\text{ k}}$$

$$36 - 3V_1 + 12 = V_1$$

$$4V_1 = 48$$

$$V_1 = 12\text{ V}$$

$$I_o = \frac{V_1}{4\text{ k} + 2\text{ k}}$$

$$= \frac{12}{6\text{ k}}$$

$$I_o = 2\text{ mA}$$

$$V_o = I_o (2\text{ k})$$

$$= 2\text{ m} (2\text{ k})$$

$$V_o = 4\text{ V}$$

3.15 Find I_o in the network in Fig. P3.15 using nodal analysis.

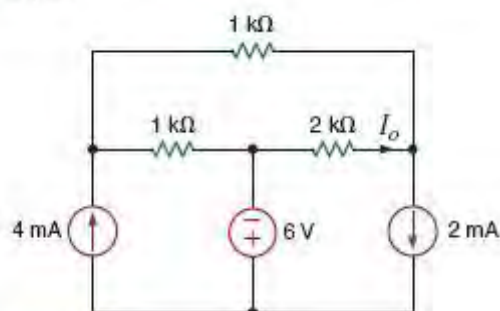


Figure P3.15

SOLUTION:

$$\frac{V_1 + 6}{1k} + \frac{V_1 - V_2}{1k} = \frac{4}{k}$$

$$\frac{V_2 - V_1}{1k} + \frac{V_2 + 6}{2k} = -2/k$$

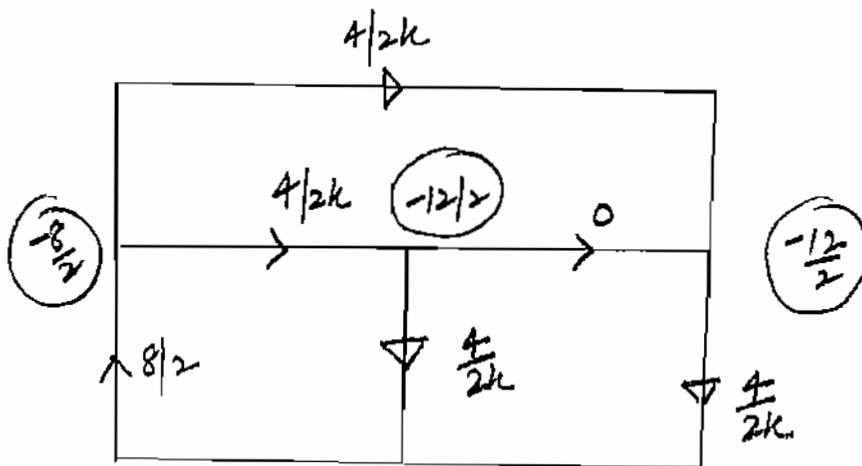
$$\begin{bmatrix} \frac{1}{k} + \frac{1}{k} & -\frac{1}{k} \\ -\frac{1}{k} & \frac{1}{k} + \frac{1}{2k} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -2/k \\ -5/k \end{bmatrix}$$

$$\Delta = \left(\frac{2}{k}\right) \left(\frac{3}{2k}\right) - \frac{1}{k^2} = \frac{2}{k^2}$$

$$\begin{aligned} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} &= \frac{1}{2} \begin{bmatrix} 3/2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 \\ -5 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} -3 & -5 \\ -2 & -10 \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} -8/2 \\ -12/2 \end{bmatrix}$$

$$I_0 = \frac{-\frac{12}{2} - \left(-\frac{12}{2}\right)}{2k} = 0$$



3.16 Find I_o in the circuit in Fig. P3.16 using nodal analysis.

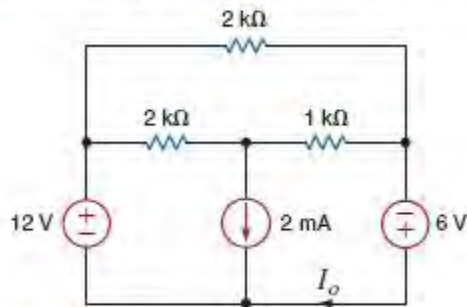


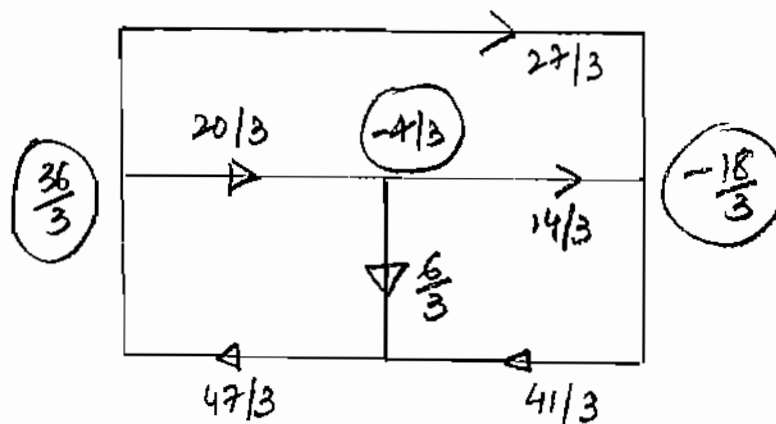
Figure P3.16

SOLUTION:

$$\frac{V_1 - 12}{2k} + \frac{2}{k} + \frac{V_1 + 6}{1k} = 0$$

$$V_1 \left(\frac{3}{2k} \right) = \frac{6}{k} - \frac{2}{k} - \frac{6}{k}$$

$$V_1 = -\frac{2}{k} \left(\frac{2k}{3} \right) = -\frac{4}{3} \text{ V}$$



$$I_o = \frac{41}{3} \text{ mA}$$

3.17 Find V_o in the circuit in Fig. P3.17 using nodal analysis.

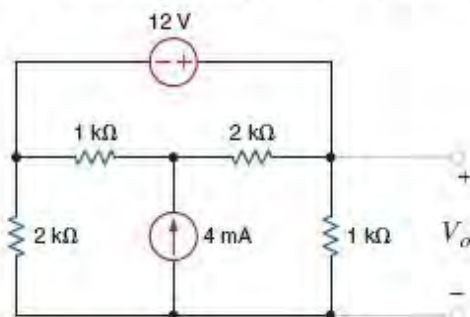


Figure P3.17

SOLUTION:

$$V_o - V_1 = 12$$

$$V_o - 12 = V_1$$

$$\frac{V_1}{2k} + \frac{V_1 - V_2}{1k} + \frac{V_o - V_2}{2k} + \frac{V_o}{1k} = 0$$

$$\frac{V_2 - V_1}{1k} + \frac{V_2 - V_o}{2k} = 4/k$$

$$\frac{V_o - 12}{2k} + \frac{V_o - 12 - V_2}{1k} + \frac{V_o}{2k} - \frac{V_2}{2k} + \frac{V_o}{1k} = 0$$

$$\frac{V_2 - V_o + 12}{1k} + \frac{V_2}{2k} - \frac{V_o}{2k} = \frac{4}{k}$$

$$\cancel{\frac{V_o}{2k}} - \frac{12}{2k} + \cancel{\frac{V_o}{1k}} - \frac{12}{1k} - \frac{V_2}{1k} + \cancel{\frac{V_o}{2k}} - \frac{V_2}{2k} + \cancel{\frac{V_o}{1k}} = 0$$

$$\frac{V_2}{1k} - \frac{V_0}{1k} + \frac{12}{k} + \frac{V_2}{2k} - \frac{V_0}{2k} = \frac{4}{k}$$

$$V_0 \left(\frac{1}{2k} + \frac{1}{k} + \frac{1}{2k} + \frac{1}{k} \right) - V_2 \left(\frac{1}{k} + \frac{1}{2k} \right) = \frac{+18}{k}$$

$$-V_0 \left(\frac{1}{k} + \frac{1}{2k} \right) + V_2 \left(\frac{1}{k} + \frac{1}{2k} \right) = -8/k$$

$$V_0 \left(\frac{3}{k} \right) - V_2 \left(\frac{3}{2k} \right) = \frac{+18}{k}$$

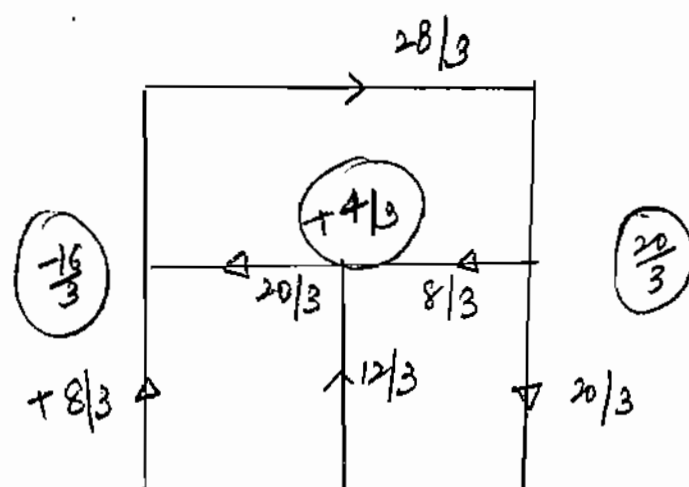
$$-V_0 \left(3/2k \right) + V_2 \left(3/2k \right) = -8/k$$

$$\Delta = \left(\frac{3}{k} \right) \left(\frac{3}{2k} \right) - \frac{9}{4k^2} = \frac{18}{4k^2} - \frac{9}{4k^2} = \frac{9}{4k^2}$$

$$\begin{bmatrix} V_0 \\ V_2 \end{bmatrix} = \frac{4k^2}{9} \begin{bmatrix} 3/2k & 3/2k \\ 3/2k & 3 \end{bmatrix} \begin{bmatrix} +18/k \\ -8/k \end{bmatrix}$$

$$= \frac{4}{9} \begin{bmatrix} +27 & -12 \\ +27 & -24 \end{bmatrix} = \frac{4}{9} \begin{bmatrix} +15 \\ +3 \end{bmatrix}$$

$$= \frac{4}{3} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \frac{20/3}{4/3}$$



3.18 Find V_o in the circuit in Fig. P3.18 using nodal analysis.

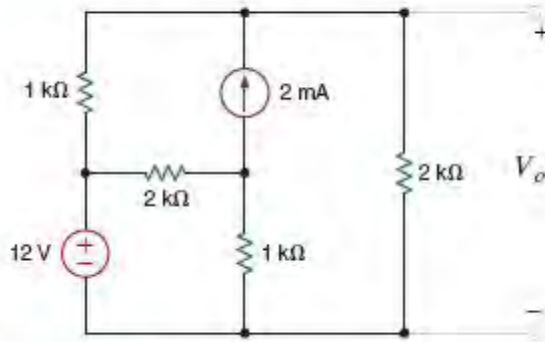
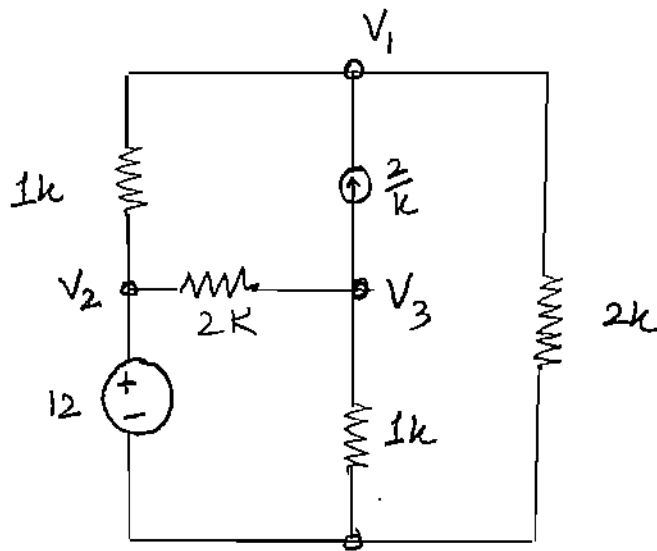


Figure P3.18

SOLUTION:

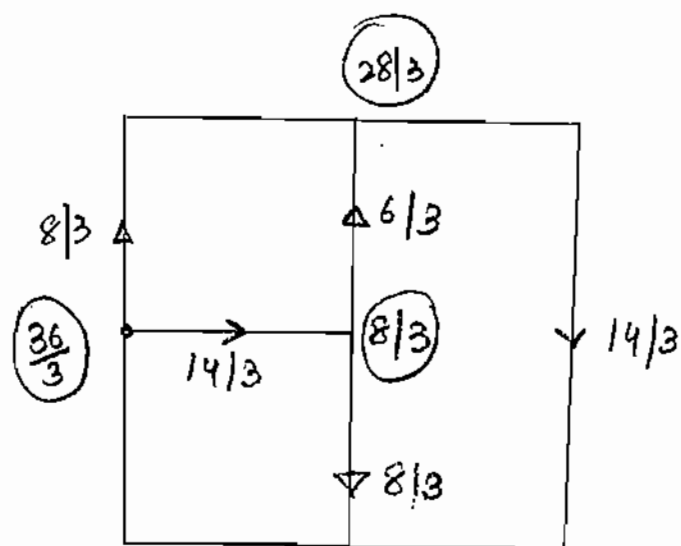


$$\frac{V_1 - 12}{1k} + \frac{V_1}{2k} = \frac{2}{k}$$

$$\frac{V_3 - 12}{2k} + \frac{V_3}{1k} = -2/k$$

$$\frac{V_1}{k} + \frac{V_1}{2k} = \frac{14}{k} \Rightarrow \frac{3}{2} V_1 = 14 \quad V_1 = \frac{28}{3} V$$

$$\frac{V_3}{2k} + \frac{V_3}{1k} = \frac{4}{k} \Rightarrow \frac{3}{2} V_3 = 4 \quad V_3 = \frac{8}{3} V$$



3.19 Find I_o in the circuit in Fig. P3.19 using nodal analysis.

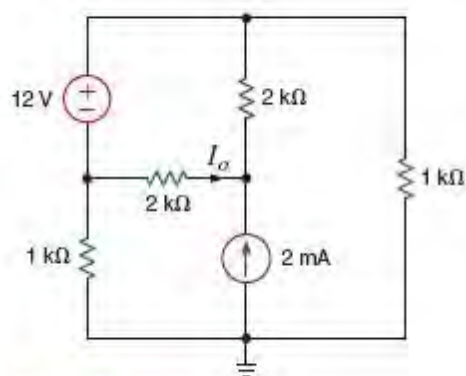
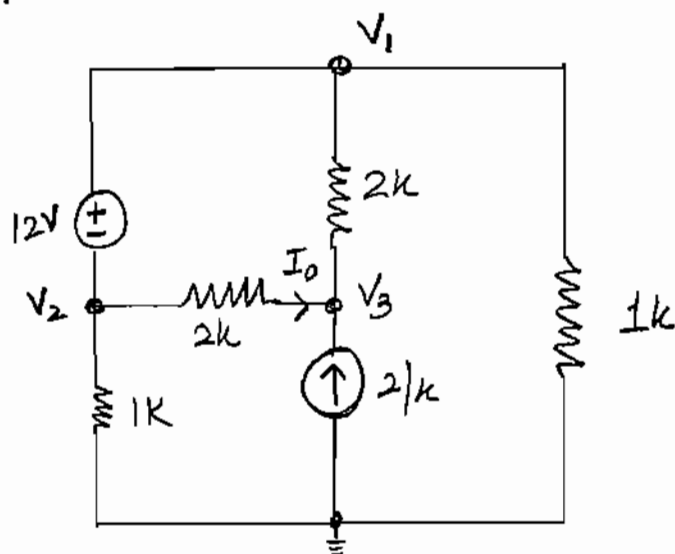


Figure P3.19

SOLUTION:



$$V_1 - V_2 = 12$$

$$\frac{V_1 - V_3}{2k} + \frac{V_1}{1k} + \frac{V_2}{1k} + \frac{V_2 - V_3}{2k} = 0$$

$$\frac{V_3 - V_1}{2k} + \frac{V_3 - V_2}{2k} = 2/k$$

$$\frac{V_1}{2k} - \frac{V_3}{2k} + \frac{V_1}{1k} + \frac{V_2}{1k} + \frac{V_2}{2k} - \frac{V_3}{2k} = 0$$

$$\frac{V_3}{2k} - \frac{V_1}{2k} + \frac{V_3}{2k} - \frac{V_2}{2k} = \frac{2}{k}$$

$$V_1 = V_2 + 12$$

$$\cancel{\frac{V_2}{2k}} + \frac{12}{2k} - \frac{V_3}{2k} + \cancel{\frac{V_2}{2k}} + \frac{12}{1k} + \cancel{\frac{V_1}{1k}} + \cancel{\frac{V_2}{2k}} - \frac{V_3}{2k} = 0$$

$$\frac{V_3}{2k} - \cancel{\frac{V_2}{2k}} - \frac{12}{2k} + \frac{V_3}{2k} - \cancel{\frac{V_2}{2k}} = \frac{2}{k}$$

$$V_2 \left(\frac{1}{2k} + \frac{1}{k} + \frac{1}{k} + \frac{1}{2k} \right) - V_3 \left(\frac{1}{2k} + \frac{1}{2k} \right) = -18/k$$

$$-V_2 \left(\frac{1}{2k} + \frac{1}{2k} \right) + V_3 \left(\frac{1}{2k} + \frac{1}{2k} \right) = \frac{8}{k}$$

$$V_2 \left(\frac{3}{k} \right) - V_3 \left(\frac{1}{k} \right) = -18/k$$

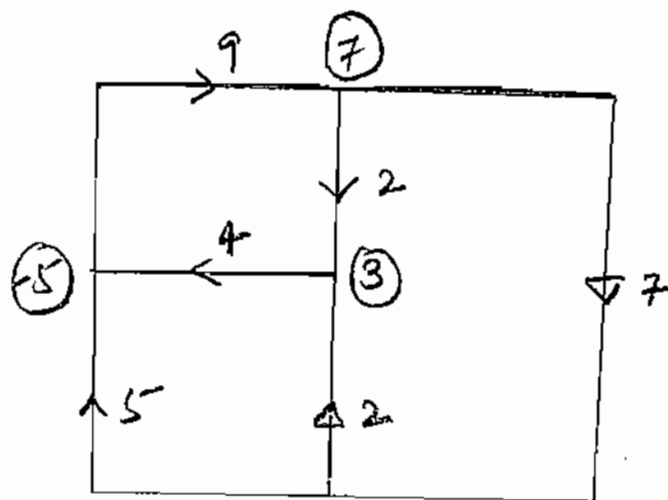
$$-V_2 \left(\frac{1}{k} \right) + V_3 \left(\frac{1}{k} \right) = \frac{8}{k}$$

$$\begin{bmatrix} 3 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -18 \\ 8 \end{bmatrix}$$

$$\Delta = 3 - 1 = 2$$

$$\begin{bmatrix} V_2 \\ V_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -18 \\ 8 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -10 \\ 6 \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \end{bmatrix}$$

$$I_0 = \frac{V_2 - V_3}{2k} = \frac{-5 - 3}{2k} = \frac{-8}{2k} = -4mA$$



3.20 Find V_o in the network in Fig. P3.20 using nodal analysis.

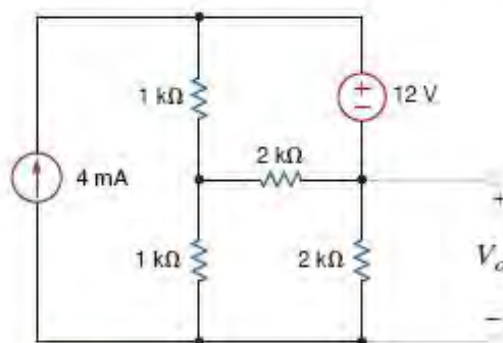
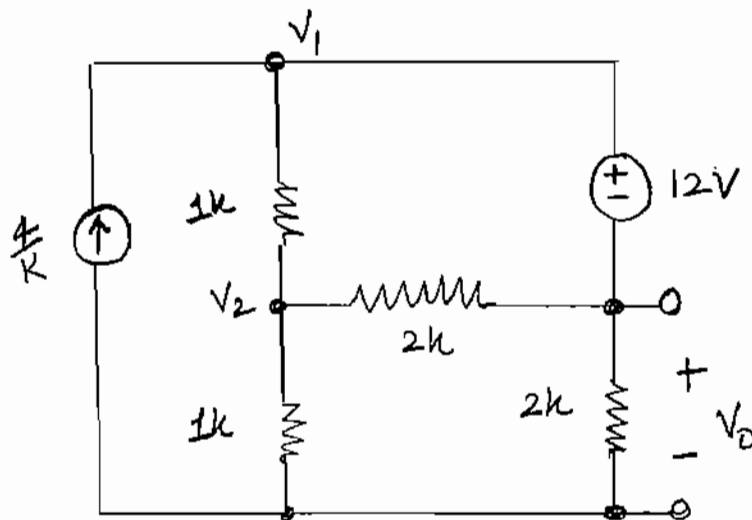


Figure P3.20

SOLUTION:



$$V_1 - V_o = 12 \Rightarrow V_1 = V_o + 12$$

$$\frac{V_1 - V_2}{1k} + \frac{V_o - V_2}{2k} + \frac{V_o}{2k} = \frac{4}{k}$$

$$\frac{V_2 - V_1}{1k} + \frac{V_2 - V_o}{2k} + \frac{V_2}{1k} = 0$$

$$\frac{V_1}{k} - \frac{V_2}{k} + \frac{V_o}{2k} - \frac{V_2}{2k} + \frac{V_o}{2k} = \frac{4}{k}$$

$$\frac{V_2}{k} - \frac{V_1}{k} + \frac{V_2}{2k} - \frac{V_0}{2k} + \frac{V_2}{k} = 0$$

$$\cancel{\frac{V_0}{k}} + \frac{12}{k} - \frac{V_2}{1k} + \cancel{\frac{V_0}{2k}} - \frac{V_2}{2k} + \cancel{\frac{V_0}{2k}} = 4/k$$

$$\frac{V_2}{k} - \cancel{\frac{V_0}{k}} - \frac{12}{k} + \frac{V_2}{2k} - \cancel{\frac{V_0}{2k}} + \frac{V_2}{k} = 0$$

$$V_0 \left(\frac{1}{k} + \frac{1}{2k} + \frac{1}{2k} \right) - V_2 \left(\frac{1}{k} + \frac{1}{2k} \right) = -8/k$$

$$-V_0 \left(\frac{1}{k} + \frac{1}{2k} \right) + V_2 \left(\frac{1}{k} + \frac{1}{2k} + \frac{1}{k} \right) = 12/k$$

$$V_0(2) - V_2(3/2) = -8$$

$$-V_0(3/2) + V_2(5/2) = 12$$

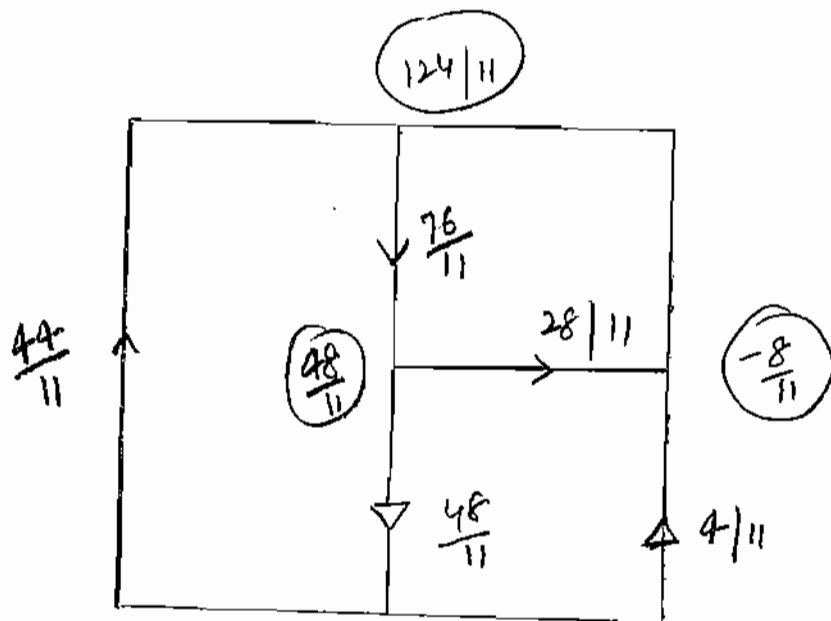
$$\Delta = 5 - \frac{9}{4} = \frac{20-9}{4} = \frac{11}{4}$$

$$\begin{bmatrix} V_0 \\ V_2 \end{bmatrix} = \frac{4}{11} \begin{bmatrix} 5/2 & 3/2 \\ 3/2 & 2 \end{bmatrix} \begin{bmatrix} -8 \\ 12 \end{bmatrix}$$

$$= \frac{4}{11} \begin{bmatrix} -20 + 18 \\ -12 + 24 \end{bmatrix} = \frac{4}{11} \begin{bmatrix} -2 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} -8/11 \\ 48/11 \end{bmatrix}$$

$$V_o = -\frac{8}{11} \text{ V}$$



3.21 Find V_o in the circuit in Fig. P3.21 using nodal analysis.

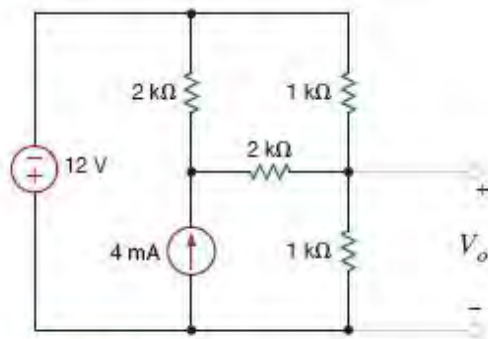
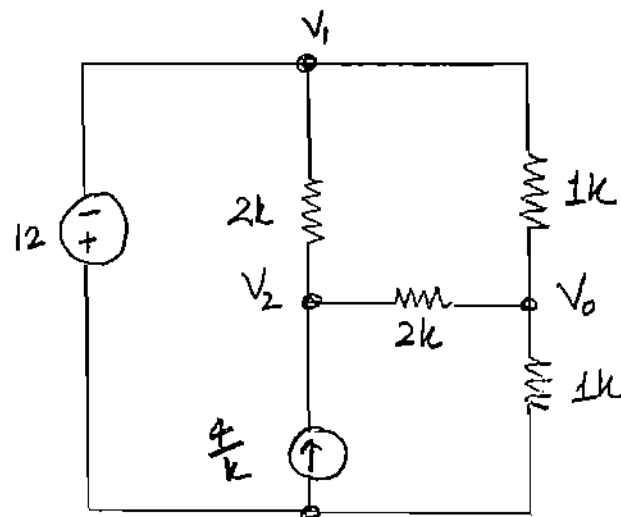


Figure P3.21

SOLUTION:



$$V_1 = -12$$

$$\frac{V_2 + 12}{2k} + \frac{V_2 - V_o}{2k} = \frac{4}{k}$$

$$\frac{V_o + 12}{1k} + \frac{V_o - V_2}{2k} + \frac{V_o}{1k} = 0$$

$$\frac{V_2}{2} + \frac{12}{2} + \frac{V_2}{2} - \frac{V_o}{2} = 4$$

$$V_o + 12 + \frac{V_o}{2} - \frac{V_2}{2} + V_o = 0$$

$$V_2 - \frac{1}{2}V_0 = -2$$

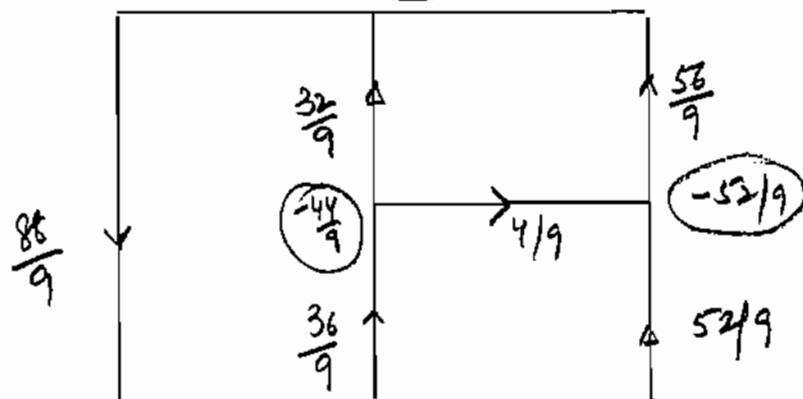
$$-\frac{1}{2}V_2 + \frac{5}{2}V_0 = -12$$

$$\begin{bmatrix} 1 & -1/2 \\ -1/2 & 5/2 \end{bmatrix} \begin{bmatrix} V_2 \\ V_0 \end{bmatrix} = \begin{bmatrix} -2 \\ -12 \end{bmatrix}$$

$$\Delta = \frac{5}{2} - \frac{1}{4} = \frac{10}{4} - \frac{1}{4} = \frac{9}{4}$$

$$\begin{aligned} \begin{bmatrix} V_2 \\ V_0 \end{bmatrix} &= \frac{4}{9} \begin{bmatrix} 5/2 & 1/2 \\ 1/2 & 1 \end{bmatrix} \begin{bmatrix} -2 \\ -12 \end{bmatrix} \\ &= \frac{4}{9} \begin{bmatrix} -5 & -6 \\ -1 & -12 \end{bmatrix} \\ &= \frac{4}{9} \begin{bmatrix} -11 \\ -13 \end{bmatrix} = \begin{bmatrix} -44/9 \\ -52/9 \end{bmatrix} \end{aligned}$$

$$V_0 = -52/9 \text{ V}$$



3.22 Find V_o in the network in Fig. P3.22 using nodal analysis.

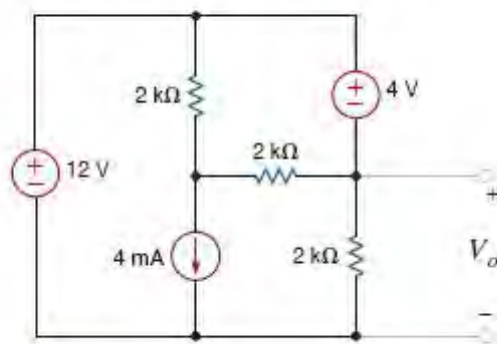


Figure P3.22

SOLUTION:

$$V_o = 12 - 4 = 8\text{ V}$$

3.23 Find I_o in the circuit in Fig. P3.23 using nodal analysis.

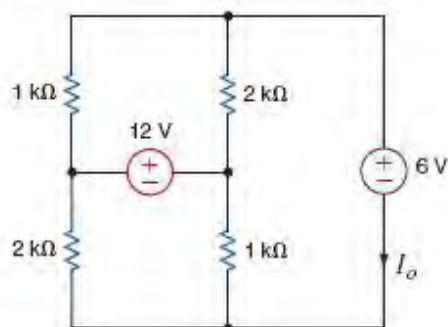
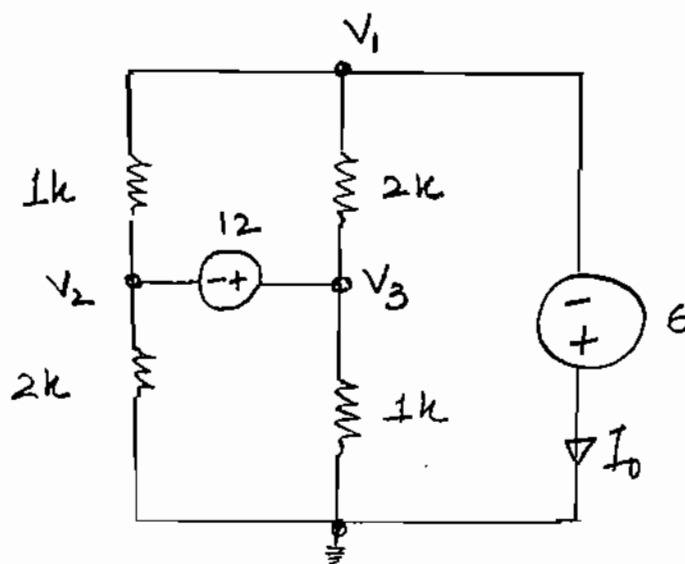


Figure P3.23

SOLUTION:



$$V_3 = V_2 + 12$$

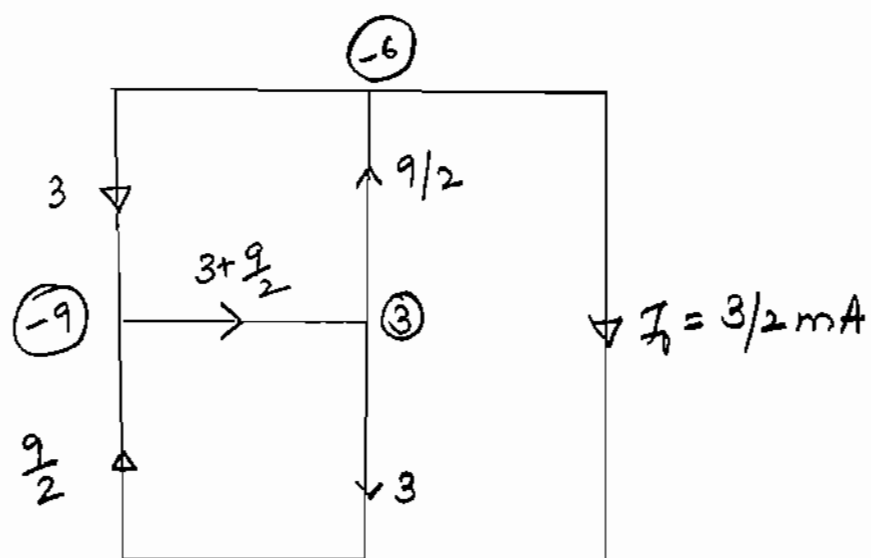
$$\frac{V_2 + 6}{1} + \frac{V_3 + 6}{2} + \frac{V_2}{2} + \frac{V_3}{1} = 0$$

$$V_2 + 6 + \frac{V_2}{2} + \frac{12}{2} + \frac{6}{2} + \frac{V_2}{2} + V_2 + 12 = 0$$

$$3V_2 = -27$$

$$V_2 = -9$$

$$V_3 = V_2 + 12 = 3$$



3.24 Find I_o in the circuit in Fig. P3.24 using nodal analysis.

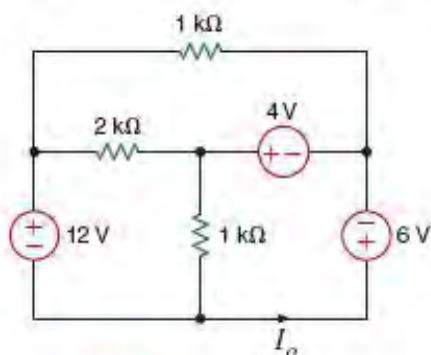
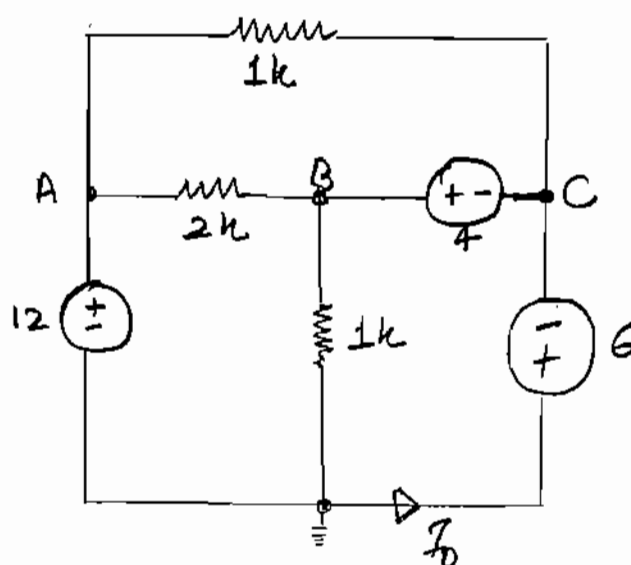


Figure P3.24

SOLUTION:



$$V_B \triangleq 4 - 6 = -2\text{ V}$$

$$I_B \triangleq = \frac{-2}{1k} = 2\text{ mA} \uparrow$$

$$V_{AB} = 12 + 6 - 4 = 14\text{ V}$$

$$I_{AB} = \frac{14}{2k} = 7\text{ mA} \rightarrow$$

$$V_{AC} = 12 + 6 = 18$$

$$I_{AC} = \frac{18}{1k} = 18mA \rightarrow$$

$$\begin{aligned} -I_b &= 7mA \rightarrow + 2mA \uparrow + 18mA \rightarrow \\ &= 27mA \end{aligned}$$

$$I_b = -27mA$$

- 3.25** Use nodal analysis to solve for the node voltages in the circuit in Fig. P3.25. Also calculate the power supplied by the 2-mA current source.

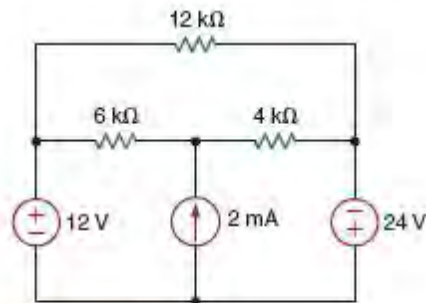
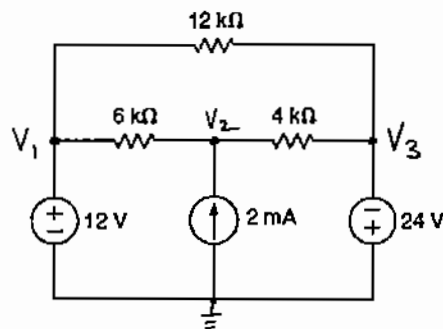


Figure P3.25

SOLUTION:



Constraint equations

$$\textcircled{1} V_1 = 12 \text{ V} \quad \textcircled{2} V_3 = -24 \text{ V}$$

$$\frac{V_2 - V_1}{6 \text{ k}\Omega} + \frac{V_2 - V_3}{4 \text{ k}\Omega} = 2 \text{ mA} \quad P_{2 \text{ mA}} = (-4.8)(2 \text{ mA}) = -9.6 \text{ mW}$$

$$2(V_2 - V_1) + 3(V_2 - V_3) = 24$$

$$-2V_1 + 5V_2 - 3V_3 = 24$$

$$-2(12) + 5V_2 - 3(-24) = 24$$

$$-24 + 5V_2 + 72 = 24$$

$$5V_2 + 48 = 24$$

$$5V_2 = -24$$

$$V_2 = \frac{-24}{5} = \underline{\underline{-4.8\text{ V}}}$$

3.26 Use nodal analysis to determine the node voltages defined in the circuit in Fig. P3.26.

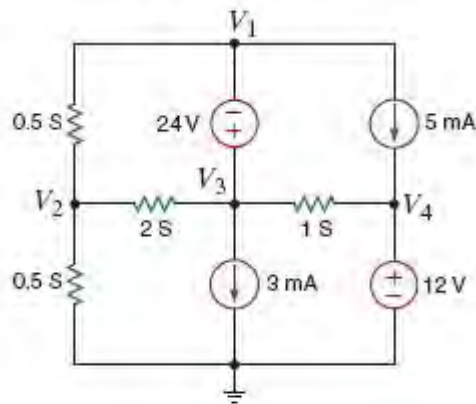


Figure P3.26

SOLUTION:

constraint equations:

$$\textcircled{1} V_4 = 12V$$

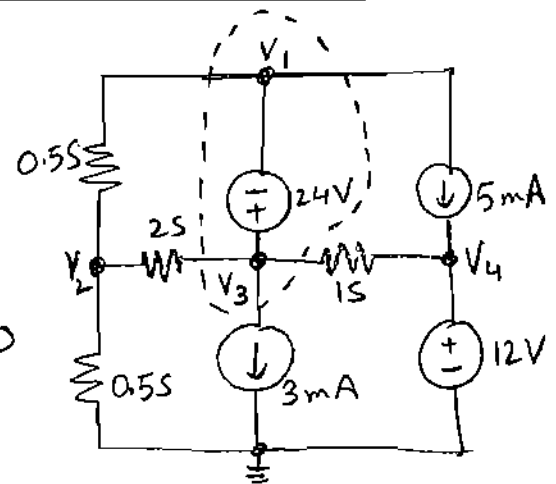
$$\textcircled{2} V_3 - V_1 = 24$$

$$\textcircled{3} -0.5V_1 + 3V_2 - 2V_3 = 0$$

KCL @ supernode:

$$0.5(V_1 - V_2) + 2(V_3 - V_2) + 1(V_3 - V_4) = -3m - 5m$$

$$\textcircled{4} 0.5V_1 - 2.5V_2 + 3V_3 - V_4 = -8m$$



$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ -0.5 & 3 & -2 & 0 \\ 0.5 & -2.5 & 3 & -1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 12 \\ 24 \\ 0 \\ -0.008 \end{bmatrix}$$

$$V_1 = -14.12 \text{ V}$$

$$V_2 = 4.23 \text{ V}$$

$$V_3 = 9.88 \text{ V}$$

$$V_4 = 12 \text{ V}$$

- 3.27 Use nodal analysis to solve for the node voltages in the circuit in Fig. P3.27. Also calculate the power supplied by the 2-mA current source.

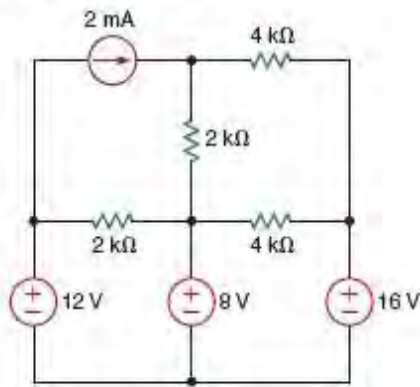
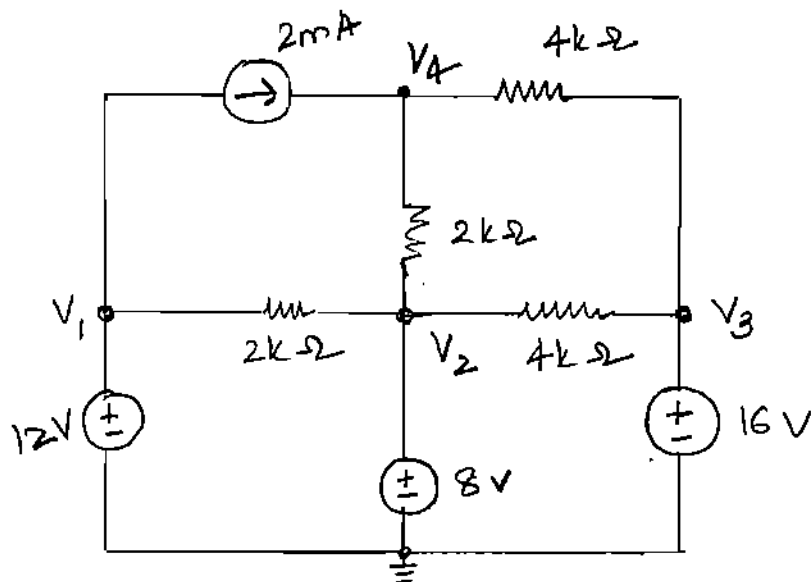


Figure P3.27

SOLUTION:



$$V_1 = 12 \text{ V}$$

$$V_2 = 8 \text{ V}$$

$$V_3 = 16 \text{ V}$$

$$\frac{V_4 - V_2}{2k} + \frac{V_4 - V_3}{4k} = 2m$$

$$2(V_4 - V_2) + V_4 - V_3 = 8$$

$$-2V_2 - V_3 + 3V_4 = 8$$

$$-2(8) - 16 + 3V_4 = 8$$

$$3V_4 = 40$$

$$V_4 = \frac{40}{3} = \underline{13.33 \text{ V}}$$

$$P_{2mA} = (V_4 - V_1) 2m$$

$$= (13.33 - 12) (2m)$$

$$= \underline{2.66 \text{ mW}}$$

3.28 Use nodal analysis to find V_o in the network in Fig. P3.28.

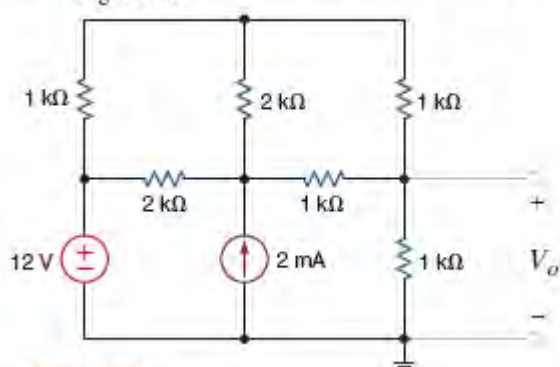
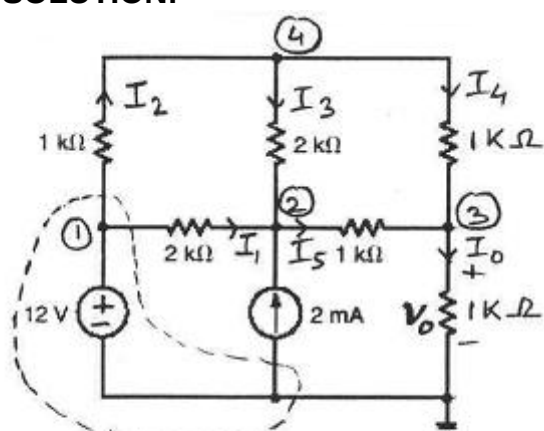


Figure P3.28

SOLUTION:



KCL at supernode: $I_o = 2\text{m} + I_1 + I_2$

$$\frac{V_3}{1\text{K}} = 2\text{m} + \frac{V_1 - V_2}{2\text{K}} + \frac{V_1 - V_4}{1\text{K}}$$

$$2V_3 = 4 + V_1 - V_2 + 2V_1 - 2V_4$$

$$\boxed{3V_1 - V_2 - 2V_3 - 2V_4 = -4}$$

KCL at ②: $I_1 + I_3 + 2\text{m} = I_5$

$$\frac{V_1 - V_2}{2\text{K}} + \frac{V_4 - V_2}{2\text{K}} + 2\text{m} = \frac{V_2 - V_3}{1\text{K}}$$

$$V_1 - V_2 + V_4 - V_2 + 4 = 2V_2 - 2V_3$$

$$\boxed{V_1 - 4V_2 + 2V_3 + V_4 = -4}$$

$$\text{KCL at } \textcircled{3}: I_5 + I_4 = I_o$$

$$\frac{V_2 - V_3}{1K} + \frac{V_4 - V_3}{1K} = \frac{V_3}{1K}$$

$$V_2 - V_3 + V_4 - V_3 = V_3$$

$$\boxed{V_2 - 3V_3 + V_4 = 0}$$

$$V_1 = 12 \text{ V}$$

$$-V_2 - 2V_3 - 2V_4 = -40$$

$$-4V_2 + 2V_3 + V_4 = -16$$

$$V_2 - 3V_3 + V_4 = 0$$

$$V_2 = 9.37 \text{ V}$$

$$V_3 = 6.17 \text{ V}$$

$$V_4 = 9.14 \text{ V}$$

$$V_o = V_3 = 6.17 \text{ V}$$

$$V_o = 6.17 \text{ V}$$

3.29 Use nodal analysis to find V_o in the circuit in Fig. P3.29.

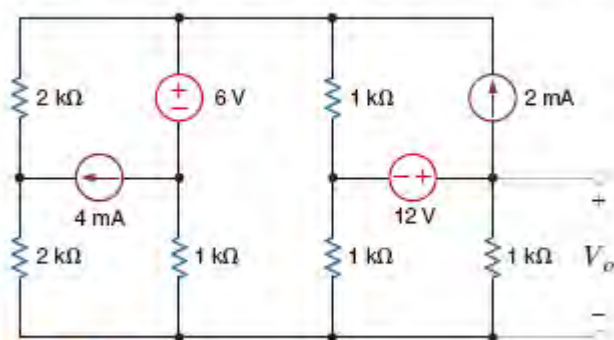
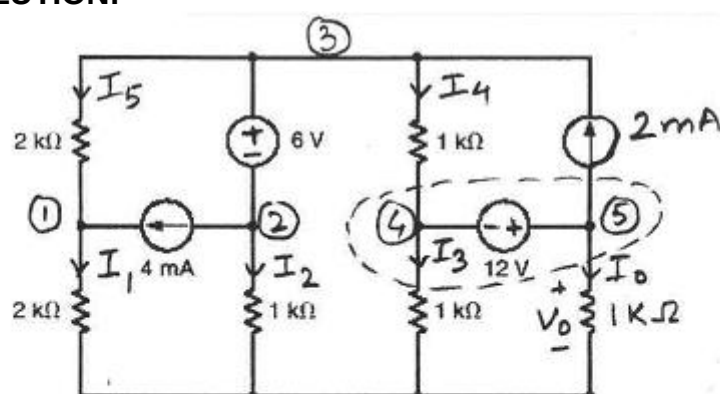


Figure P3.29

SOLUTION:



$$\text{KCL at } ①: I_5 + 4\text{m} = I_1$$

$$\frac{V_3 - V_1}{2\text{k}} + 4\text{m} = \frac{V_1}{2\text{k}}$$

$$V_3 - V_1 + 8 = V_1$$

$$\boxed{-2V_1 + V_3 = -8}$$

$$\text{KCL at reference node: } I_1 + I_2 + I_3 + I_o = 0$$

$$\frac{V_1}{2\text{k}} + \frac{V_2}{1\text{k}} + \frac{V_4}{1\text{k}} + \frac{V_5}{1\text{k}} = 0$$

$$\boxed{V_1 + 2V_2 + 2V_4 + 2V_5 = 0}$$

$$\boxed{-V_4 + V_5 = 12}$$

$$\boxed{-V_2 + V_3 = 6}$$

$$\text{KCL at supernode: } I_4 = 2\text{m} + I_3 + I_o$$

$$\frac{V_3 - V_4}{1K} = 2m + \frac{V_4}{1K} + \frac{V_5}{1K}$$

$$V_3 - V_4 = 2 + V_4 + V_5$$

$$V_3 - 2V_4 - V_5 = 2$$

$$-2V_1 + 0V_2 + V_3 + 0V_4 + 0V_5 = -8$$

$$V_1 + 2V_2 + 0V_3 + 2V_4 + 2V_5 = 0$$

$$0V_1 + 0V_2 + 0V_3 - V_4 + V_5 = 12$$

$$0V_1 - V_2 + V_3 + 0V_4 + 0V_5 = 6$$

$$0V_1 + 0V_2 + V_3 - 2V_4 - V_5 = 2$$

$$V_1 = 4.348 \text{ V}$$

$$V_2 = -5.304 \text{ V}$$

$$V_3 = 0.696 \text{ V}$$

$$V_4 = -4.435 \text{ V}$$

$$V_5 = 7.565 \text{ V}$$

$$V_0 = V_5 = 7.565 \text{ V}$$

$$V_0 = 7.565 \text{ V}$$

3.30 Use nodal analysis to find V_o in the circuit in Fig. P3.30.

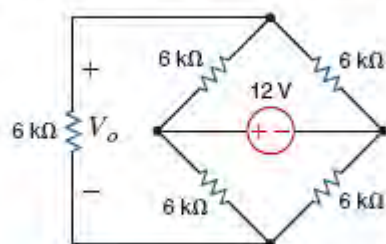
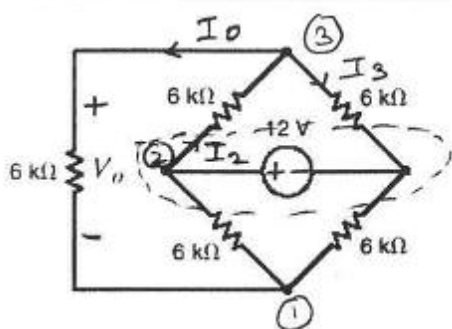


Figure P3.30

SOLUTION:



KCL at supernode: $I_1 + I_4 + I_3 = I_2$

$$\frac{V_1}{6k} + \frac{V_1 - V_2}{6k} + \frac{V_3}{6k} = \frac{V_2 - V_3}{6k}$$

$$V_1 + V_1 - V_2 + V_3 = V_2 - V_3$$

$$\boxed{2V_1 - 2V_2 + 2V_3 = 0}$$

KCL at ①: $I_0 = I_1 + I_4$

$$\frac{V_3 - V_1}{6k} = \frac{V_1}{6k} + \frac{V_1 - V_2}{6k}$$

$$V_3 - V_1 = V_1 + V_1 - V_2$$

$$\boxed{3V_1 - V_2 - V_3 = 0}$$

$$\boxed{V_2 = 12V}$$

$$\boxed{2V_1 + 2V_3 = 24}$$

$$\boxed{3V_1 - V_3 = 12}$$

$$2V_1 + 2V_3 = 24$$

$$3V_1 - V_3 = 12$$

$$\begin{aligned} 2V_1 + 2V_3 &= 24 \\ 3V_1 - V_3 &= 12 \end{aligned}$$

$$V_1 = 6 \text{ V}$$

$$V_3 = 6 \text{ V}$$

$$V_o = V_3 - V_1 = 6 - 6$$

$$V_o = 0 \text{ V}$$

It should be noted that this circuit is a balanced Wheatstone bridge. Therefore, the output voltage is zero.

3.31 Find V_o in the network in Fig. P3.31.

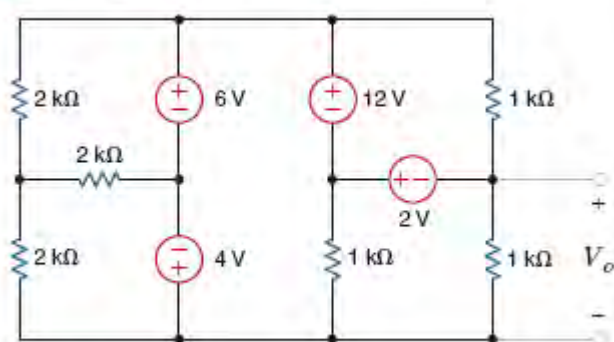
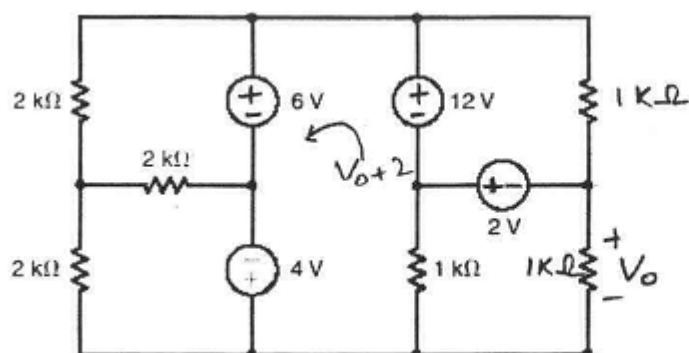


Figure P3.31

SOLUTION:



KVL in the loop indicated:

$$\begin{aligned}
 V_o + 2 + 12 - 6 + 4 &= 0 \\
 V_o &= 0 - 2 - 12 + 6 - 4 \\
 V_o &= -12 \text{ V}
 \end{aligned}$$

3.32 Find I_o in the network in Fig. P3.32 using nodal analysis.

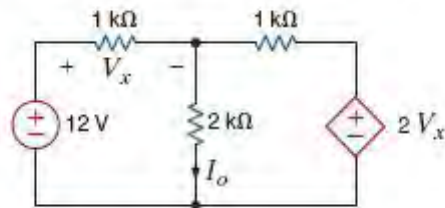


Figure P3.32

SOLUTION:

$$\frac{V_o - 12}{1k} + \frac{V_o}{2k} + \frac{V_o - 2V_x}{1k} = 0$$

$$V_x = 12 - V_o$$

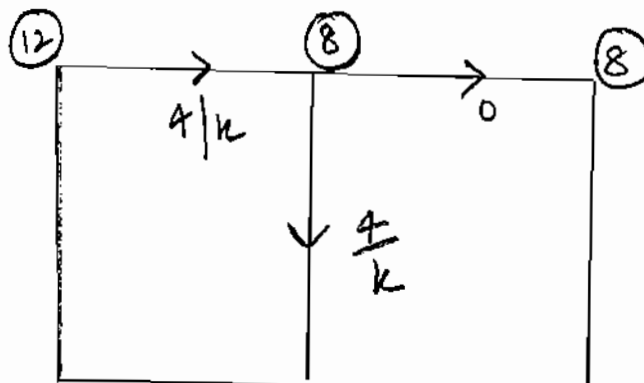
$$V_o - 12 + \frac{V_o}{2} + V_o - 2(12 - V_o) = 0$$

$$V_o - 12 + \frac{V_o}{2} + V_o - 24 + 2V_o = 0$$

$$\frac{9}{2}V_o = 36$$

$$V_o = 8V$$

$$I_o = 4mA$$



3.33 Find I_o in the network in Fig. P3.33 using nodal analysis.

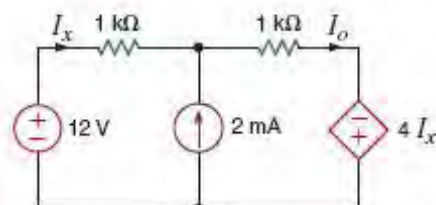
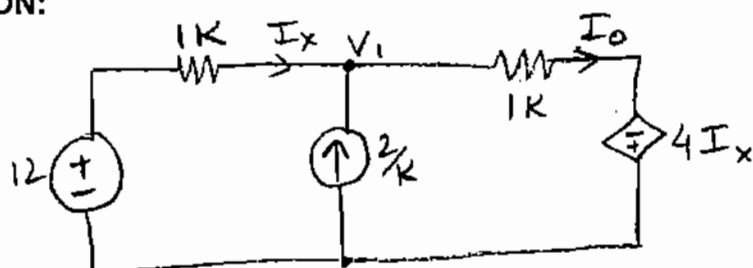


Figure P3.33

SOLUTION:



$$\frac{V_1 - 12}{1k} - \frac{2}{k} + \frac{V_1 + 4I_x}{1k} = 0$$

$$I_x = \frac{12 - V_1}{1k}$$

$$V_1 - 12 - 2 + V_1 + 4 \left(\frac{12 - V_1}{k} \right) = 0$$

$$2V_1 - 14 + \frac{48}{k} - \frac{4V_1}{k} = 0$$

$$\left(2 - \frac{4}{k} \right) V_1 = 14 - \frac{48}{k}$$

$$(2 - 0.004) V_1 = 14 - 0.048$$

$$1.996 V_1 = 13.952$$

$$V_1 = 6.99$$

$$I_o = I_x + \frac{2}{k} = \frac{12 - 6.99}{1k} + \frac{2}{k} = 5.01 + 2 = 7.01 \text{ mA}$$

3-34 Find V_o in the network in Fig. P3.34 using nodal analysis.

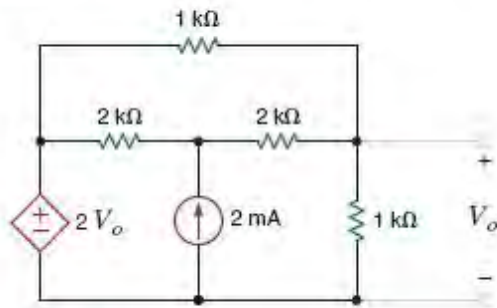
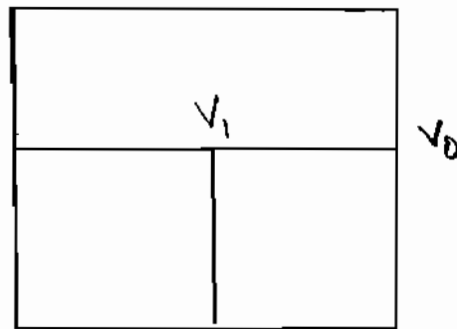


Figure P3.34

SOLUTION:



$$\frac{V_1 - 2V_0}{2k} - \frac{2}{k} + \frac{V_1 - V_0}{2k} = 0$$

$$\frac{V_0 - 2V_0}{1k} + \frac{V_0 - V_1}{2k} + \frac{V_0}{1k} = 0$$

$$\frac{V_1}{2} - V_0 + \frac{V_1}{2} - \frac{V_0}{2} = 2$$

$$-V_0 + \frac{V_0}{2} - \frac{V_1}{2} + V_0 = 0$$

$$V_1 - \frac{3}{2}V_0 = 2$$

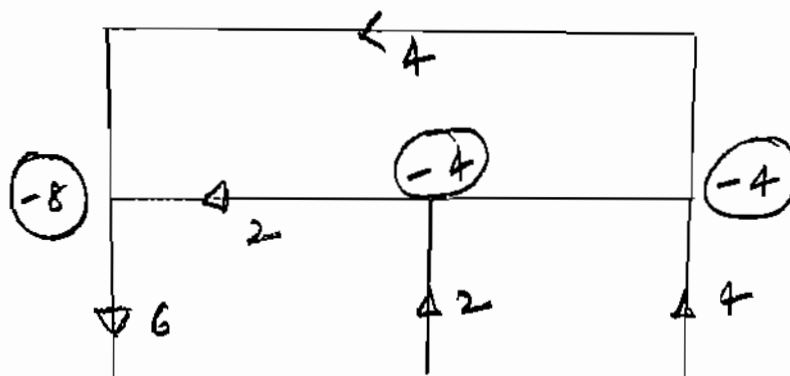
$$-\frac{1}{2}V_1 + \frac{V_0}{2} = 0$$

$$\begin{bmatrix} 1 & -3/2 \\ -1/2 & +1/2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\Delta = \frac{1}{2} - \frac{1}{2} \left(\frac{3}{2} \right) = \frac{1}{2} - \frac{3}{4} = -\frac{1}{4}$$

$$\begin{bmatrix} V_1 \\ V_0 \end{bmatrix} = -4 \begin{bmatrix} 1/2 & 3/2 \\ 1/2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= -4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -4 \\ -4 \end{bmatrix}$$



3.35 Find V_o in the circuit in Fig. P3.35 using nodal analysis.

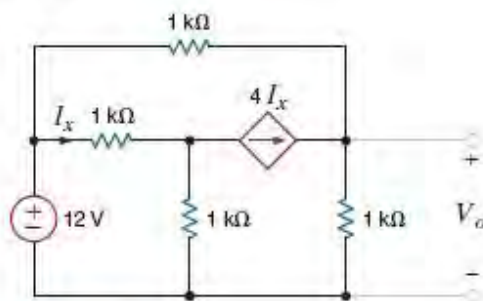


Figure P3.35

SOLUTION:

$$\frac{V_1 - 12}{k} + \frac{V_1}{k} + 4I_x = 0$$

$$\frac{V_o - 12}{k} + \frac{V_o}{k} = 4I_x \quad I_x = \frac{12 - V_1}{k}$$

$$V_1 - 12 + V_1 + 48 - 4V_1 = 0$$

$$V_o - 12 + V_o = 48 - 4V_1$$

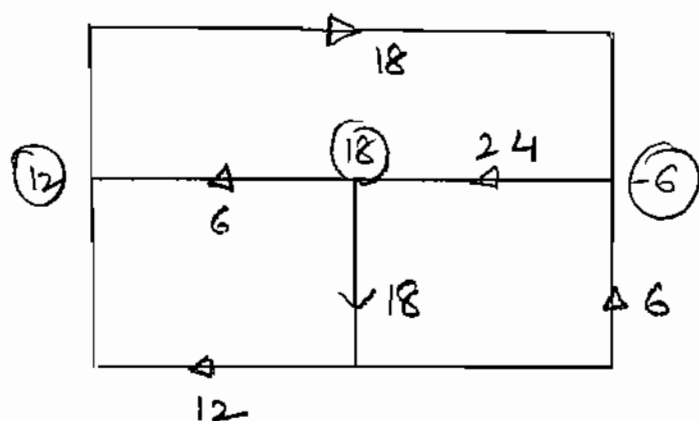
$$-2V_1 + 36 = 0$$

$$2V_o + 4V_1 = 60$$

$$V_1 = 18V$$

$$2V_o = 60 - 72 = -12$$

$$V_o = -6V$$



3.36 Find V_o in the network in Fig. P3.36 using nodal analysis.

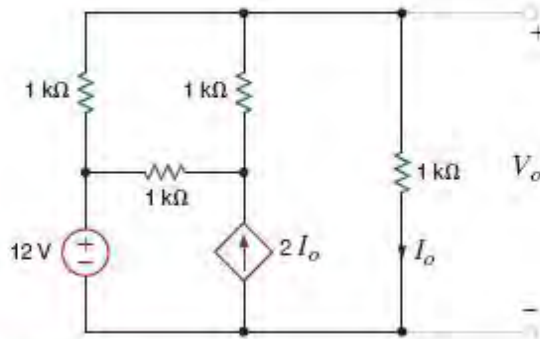


Figure P3.36

SOLUTION:

$$\frac{V_0 - 12}{k} + \frac{V_0 - V_1}{1k} + \frac{V_0}{k} = 0$$

$$\frac{V_1 - V_0}{k} + \frac{V_1 - 12}{k} = 2I_o \quad I_o = \frac{V_0}{k}$$

$$V_0 + V_0 - V_1 + V_0 = 12$$

$$V_1 - V_0 + V_1 = 12 + 2V_0$$

$$3V_0 - V_1 = 12$$

$$-3V_0 + 2V_1 = 12$$

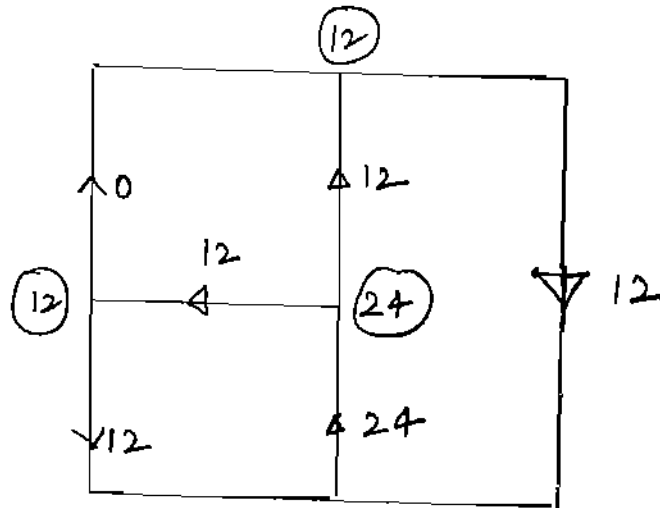
$$\Delta = (3)(2) - 3 = 3$$

$$\begin{bmatrix} V_0 \\ V_1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 2 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 12 \\ 12 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 24 & +12 \\ 36 & +36 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 36 \\ 72 \end{bmatrix}$$

$$= \begin{bmatrix} 12 \\ 24 \end{bmatrix}$$



- 3.37 Find I_o in the network in Fig. P3.37 using nodal analysis.

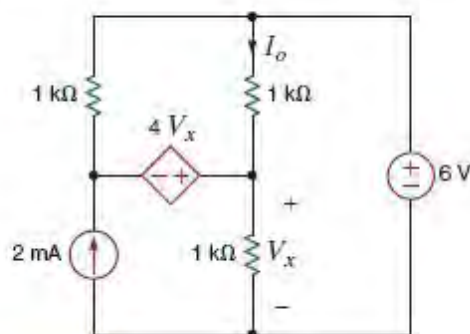


Figure P3.37

SOLUTION:

$$\frac{V_1 - 6}{k} + \frac{V_x - 6}{k} + \frac{V_x}{k} = \frac{2}{k}$$

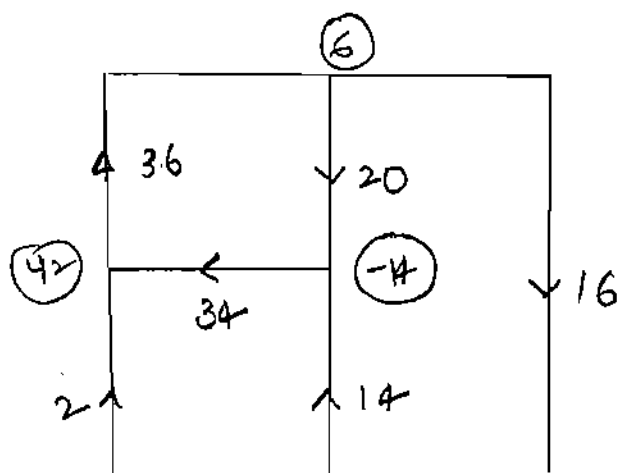
$$V_x - V_1 = 4V_x \quad V_1 = -3V_x$$

$$-3V_x - 6 + V_x - 6 + V_x = 2$$

$$-V_x = 14$$

$$V_x = -14$$

$$I_o = 20 \text{ mA}$$



3.38 Find I_o in the network in Fig. P3.38 using nodal analysis.

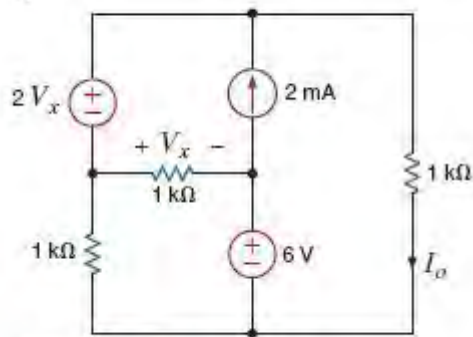


Figure P3.38

SOLUTION:

$$\frac{V_1}{k} + \frac{V_2}{k} + \frac{V_2 - 6}{k} = \frac{2}{k}$$

$$V_2 - V_1 = 2V_x$$

$$V_x = V_2 - 6$$

$$V_2 - V_1 = 2V_2 - 12$$

$$-V_1 - V_2 = -12$$

$$V_1 + V_2 = 12$$

$$V_1 + V_2 + V_2 = 8$$

$$V_1 + V_2 = 12$$

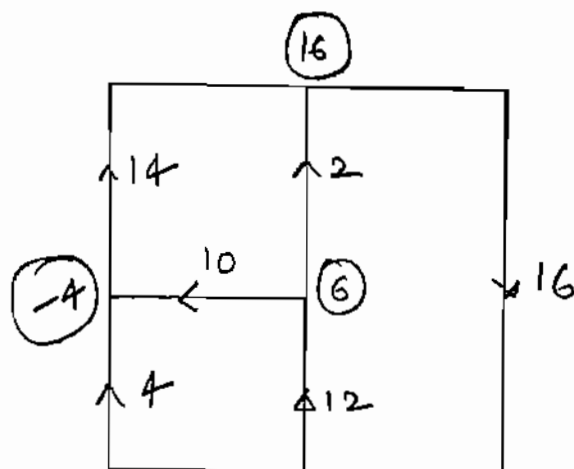
$$V_1 + 2V_2 = 8$$

$$\Delta = 2 - 1 = 1$$

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 12 \\ 8 \end{bmatrix}$$

$$= \begin{bmatrix} 24 & -8 \\ -12 & +8 \end{bmatrix} = \begin{bmatrix} 16 \\ -4 \end{bmatrix}$$

$$I_D = \frac{V_1}{1k} = 16 \text{ mA}$$



3.39 Find V_o in the network in Fig. P3.39 using nodal analysis.

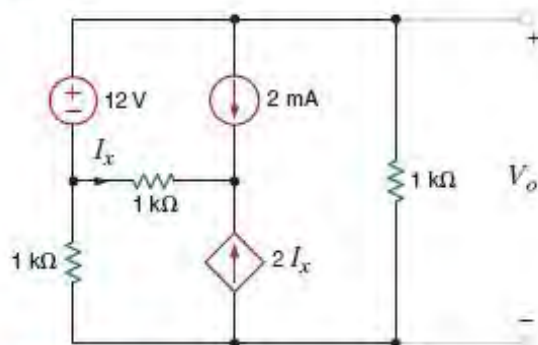


Figure P3.39

SOLUTION:

$$\frac{V_o}{k} + \frac{2}{k} + \frac{V_1}{k} + \frac{V_1 - V_2}{k} = 0$$

$$\frac{V_2 - V_1}{k} = \frac{2}{k} + 2I_x$$

$$I_x = \frac{V_1 - V_2}{k}$$

$$V_o - V_1 = 12$$

$$V_1 = V_o - 12$$

$$V_o + 2 + \frac{V_o - 12}{k} + \frac{V_o - 12 - V_2}{k} = 0$$

$$V_2 - V_o + 12 = 2 + 2V_1 - 2V_2$$

$$3V_o - V_2 = 22$$

$$V_2 - V_o + 12 = 2 + 2V_o - 24 - 2V_2$$

$$3V_o - V_2 = 22$$

$$-3V_o + 3V_2 = -34$$

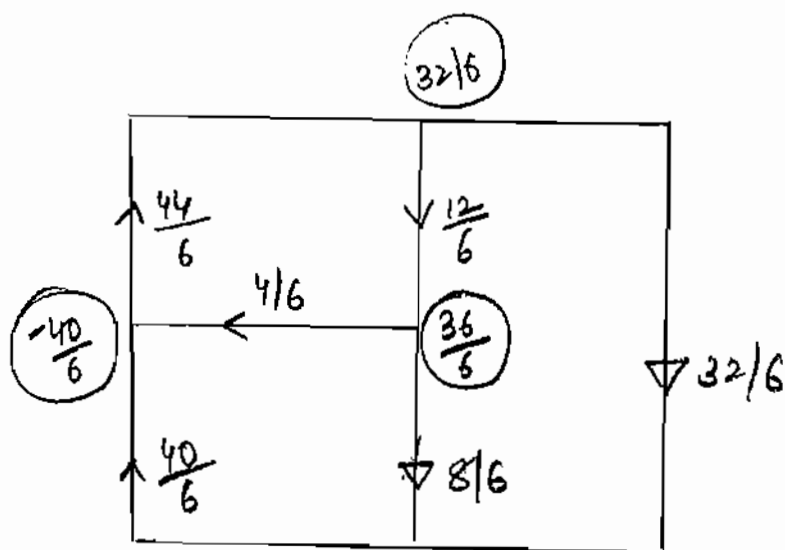
$$\Delta = 9 - 3 = 6$$

$$\begin{bmatrix} V_0 \\ V_2 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 3 & 1 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} 22 \\ -34 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 66 - 34 \\ 66 - 102 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 32 \\ 36 \end{bmatrix}$$

$$= \begin{array}{c} 32/6 \\ 36/6 \end{array}$$



3.40 Use nodal analysis to find V_o in the circuit in Fig. P3.40.

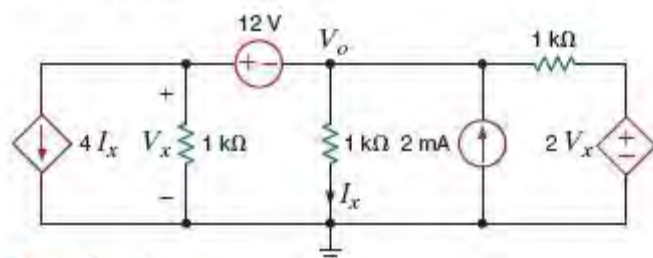


Figure P3.40

SOLUTION:

$$\frac{V_o + 12}{k} + \frac{V_o}{k} + \frac{V_o - 2V_x}{k} + 4I_x = \frac{2}{k}$$

$$V_x = V_o + 12$$

$$I_x = \frac{V_o}{k}$$

$$V_o + 12 + V_o + V_o - 2(V_o + 12) + 4V_o = 2$$

$$V_o + 12 + V_o + V_o - 2V_o - 24 + 4V_o = 2$$

$$5V_o = 14$$

$$V_o = \frac{14}{5} \text{ V}$$

3.41 Use nodal analysis to find V_o in the circuit in Fig. P3.41.

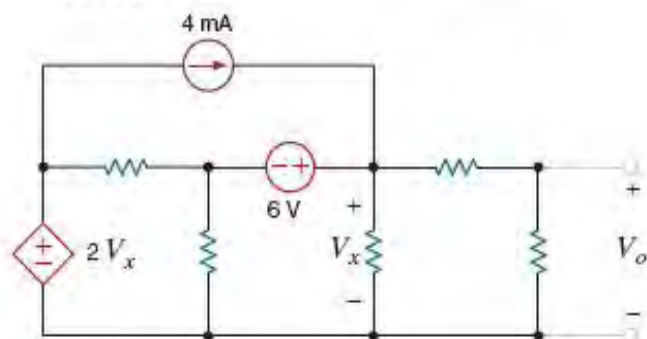
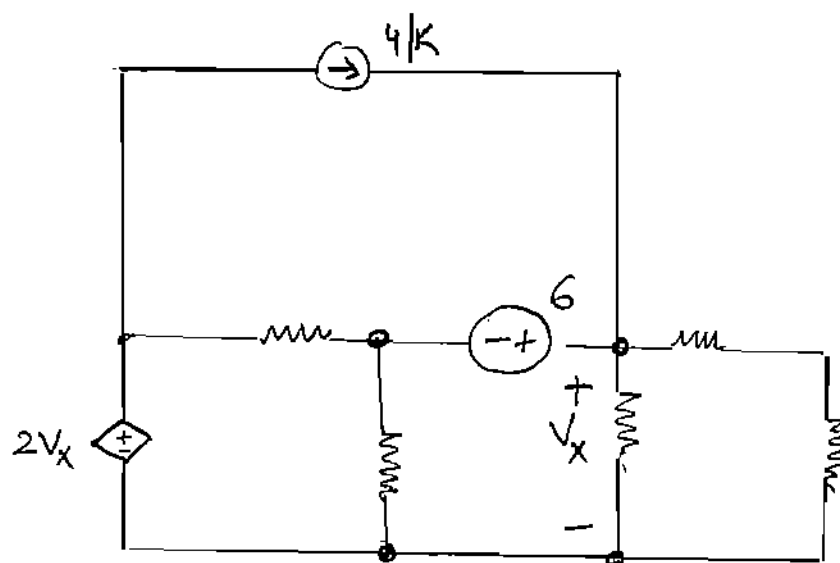


Figure P3.41

SOLUTION:



$$\frac{V_x - 6 - 2V_x}{k} + \frac{V_x - 6}{k} + \frac{V_x}{k} + \frac{V_x}{2k} = \frac{4}{k}$$

$$V_x - 6 - 2V_x + V_x - 6 + V_x + \frac{V_x}{2} = 4$$

$$\frac{3}{2} V_x = 16$$

$$V_x = \frac{32}{3} \text{ V}$$

$$V_0 = \frac{1}{2} V_x = \frac{16}{3} \text{ V}$$

3.42 Use nodal analysis to find V_o in the network in Fig. P3.42.

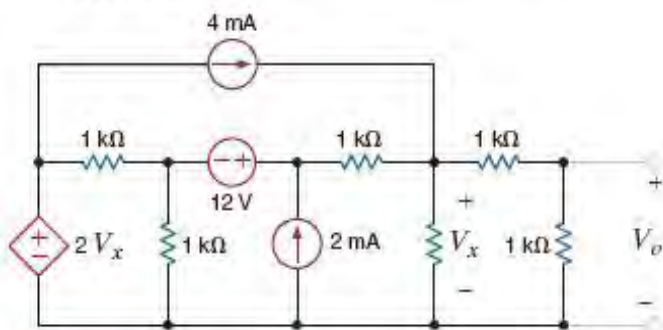
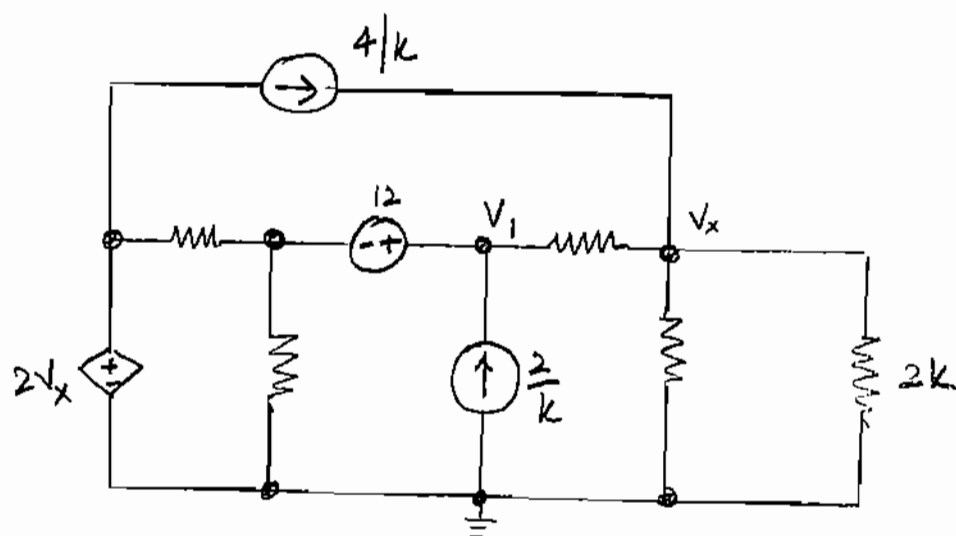


Figure P3.42

SOLUTION:



$$\frac{V_1 - 12 - 2V_x}{k} + \frac{V_1 - 12}{k} + \frac{V_1 - V_x}{k} = \frac{2}{k}$$

$$\frac{V_x - V_1}{k} + \frac{V_x}{k} + \frac{V_x}{2k} = \frac{4}{k}$$

$$V_1 - 12 - 2V_x + V_1 - 12 + V_1 - V_x = 2$$

$$V_x - V_1 + V_x + \frac{V_x}{2} = 4$$

$$3V_1 - 3V_x = 26$$

$$-V_1 + 5/2 V_x = 4$$

$$\Delta = 3(5/2) - 3 = \frac{15}{2} - \frac{6}{2} = \frac{9}{2}$$

$$\begin{bmatrix} V_1 \\ V_x \end{bmatrix} = \frac{2}{9} \begin{bmatrix} 5/2 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 26 \\ 4 \end{bmatrix}$$

$$V_x = \frac{2}{9} [26 + 12]$$

$$= \frac{2 \times 38}{9} = \frac{76}{9} \text{ V}$$

$$V_o = \frac{1}{2} V_x = \frac{38}{9} \text{ V}$$

3.43 Use nodal analysis to find V_o in the circuit in Fig. P3.43.

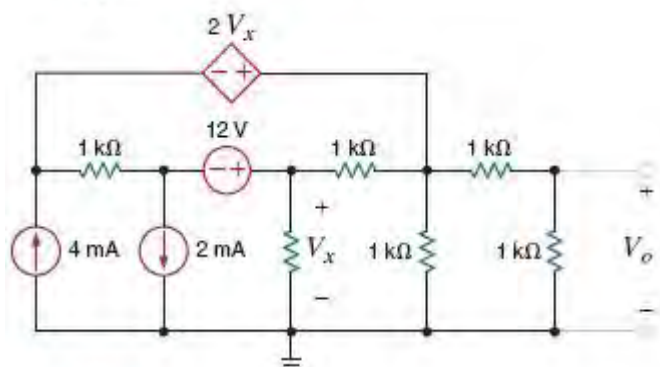
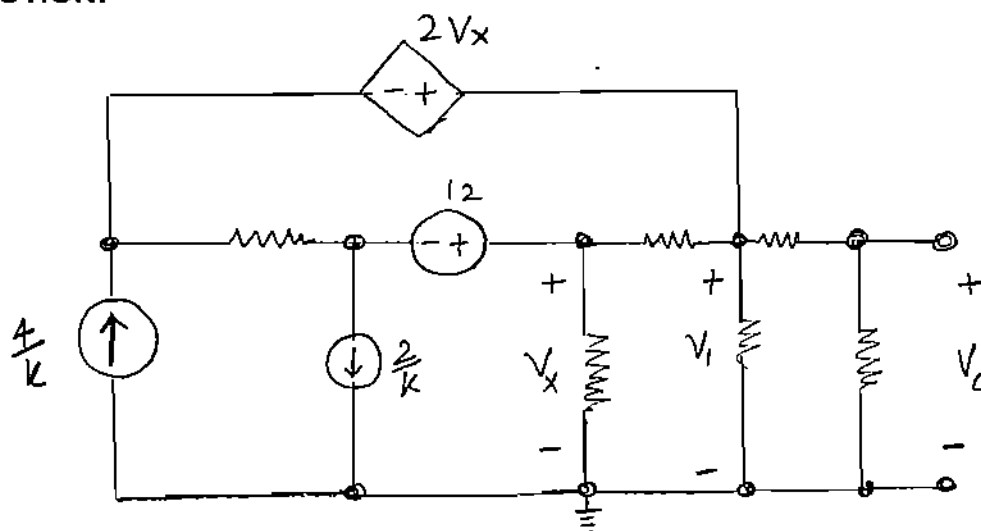


Figure P3.43

SOLUTION:



$$\frac{V_1 - 2V_x - (V_x - 12)}{k} + \frac{V_1 - V_x}{k} + \frac{V_1}{k} + \frac{V_1}{2k} = \frac{4}{k}$$

$$\frac{V_x - 12 - (V_1 - 2V_x)}{k} + \frac{V_x}{k} + \frac{V_x - V_1}{k} = -\frac{2}{k}$$

$$V_1 - 2V_x - V_x + 12 + V_1 - V_x + V_1 + \frac{V_1}{2} = 4$$

$$V_x - 12 - V_1 + 2V_x + V_x + V_x - V_1 = -2$$

$$\frac{7}{2}V_1 - 4V_x = -8$$

$$-2V_1 + 5V_x = 10$$

$$\Delta = 5\left(\frac{7}{2}\right) - 8 = \frac{35}{2} - \frac{16}{2} = \frac{19}{2}$$

$$\begin{bmatrix} V_1 \\ V_x \end{bmatrix} = \frac{2}{19} \begin{bmatrix} 5 & 4 \\ 2 & 7/2 \end{bmatrix} \begin{bmatrix} -8 \\ 10 \end{bmatrix}$$

$$V_1 = \frac{2}{19} [-40 + 40] = 0$$

$$\therefore V_0 = 0$$

3.44 Use nodal analysis to find I_o in the circuit in Fig. P3.44.

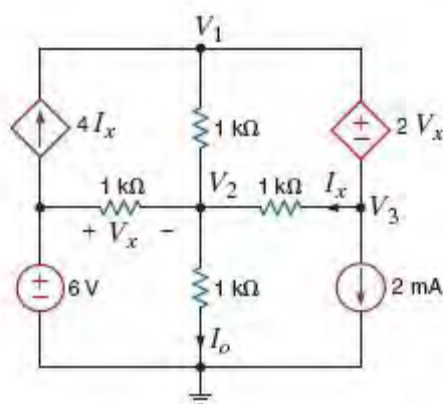


Figure P3.44

SOLUTION:

$$\frac{V_1 - V_2}{k} + \frac{V_3 - V_2}{k} - 4I_x = -\frac{2}{k}$$

$$I_x = \frac{V_3 - V_2}{k}$$

$$V_1 - V_3 = 2V_x = 2(6 - V_2)$$

$$\frac{V_2 - 6}{k} + \frac{V_2 - V_1}{k} + \frac{V_2 - V_3}{k} + \frac{V_2}{k} = 0$$

$$V_1 - V_2 + V_3 - V_2 - 4V_3 + 4V_2 = -2$$

$$V_1 - V_3 + 2V_2 = 12$$

$$V_2 + V_2 - V_1 + V_2 - V_3 + V_2 = 6$$

$$V_1 + 2V_2 - 3V_3 = -2$$

$$V_1 + 2V_2 - V_3 = 12$$

$$-V_1 + 4V_2 - V_3 = 6$$

$$V_1 = -2 - 2V_2 + 3V_3$$

$$-2 - 2V_2 + 3V_3 + 2V_2 - V_3 = 12$$

$$2 + 2V_2 - 3V_3 + 4V_2 - V_3 = 6$$

$$2V_3 = 14$$

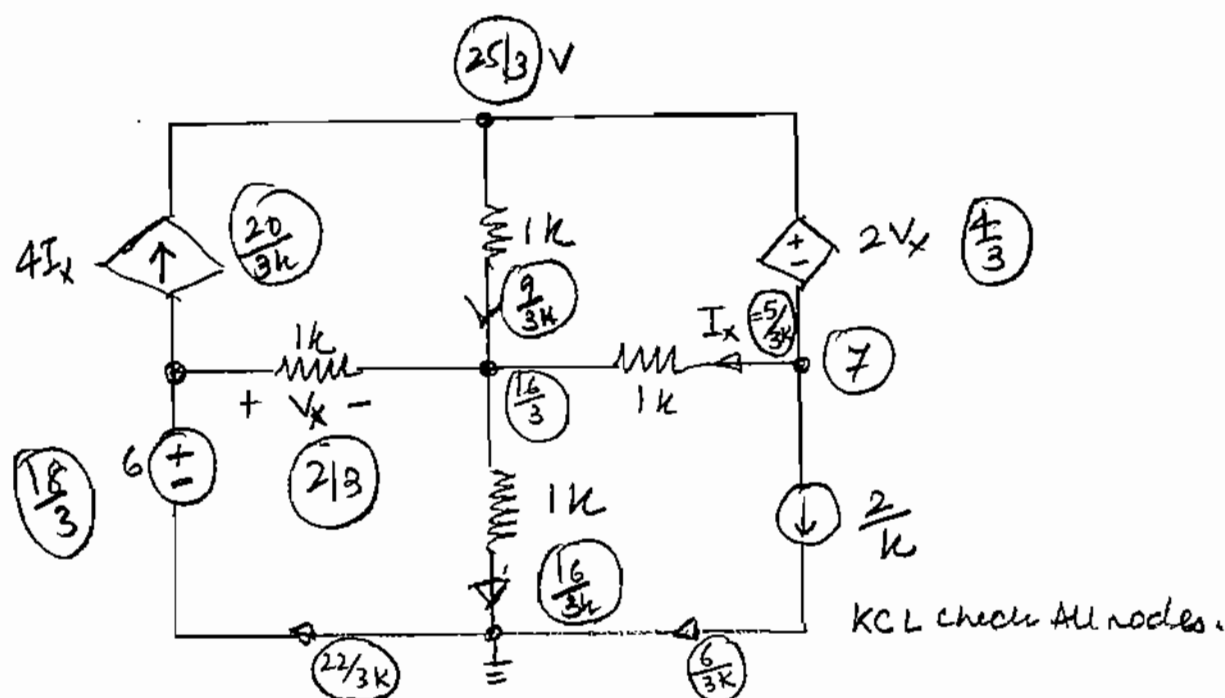
$$V_3 = 7V$$

$$2 + 2V_2 - 21 + 4V_2 - 7 = 6$$

$$6V_2 = 32$$

$$V_2 = \frac{16}{3} V$$

$$I_o = \frac{16/3}{1k} = \frac{16}{3} mA$$



3.45 Find I_o in the circuit in Fig. P3.45 using nodal analysis.

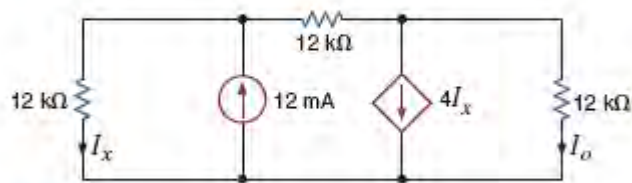
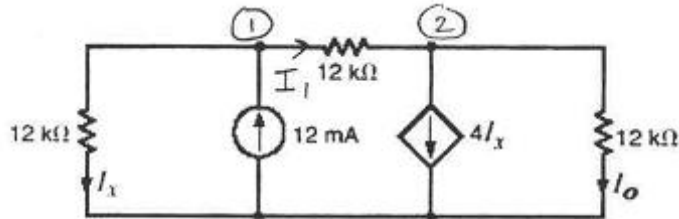


Figure P3.45

SOLUTION:



$$\text{KCL at } \textcircled{1} : 12 \text{ m} = I_x + I_1$$

$$\frac{V_1}{12 \text{ K}} + \frac{V_1 - V_2}{12 \text{ K}} = 12 \text{ m}$$

$$V_1 + V_1 - V_2 = 144$$

$$\boxed{2V_1 - V_2 = 144}$$

$$\text{KCL at } \textcircled{2} : I_1 = 4I_x + I_o$$

$$I_x = \frac{V_1}{12 \text{ K}}$$

$$I_1 = 4 \left(\frac{V_1}{12 \text{ K}} \right) + I_o$$

$$\frac{V_1 - V_2}{12 \text{ K}} = \frac{V_1}{3 \text{ K}} + \frac{V_2}{12 \text{ K}}$$

$$V_1 - V_2 = 4V_1 + V_2$$

$$\boxed{3V_1 + 2V_2 = 0}$$

$$2V_1 - V_2 = 144$$

$$3V_1 + 2V_2 = 0$$

$$V_1 = 41.14 \text{ V}$$

$$V_2 = -61.71 \text{ V}$$

$$I_0 = \frac{V_2}{12\text{K}} = \frac{-61.71}{12\text{K}}$$

$$I_0 = -5.14 \text{ mA}$$

3.46 Find V_o in the circuit in Fig. P3.46.

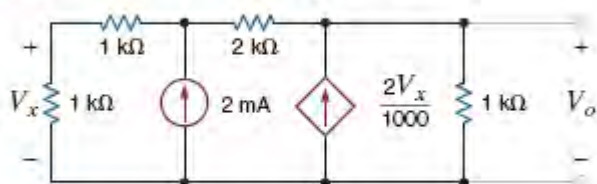
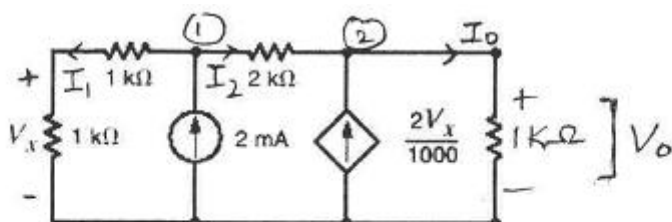


Figure P3.46

SOLUTION:



$$\text{KCL at } \textcircled{1} : 2 \text{ m} = I_1 + I_2$$

$$\frac{V_1}{1\text{K} + 1\text{K}} + \frac{V_1 - V_2}{2\text{K}} = 2\text{m}$$

$$V_1 + V_1 - V_2 = 4$$

$$\boxed{2V_1 - V_2 = 4}$$

$$\text{KCL at } \textcircled{2} : I_2 + \frac{2V_x}{1000} = I_o$$

$$\frac{V_1 - V_2}{2\text{K}} + \frac{2V_x}{1000} = \frac{V_2}{1\text{K}}$$

$$V_x = I_1(1\text{K}) = \left(\frac{V_1}{1\text{K} + 1\text{K}}\right)(1\text{K})$$

$$V_x = \frac{V_1}{2}$$

$$\frac{V_1 - V_2}{2\text{K}} + \frac{2\left(\frac{V_1}{2}\right)}{1000} = \frac{V_2}{1\text{K}}$$

$$V_1 - V_2 + 2V_1 = 2V_2$$

$$\boxed{3V_1 - 3V_2 = 0}$$

$$2V_1 - V_2 = 4$$

$$3V_1 - 3V_2 = 0$$

$$V_1 = 4 \text{ V}$$

$$V_2 = 4 \text{ V}$$

$$V_o = V_2 = 4 \text{ V}$$

- 3.47** Use nodal analysis to find V_o in the circuit in Fig. P3.47. In addition, find all branch currents and check your answers using KCL at every node.

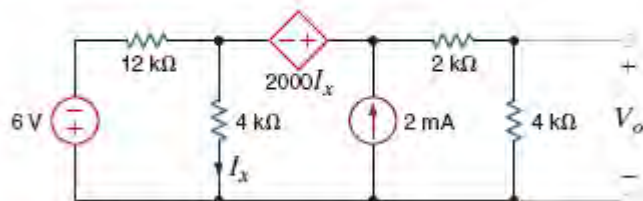
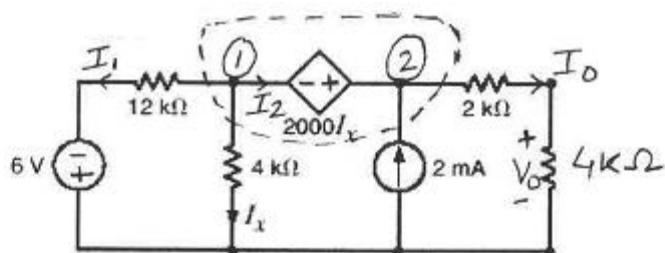


Figure P3.47

SOLUTION:



$$\text{KCL at supernode: } I_1 + I_x + I_o = 2\text{m}$$

$$\frac{V_1 - (-6)}{12\text{K}} + \frac{V_1}{4\text{K}} + \frac{V_2}{2\text{K} + 4\text{K}} = 2\text{m}$$

$$V_1 + 6 + 3V_1 + 2V_2 = 24$$

$$\boxed{4V_1 + 2V_2 = 18}$$

$$V_2 - V_1 = 2000 I_x$$

$$I_x = \frac{V_1}{4\text{K}}$$

$$-V_1 + V_2 = \frac{V_1}{2\text{K}}$$

$$\boxed{-3V_1 + 2V_2 = 0}$$

$$\begin{aligned} 4V_1 + 2V_2 &= 18 \\ -3V_1 + 2V_2 &= 0 \end{aligned}$$

$$V_1 = 2.57\text{ V}$$

$$V_2 = 3.86\text{ V}$$

$$I_0 = \frac{V_2}{2K + 4K}$$
$$= \frac{3.86}{6K}$$

$$I_0 = 0.643 \text{ mA}$$

$$V_0 = I_0(4K) = 0.643 \text{ m}(4K)$$

$$V_0 = 2.57 \text{ V}$$

$$\text{KCL at } \textcircled{1}: I_1 + I_2 + I_x = 0$$
$$I_1 = \frac{V_1 - (-6)}{12K}$$

$$I_1 = \frac{2.57 + 6}{12K}$$

$$I_1 = 0.714 \text{ mA}$$

$$I_x = \frac{V_1}{4K}$$
$$= \frac{2.57}{4K}$$

$$I_x = 0.643 \text{ mA}$$

$$I_2 = -I_x - I_1 = -1.357 \text{ mA}$$

$$\text{CHECK: } I_1 + I_2 + I_x = 0$$

$$0.714 \text{ m} - 1.357 \text{ m} + 0.643 \text{ m} = 0$$
$$0 = 0$$

$$\text{KCL at } \textcircled{2}: I_2 + 2m = I_o$$

$$I_2 + 2m - I_o = 0$$

$$-1.357m + 2m - 0.643m = 0$$

$$0 = 0$$

$$\text{KCL at reference node: } I_1 + I_x + I_o = 2m$$

$$0.714m + 0.643m + 0.643m = 2m$$

$$2m = 2m$$

3.48 Determine V_o in the network in Fig. P3.48 using nodal analysis.

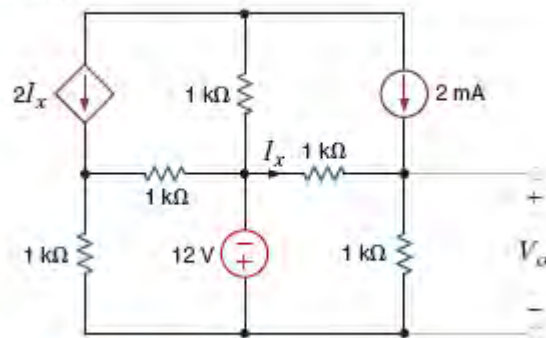
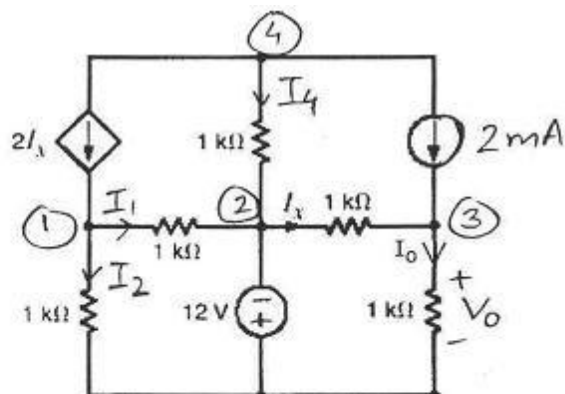


Figure P3.48

SOLUTION:



$$\text{KCL at } (4) : 2I_x + I_4 + 2\text{mA} = 0$$

$$I_x = \frac{V_2 - V_3}{1\text{K}}$$

$$2 \left[\frac{V_2 - V_3}{1\text{K}} \right] + \frac{V_4}{1\text{K}} + 2\text{mA} = 0$$

$$2V_2 - 2V_3 + V_4 + 2 = 0$$

$$\boxed{2V_2 - 2V_3 + V_4 = -2}$$

$$\text{KCL at } \textcircled{3}: I_x + 2m = I_0$$

$$\frac{V_2 - V_3}{1k} + 2m = \frac{V_3}{1k}$$

$$V_2 - V_3 + 2 = V_3$$

$$\boxed{V_2 - 2V_3 = -2}$$

$$\text{KCL at } \textcircled{1}: 2I_x = I_1 + I_2$$

$$2 \left[\frac{V_2 - V_3}{1k} \right] = \frac{V_1 - V_2}{1k} + \frac{V_1}{1k}$$

$$2V_2 - 2V_3 = V_1 - V_2 + V_1$$

$$\boxed{2V_1 - 3V_2 + 2V_3 = 0}$$

$$V_2 = -12V$$

$$-12 - 2V_3 = -2$$

$$-2V_3 = 10$$

$$V_3 = -5V$$

$$V_0 = V_3 = -5V$$

$$V_0 = -5V$$

3.49 Use nodal analysis to find V_1 , V_2 , V_3 , and V_4 in the circuit in Fig. P3.49.

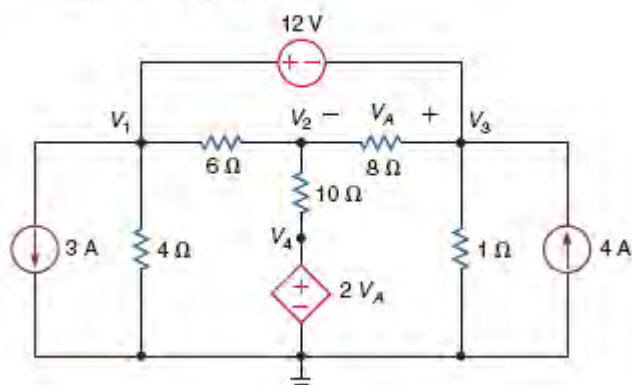
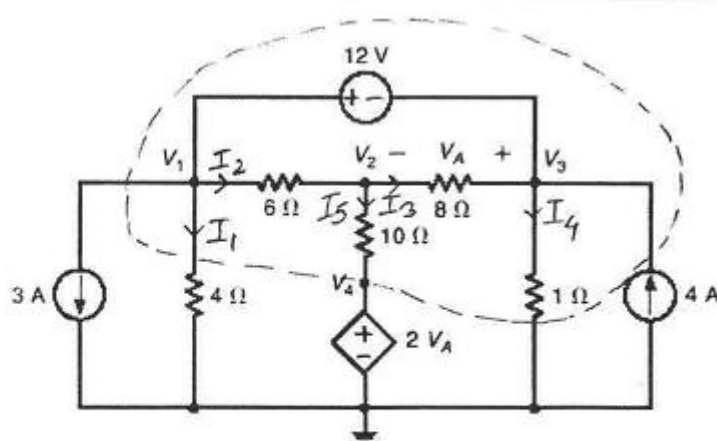


Figure P3.49

SOLUTION:



$$\text{KCL at (2): } I_2 = I_5 + I_3$$

$$\frac{V_1 - V_2}{6} = \frac{V_2 - V_4}{10} + \frac{V_2 - V_3}{8}$$

$$5V_1 - 5V_2 = 3V_2 - 3V_4 + 3.75V_2 - 3.75V_3$$

$$\boxed{5V_1 - 11.75V_2 + 3.75V_3 + 3V_4 = 0}$$

$$\text{KCL at supernode: } 3 + I_1 + I_5 + I_4 = 4$$

$$\frac{V_1}{4} + \frac{V_2 - V_4}{10} + \frac{V_3}{1} = 1$$

$$5V_1 + 2V_2 - 2V_4 + 20V_3 = 20$$

$$\boxed{5V_1 + 2V_2 + 20V_3 - 2V_4 = 20}$$

$$V_1 - V_3 = 12$$

$$V_4 = 2V_A$$

$$V_A = V_3 - V_2$$

$$V_4 = 2(V_3 - V_2)$$

$$\boxed{-2V_2 + 2V_3 - V_4 = 0}$$

$$5V_1 - 11.75V_2 + 3.75V_3 + 3V_4 = 0$$

$$5V_1 + 2V_2 + 20V_3 - 2V_4 = 20$$

$$V_1 + 0V_2 - V_3 + 0V_4 = 12$$

$$0V_1 - 2V_2 + 2V_3 - V_4 = 0$$

$$V_1 = 9.68 \text{ V}$$

$$V_2 = 1.45 \text{ V}$$

$$V_3 = -2.32 \text{ V}$$

$$V_4 = -7.54 \text{ V}$$

3.50 Use nodal analysis to determine the node voltages defined in the circuit in Fig. P3.50.

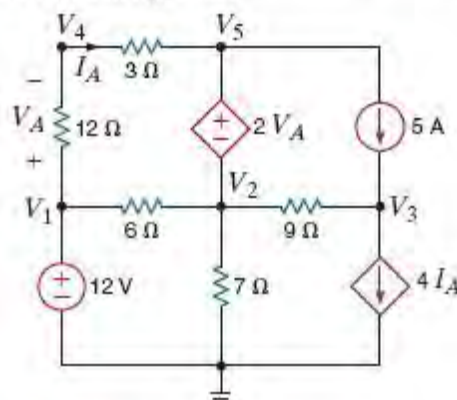
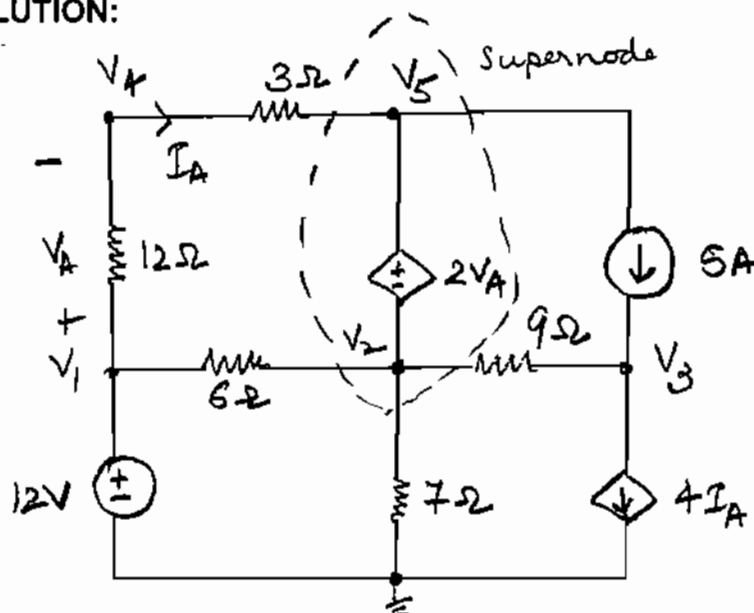


Figure P3.50

SOLUTION:



5 node voltages $\rightarrow$ 5 equations

$$\left. \begin{aligned} \textcircled{1} \quad & V_1 = 12 \text{ V} \\ & V_5 - V_2 = 2V_A = 2(V_1 - V_4) \\ \textcircled{2} \quad & 2V_1 + V_2 - 2V_4 - V_5 = 0 \end{aligned} \right\} \text{constraint equations}$$

Node 3:

$$\frac{V_3 - V_2}{9} = 5 - 4I_A$$

$$I_A = \frac{V_4 - V_5}{3}$$

$$\frac{V_3 - V_2}{9} = 5 - \frac{4}{3}(V_4 - V_5)$$

$$\textcircled{3} \quad -\frac{1}{9}V_2 + \frac{1}{9}V_3 + \frac{4}{3}V_4 - \frac{4}{3}V_5 = 5$$

Node 4:

$$\frac{V_4 - V_5}{3} + \frac{V_4 - V_1}{12} = 0$$

$$\textcircled{4} \quad -\frac{1}{12}V_1 + \frac{5}{12}V_4 - \frac{1}{3}V_5 = 0$$

KCL @ supernode:

$$\frac{V_5 - V_4}{3} + \frac{V_2 - V_1}{6} + \frac{V_1}{7} + \frac{V_2 - V_3}{9} = -5$$

$$\textcircled{5} \quad -\frac{1}{6}V_1 + \left(\frac{1}{6} + \frac{1}{7} + \frac{1}{9}\right)V_2 - \frac{1}{9}V_3 - \frac{1}{3}V_4 + \frac{1}{3}V_5 = -5$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & -2 & -1 \\ 0 & -1/9 & 1/9 & 4/3 & -4/3 \\ -1/12 & 0 & 0 & 5/12 & -1/3 \\ -1/6 & 1/6 + 1/7 + 1/9 & -1/9 & -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 5 \\ 0 \\ -5 \end{bmatrix}$$

$$V_1 = 12V \quad V_2 = 4.63V \quad V_3 = 42.83V$$
$$V_4 = 9.73V \quad V_5 = 9.17V$$

3.51 Use nodal analysis to determine the node voltages defined in the circuit in Fig. P3.51.

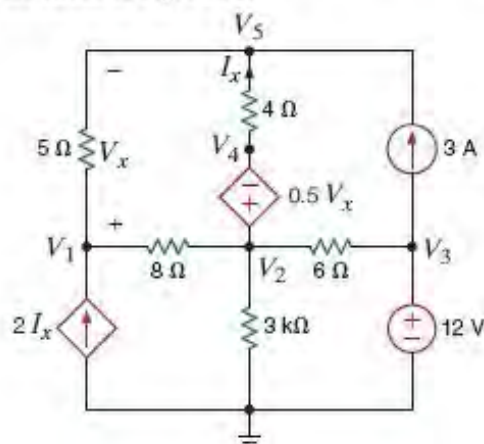
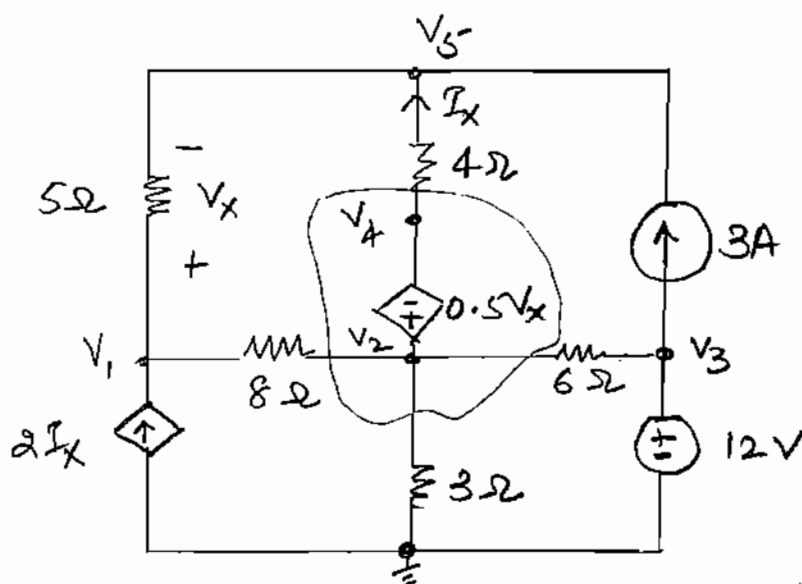


Figure P3.51

SOLUTION:



Constraint equations

$$(1) V_3 = 12 \text{ V} \quad V_2 - V_4 = 0.5V_x = 0.5(V_1 - V_5)$$

$$(2) 0.5V_1 - V_2 + V_4 - 0.5V_5 = 0$$

$$\frac{V_1 - V_2}{8} + \frac{V_1 - V_5}{5} = 2I_x = 2\left(\frac{V_4 - V_5}{4}\right) = \frac{1}{2}V_4 - \frac{1}{2}V_5$$

$$(3) \quad 0.325V_1 - 0.125V_2 - 0.5V_4 + 0.3V_5 = 0$$

$$\frac{V_5 - V_1}{5} + \frac{V_5 - V_4}{4} = 3$$

$$(4) \quad -0.2V_1 - 0.25V_4 + 0.45V_5 = 3$$

KCL @ supernode:

$$\frac{V_2 - V_1}{8} + \frac{V_2}{3} + \frac{V_2 - V_3}{6} + \frac{V_4 - V_5}{4} = 0$$

$$-0.125V_1 + 0.625V_2 - 0.167V_3 + 0.25V_4 - 0.25V_5 = 0$$

$$(5) \quad -0.25V_5 = 0$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0.5 & -1 & 0 & 1 & -0.5 \\ 0.325 & -0.125 & 0 & -0.5 & 0.3 \\ -0.2 & 0 & 0 & -0.25 & 0.45 \\ -0.125 & 0.625 & -0.167 & 0.25 & -0.25 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix}$$

$$= \begin{bmatrix} 12 \\ 0 \\ 0 \\ 3 \\ 0 \end{bmatrix}$$

$$V_1 = 5.09 \text{ V}$$

$$V_2 = 5.88 \text{ V} \quad V_3 = 12 \text{ V}$$

$$V_4 = 10.8 \text{ V}$$

$$V_5 = 14.93 \text{ V}$$

3.52 Use nodal analysis to determine the node voltages defined in the circuit in Fig. P3.52.

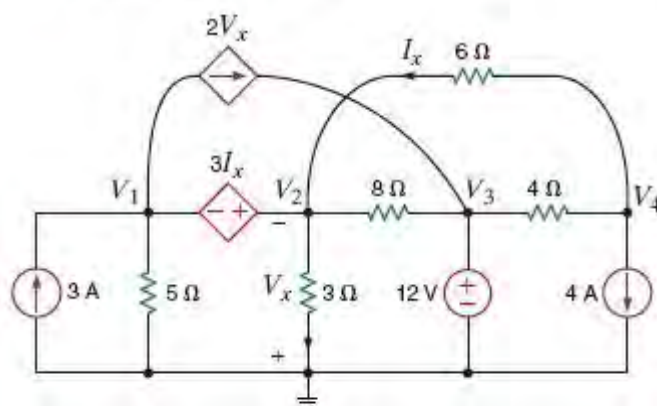
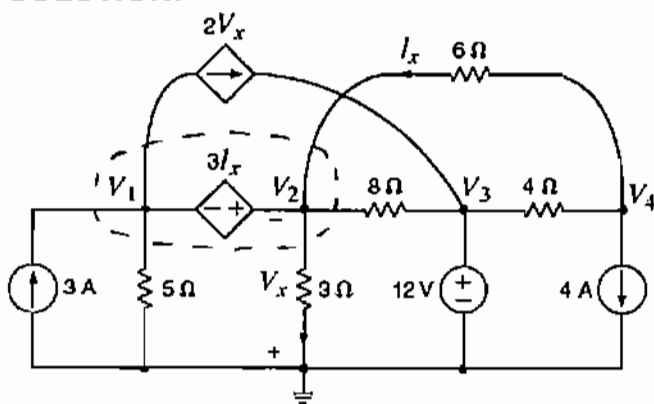


Figure P3.52

SOLUTION:



Constraint equations:

$$\textcircled{1} V_3 = 12 \text{ V}$$

$$V_2 - V_1 = 3 I_x = 3 \left(\frac{V_4 - V_2}{6} \right) = 0.5 V_4 - 0.5 V_2$$

$$\textcircled{2} V_1 - 1.5 V_2 + 0.5 V_4 = 0$$

$$\frac{V_4 - V_3}{4} + \frac{V_4 - V_2}{6} = -4$$

$$-\frac{1}{6} V_2 - \frac{1}{4} V_3 + \left(\frac{1}{6} + \frac{1}{4} \right) V_4 = -4$$

$$\textcircled{3} -\frac{1}{6}V_2 - \frac{1}{4}V_3 + \frac{5}{12}V_4 = -4$$

KCL @ supernode:

$$\frac{V_1}{5} + \frac{V_2}{3} + \frac{V_2 - V_3}{8} + \frac{V_2 - V_4}{6} = 3 - 2V_x$$

$$\frac{V_1}{5} + \frac{V_2}{3} + \frac{V_2 - V_3}{8} + \frac{V_2 - V_4}{6} = 3 + 2V_2$$

$$\frac{1}{5}V_1 + \left(\frac{1}{3} + \frac{1}{8} + \frac{1}{6} - 2\right)V_2 - \frac{1}{8}V_3 - \frac{1}{6}V_4 = 3$$

$$\textcircled{4} 0.2V_1 - 1.375V_2 - 0.125V_3 - 0.167V_4 = 3$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & -1.5 & 0 & 0.5 \\ 0 & -0.167 & -0.25 & 0.417 \\ 0.2 & -1.375 & -0.125 & -0.167 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ -4 \\ 3 \end{bmatrix}$$

$$V_1 = -3.05V$$

$$V_3 = 12V$$

$$V_2 = -3.27V$$

$$V_4 = -3.71V$$

3.53 Find I_o in the network in Fig. P3.53 using mesh analysis.

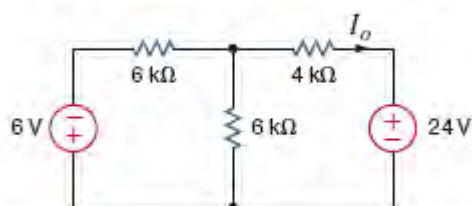
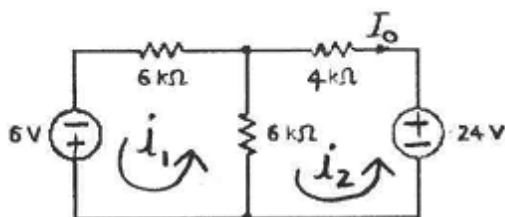


Figure P3.53

SOLUTION:



Let the loop current be I_1 & I_2 mA
as indicated in the figure

KVL for loop 1 :

$$6 = 6I_1 + 6(I_1 - I_2)$$

$$\Rightarrow 6 = 12I_1 - 6I_2 \quad \text{--- (1)}$$

KVL for the loop 2 :

$$24 = 4I_2 + 6(I_2 - I_1)$$

$$\Rightarrow 24 = -6I_1 + 10I_2 \quad \text{--- (2)}$$

Solving equation ① & ② gives

$$I_2 = \frac{27}{7} \text{ mA}$$

$$\therefore I_o = -I_2$$

$$I_o = -\frac{27}{7} \text{ mA} = -3.857 \text{ mA}$$

$$\boxed{I_o = -3.857 \text{ mA}}$$

3.54 Find I_o in the circuit in Fig. P3.54.

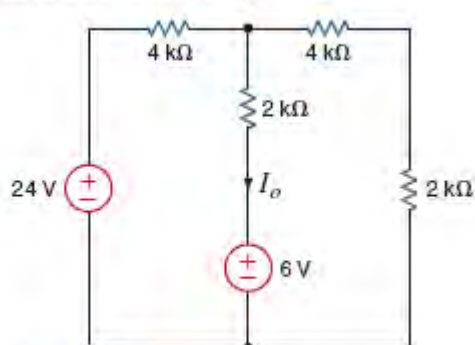
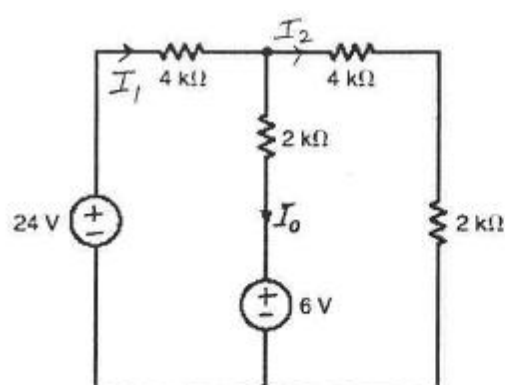


Figure P3.54

SOLUTION:



KCL at ①: $I_1 = I_o + I_2$

$$\frac{24 - V_1}{4K} = \frac{V_1 - 6}{2K} + \frac{V_1}{4K + 2K}$$

$$72 - 3V_1 = 6V_1 - 36 + 2V_1$$

$$11V_1 = 108$$

$$V_1 = 9.82 \text{ V}$$

$$I_o = \frac{V_1 - 6}{2K}$$

$$= \frac{9.82 - 6}{2K}$$

$$I_o = 1.91 \text{ mA}$$

3.55 Find V_o in the network in Fig. P3.55 using mesh analysis.

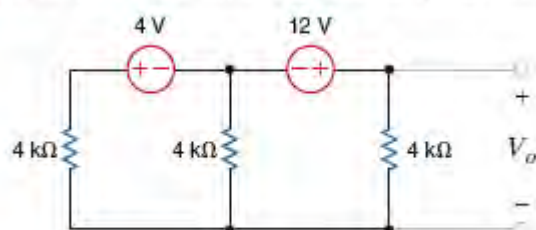
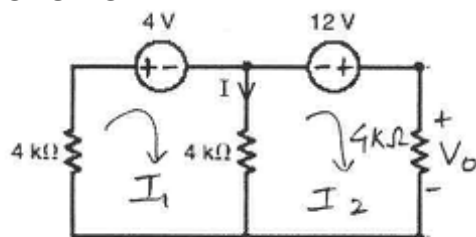


Figure P3.55

SOLUTION:



$$\text{KCL: } I_1 = I + I_2$$

$$I = I_1 - I_2$$

$$\text{KVL left loop: } 4kI_1 + 4 + 4kI = 0$$

$$4kI_1 + 4k(I_1 - I_2) = -4$$

$$\boxed{8kI_1 - 4kI_2 = -4}$$

$$\text{KVL right loop: } 12 = 4kI_2 + 4k(-I)$$

$$4kI_2 - 4k(I_1 - I_2) = 12$$

$$\boxed{-4kI_1 + 8kI_2 = 12}$$

$$8kI_1 - 4kI_2 = -4$$

$$-4kI_1 + 8kI_2 = 12$$

$$I_1 = 0.333 \text{ mA}$$

$$I_2 = 1.67 \text{ mA}$$

$$V_o = 4kI_2 = 4k(1.67 \text{ m})$$

$$V_o = 6.68 \text{ V}$$

3.56 Find I_o in the circuit in Fig. P3.56.

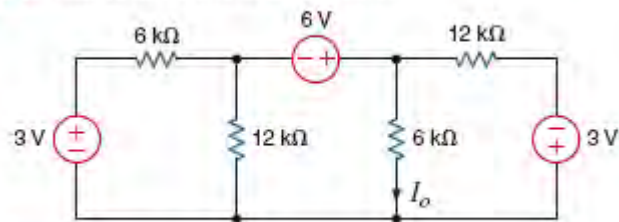
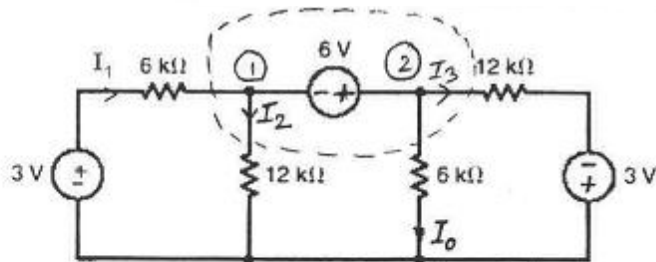


Figure P3.56

SOLUTION:



$$\text{KCL at supernode: } I_1 = I_2 + I_o + I_3$$

$$\frac{3 - V_1}{6K} = \frac{V_1}{12K} + \frac{V_2}{6K} + \frac{V_2 - (-3)}{12K}$$

$$6 - 2V_1 = V_1 + 2V_2 + V_2 + 3$$

$$\boxed{3V_1 + 3V_2 = 3}$$

$$V_2 - V_1 = 6$$

$$\boxed{-V_1 + V_2 = 6}$$

$$3V_1 + 3V_2 = 3$$

$$-V_1 + V_2 = 6$$

$$V_1 = -2.5 \text{ V}$$

$$V_2 = 3.5 \text{ V}$$

$$I_o = \frac{V_2}{6K} = \frac{3.5}{6K}$$

$$I_o = 0.5833 \text{ mA}$$

- 3.57 Use mesh analysis to find the power dissipated in the $6\text{-k}\Omega$ resistor in Fig. P3.57.

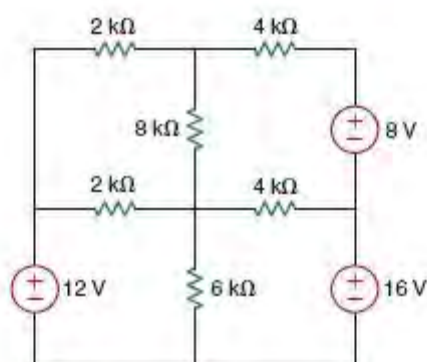
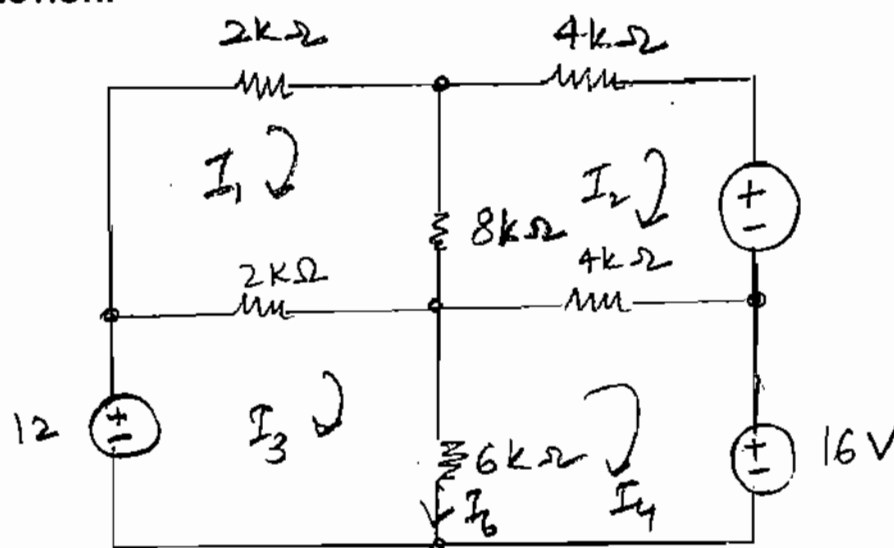


Figure P3.57

SOLUTION:



$$\begin{bmatrix} 12000 & -8000 & -2000 & 0 \\ -8000 & 16000 & 0 & -4000 \\ -2000 & 0 & 8000 & -6000 \\ 0 & -4000 & -6000 & 10000 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -8 \\ 12 \\ -16 \end{bmatrix}$$

$$I_1 = -1.7\text{mA}$$

$$I_3 = -1.4\text{mA}$$

$$I_2 = -2.2\text{mA}$$

$$I_4 = -3.3\text{mA}$$

$$\begin{aligned} I_6 &= I_3 - I_4 = -1.4\text{mA} - (-3.3\text{mA}) \\ &= 1.9\text{mA} \end{aligned}$$

$$P_{6k} = (1.9\text{mA})^2 (6k) = \underline{21.66\text{mW}}$$

3.58 Find V_o in the circuit in Fig. P3.58 using mesh analysis.

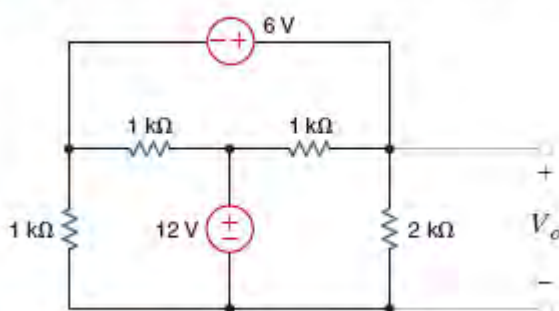
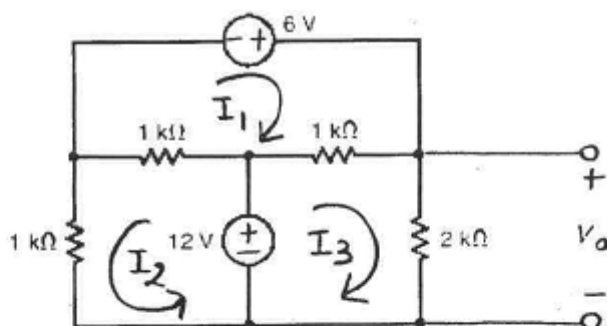


Figure P3.58

SOLUTION:



Let the mesh/loop currents be I_1 , I_2 & I_3 as shown in the diagram

KVL in loop 1 :

$$6 = (I_1)1 - I_3 + 1(I_1 + I_2)$$

$$\Rightarrow 6 = 2I_1 + I_2 - I_3 \quad \text{--- ①}$$

KVL in the loop 2 :

$$12 = 2I_2 + I_1 \quad \text{--- ②}$$

3.59 Use loop analysis to find V_o in the network in Fig. P3.59.

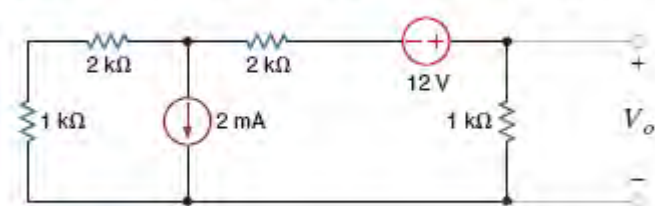
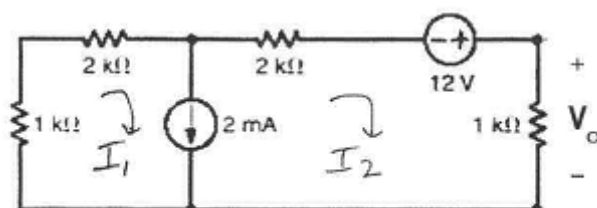


Figure P3.59

SOLUTION:



$$\text{KCL : } I_1 = 2\text{ m} + I_2$$

$$\boxed{I_1 - I_2 = 2\text{ m}}$$

$$\text{KVL : } 12 = 1\text{K}I_1 + 2\text{K}I_1 + 2\text{K}I_2 + 1\text{K}I_2$$

$$\boxed{3\text{K}I_1 + 3\text{K}I_2 = 12}$$

$$I_1 - I_2 = 2\text{ m}$$

$$3\text{K}I_1 + 3\text{K}I_2 = 12$$

$$I_1 = 3\text{ mA}$$

$$I_2 = 1\text{ mA}$$

$$V_o = 1\text{K}(I_2)$$

$$V_o = 1\text{K}(1\text{ m})$$

$$V_o = 1\text{ V}$$

3.60 Use mesh analysis to find V_o in the network in Fig. P3.60.

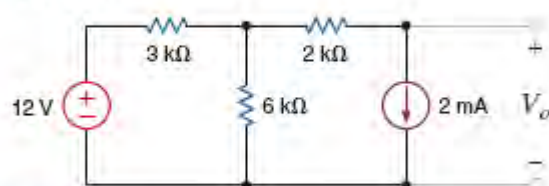
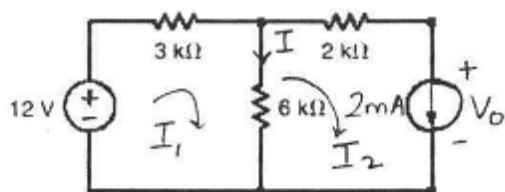


Figure P3.60

SOLUTION:



$$\begin{aligned} \text{KCL: } I_1 &= I + I_2 \\ I &= I_1 - I_2 \\ \boxed{I} &= \boxed{I_1 - 2\text{m}} \end{aligned}$$

$$\begin{aligned} \text{KVL: } 12 &= 3\text{K}I_1 + 6\text{K}I_1 \\ 6\text{K}(I_1 - 2\text{m}) + 3\text{K}I_1 &= 12 \\ 9\text{K}I_1 &= 24 \\ I_1 &= 2.67\text{ mA} \end{aligned}$$

$$\begin{aligned} \text{KVL outerloop: } 12 &= 3\text{K}I_1 + 2\text{K}I_2 + V_o \\ V_o &= 12 - 3\text{K}(2.67) - 2\text{K}(2\text{m}) \\ V_o &= 0\text{ V} \end{aligned}$$

3.61 Find V_o in the network in Fig. P3.61 using loop analysis.

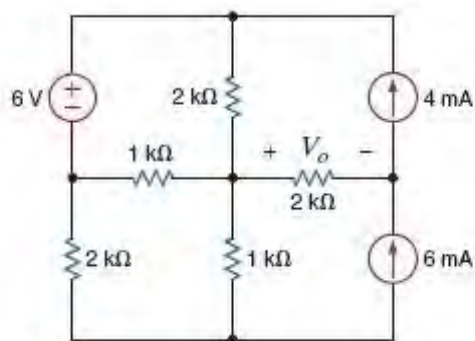


Figure P3.61

SOLUTION:

$$V_o = -(2k) \left(\frac{2}{k} \right) = -4V$$

3.62 Find V_o in the circuit in Fig. P3.62 using loop analysis.

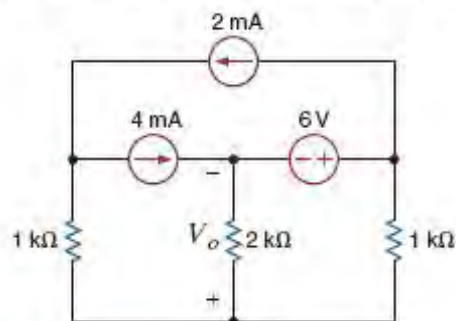
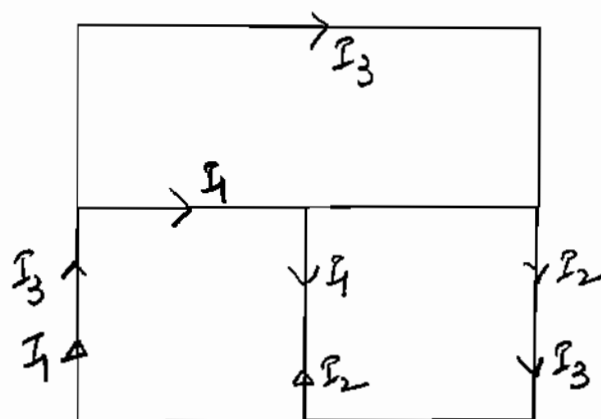
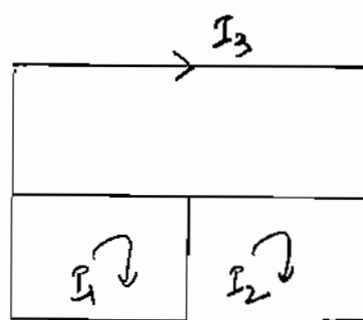


Figure P3.62

SOLUTION:



$$I_1 = \frac{4}{k}$$

$$I_3 = -2/k$$

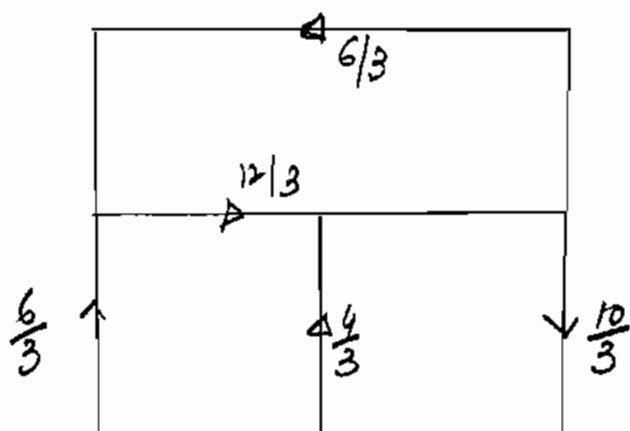
$$2k(I_2 - I_1) + 1k(I_2 + I_3) = 6$$

$$2kI_2 - 2kI_1 + 1kI_2 + 1kI_3 = 6$$

$$3kI_2 - 8 - 2 = 6$$

$$I_2 = 16/3kA$$

$$V_o = 2k(I_1 - I_2) = 2k\left(\frac{12}{3k} - \frac{16}{3k}\right) = -\frac{8}{3}V$$



3.63 Find I_o in the circuit in Fig. P3.63 using loop analysis.

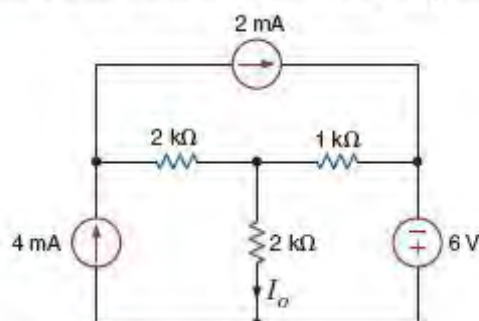
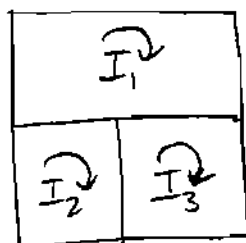


Figure P3.63

SOLUTION:

$$2k(I_3 - I_2) + 1k(I_3 - I_1) - 6 = 0$$

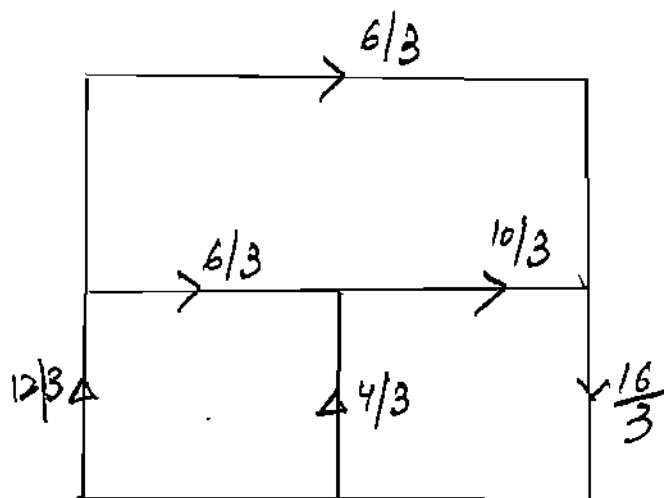


$$2kI_3 - 2kI_2 + 1kI_3 - 1kI_1 = 6$$

$$3kI_3 = 6 + 8 + 2$$

$$I_3 = 16/3k$$

$$I_o = I_2 - I_3 = \frac{12}{3k} - \frac{16}{3k} = -\frac{4}{3}kA$$



3.64 Find I_o in the network in Fig. P3.64 using loop analysis.

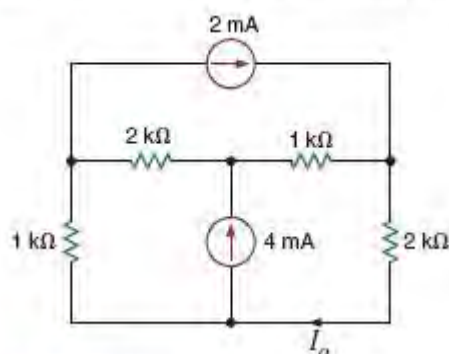
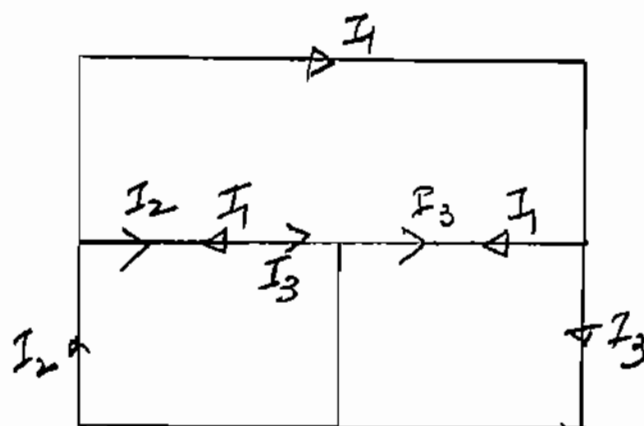
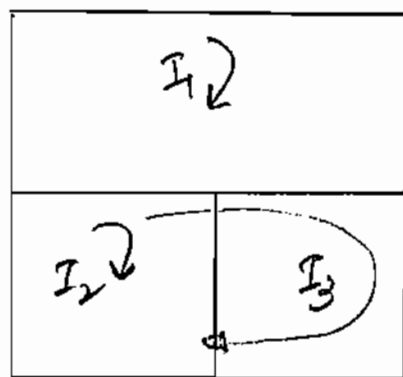


Figure P3.64

SOLUTION:



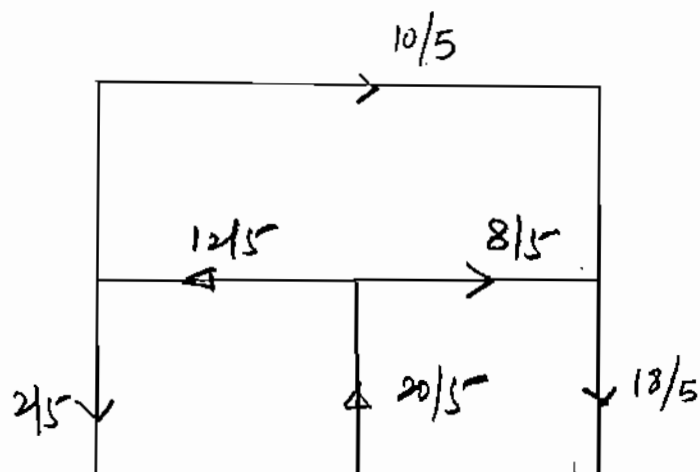
$$I_1 = \frac{2}{k} \quad I_2 = -\frac{4}{k}$$

$$1kI_2 + 2k(I_2 + I_3 - I_1) + 1k(I_3 - I_1) + I_3(2k) = 0$$

$$1kI_2 + 2kI_2 + 2kI_3 - 2kI_1 + 1kI_3 - 1kI_1 + 2kI_3 = 0$$

$$3k I_2 - 3k I_1 + 5k I_3 = 0$$

$$-12 - 6 + 5k I_3 = 0 \quad I_3 = \frac{18}{5k} \text{ A}$$



3.65 Find V_o in the network in Fig. P3.65 using loop analysis.

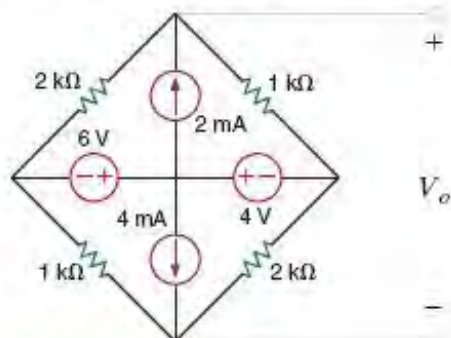
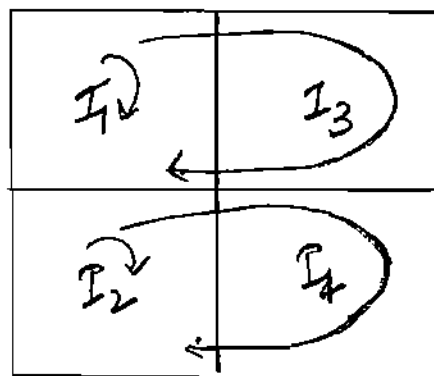


Figure P3.65

SOLUTION:



$$I_1 = -2/k$$

$$I_2 = 4/k$$

$$2k(I_1 + I_3) + 1kI_3 + 2 = 0$$

$$2k(I_4) + 1k(I_2 + I_4) - 2 = 0$$

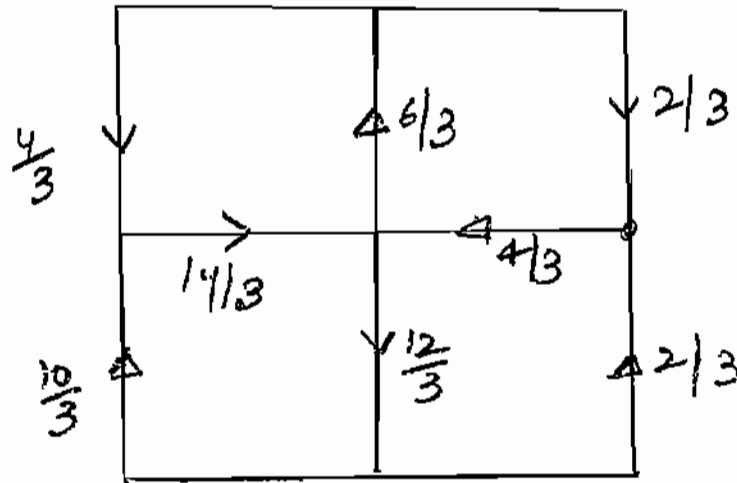
$$2kI_1 + 3kI_3 = -2$$

$$3kI_4 + 1kI_2 = 2$$

$$3kI_3 = 2 \quad I_3 = 2/3k$$

$$3kI_4 = -2 \quad I_4 = -2/3k$$

$$V_0 = 1kI_3 + 2kI_4 = \frac{2}{3} - \frac{4}{3} = -\frac{2}{3}V$$



3.66 Find V_o in the circuit in Fig. P3.66 using loop analysis.

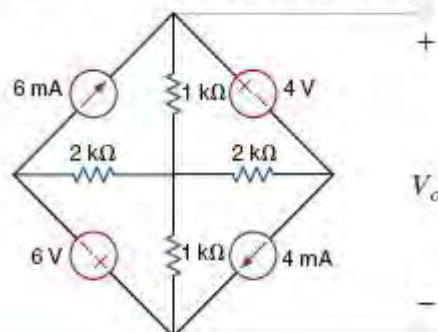
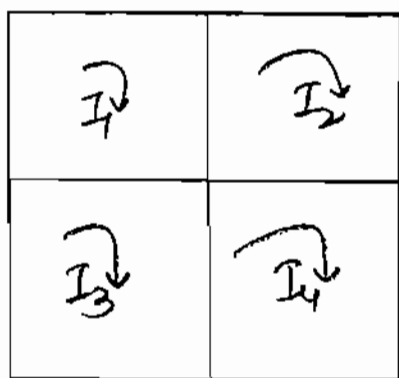


Figure P3.66

SOLUTION:



$$I_1 = \frac{6}{k}$$

$$I_4 = \frac{4}{k}$$

$$1k(I_2 - I_1) + 2k(I_2 - I_4) = -4$$

$$2k(I_3 - I_1) + 1k(I_3 - I_4) = -6$$

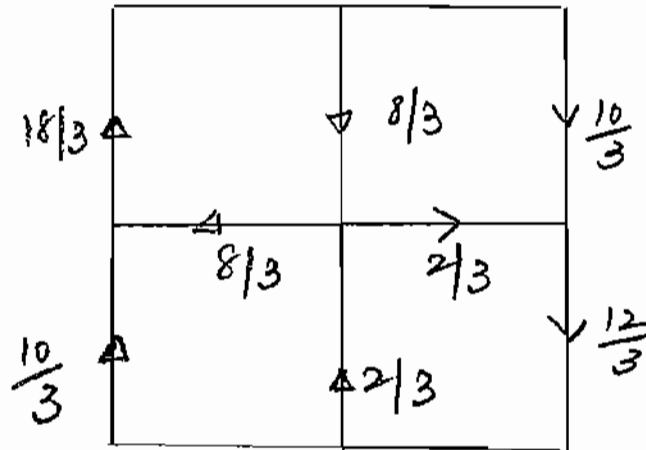
$$3kI_2 = -4 + 6 + 8 \quad I_2 = \frac{10}{3k}$$

$$3kI_3 = -6 + 12 + 4 \quad I_3 = \frac{10}{3k}$$

$$V_o = 4 + 2k(I_2 - I_4) + 1k(I_3 - I_4)$$

$$= 4 + \frac{20}{3} - \frac{24}{3} + \frac{10}{3} - \frac{12}{3}$$

$$= \frac{12}{3} - \frac{6}{3} = \frac{6}{3} = 2V$$



3.67 Find I_o in the network in Fig. P3.67 using loop analysis.

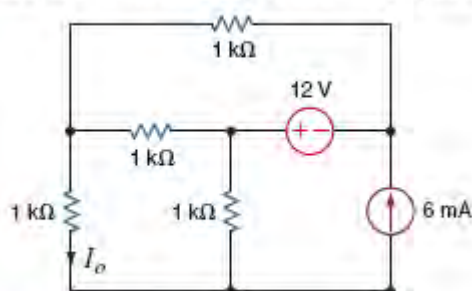
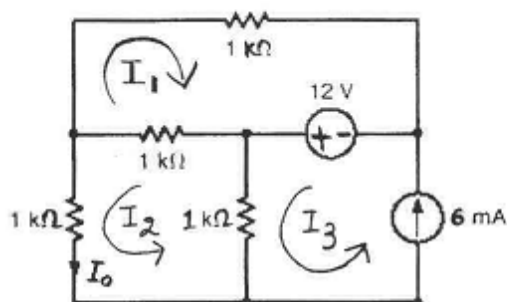


Figure P3.67

SOLUTION:



Let I_1 , I_2 and I_3 be the loop currents (in mA).

KVL in the first loop:

$$12 = 1(I_1 + I_2) + 1 \times I_1$$

$$\Rightarrow 12 = 2I_1 + I_2 \quad \text{--- (1)}$$

From the third loop:

$$I_3 = 6 \text{ mA} \quad \text{--- (2)}$$

KVL in the second loop:

$$1(I_1 + I_2) + 1(I_2) + 1(I_2 - I_3) = 0$$

$$\Rightarrow I_1 + 3I_2 - I_3 = 0 \quad \text{--- (3)}$$

From equation (2) & (3), we get:

$$I_1 + 3I_2 = I_3 = 6 \quad \text{--- (4)}$$

From equation (1) & (4):

$$12 = 2I_1 + I_2 \quad \text{--- (1)}$$

$$6 = I_1 + 3I_2$$

$$\Rightarrow 12 = 2I_1 + 6I_2 \quad \text{--- (5)}$$

$$5I_2 = 0 \quad \{ \text{eq. (5)} - \text{eq. (1)} \}$$

$$\text{or } \boxed{I_2 = 0 \text{ mA}}$$

3.68 Find I_o in the network in Fig. P3.68 using loop analysis.

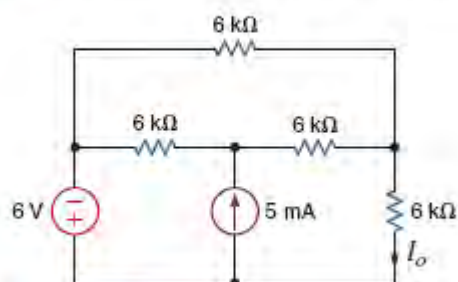
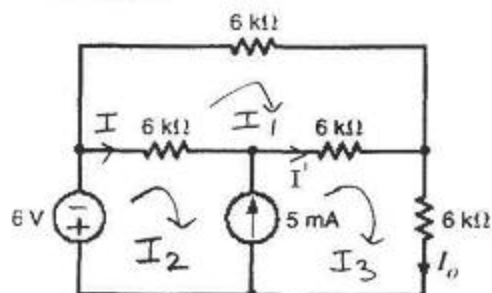


Figure P3.68

SOLUTION:



$$I_3 = I_o$$

$$\text{KCL: } I_2 = I + I_1$$

$$I = I_2 - I_1$$

$$\text{KCL: } I_1 + I' = I_3$$

$$I' = I_3 - I_1$$

KVL: KVL in the top loop:

$$18I_1 - 6I_2 - 6I_3 = 0$$

$$\boxed{3I_1 - I_2 - I_3 = 0}$$

$$\text{KVL: } 6 + 6KI + 6KI' + 6KI_o = 0$$

$$6K(I_2 - I_1) + 6K(I_3 - I_1) + 6KI_3 = -6$$

$$\boxed{-12KI_1 + 6KI_2 + 12KI_3 = -6} \Rightarrow \boxed{2I_1 - I_2 - 2I_3 = 1}$$

$$\text{KCL: } I_3 = 5\text{m} + I_2$$

$$\boxed{-I_2 + I_3 = 5\text{m}}$$

upon solving these equations we get, $i_3 = 2/5 = 0.4 \text{ mA}$

% MATLAB Code and Solution for Problem 3.65

$$R = [3, -1, -1; 2, -1, -2; 0, -2, 1];$$

$$V = [0; 1; 5]$$

$$I_{matrix} = \text{inv}(R) * V$$

>>

$I_{matrix} =$

- 0.0016

- 0.0044

0.0006

3.69 Use loop analysis to find V_o in the circuit in Fig. P3.69.

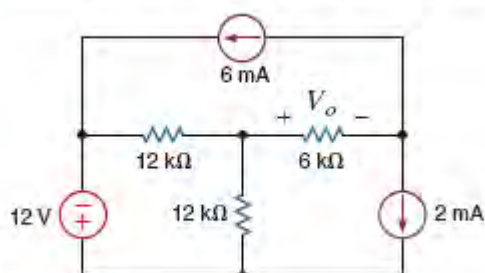
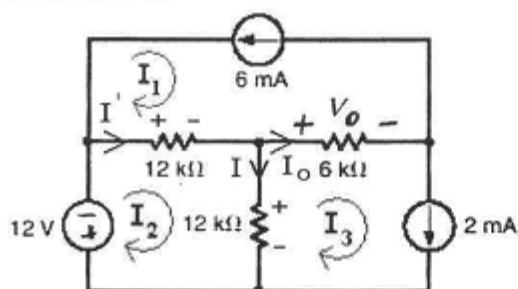


Figure P3.69

SOLUTION:



$$I_1 = -6 \text{ mA} \quad \text{and} \quad I_3 = 2 \text{ mA}$$

$$\begin{aligned} \text{KCL: } 6\text{m} + I_2 &= I' \\ I' &= 6\text{m} + I_2 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I + I_3 &= I_2 \\ I &= I_2 - I_3 \end{aligned}$$

$$\begin{aligned} \text{KVL: } -12 &= 12\text{K}I' + 12\text{K}I \\ 12\text{K}(6\text{m} + I_2) + 12\text{K}(I_2 - I_3) &= -12 \\ 72 + 12\text{K}I_2 + 12\text{K}I_2 - 12\text{K}I_3 &= -12 \\ \boxed{24\text{K}I_2 - 12\text{K}I_3} &= -84 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_1 + I_o &= I_3 \\ I_o &= I_3 - I_1 \\ I_o &= 2\text{m} - (-6\text{m}) \\ I_o &= 8 \text{ mA} \end{aligned}$$

$$V_o = I_o (6K)$$

$$V_o = 8m (6K)$$

$$V_o = 48 V$$

3.70 Using loop analysis, find V_o in the network in Fig. P3.70.

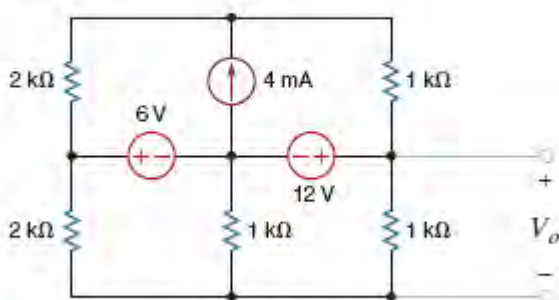
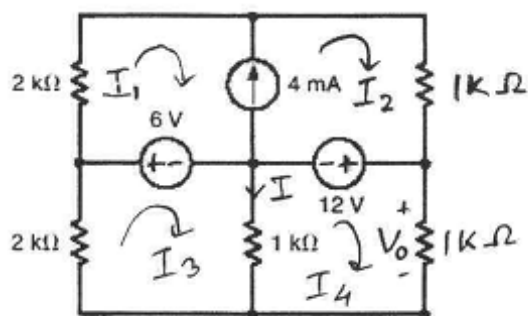


Figure P3.70

SOLUTION:



$$\text{KCL: } I + I_4 = I_3$$

$$I = I_3 - I_4$$

$$\text{KVL: } 6 = 2\text{K}I_1 + 1\text{K}I_2 + 12$$

$$\boxed{2\text{K}I_1 + 1\text{K}I_2 = -6}$$

$$\text{KVL: } 12 = 6 + 2\text{K}I_3 + 1\text{K}I_4$$

$$\boxed{2\text{K}I_3 + 1\text{K}I_4 = 6}$$

$$\text{KVL: } 2\text{K}I_3 + 6 + KI = 0$$

$$2\text{K}I_3 + 1\text{K}(I_3 - I_4) = -6$$

$$\boxed{3\text{K}I_3 - 1\text{K}I_4 = -6}$$

$$2\text{K}I_3 + 1\text{K}I_4 = 6$$

$$3\text{K}I_3 - 1\text{K}I_4 = -6$$

$$I_3 = 0\text{A}$$

$$I_4 = 6\text{mA}$$

$$V_o = 1\text{K}(I_4) = 6\text{V}$$

3.71 Find I_o in the circuit in Fig. P3.71.

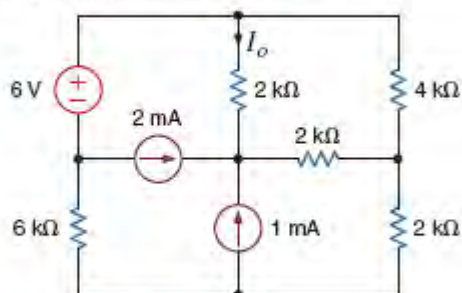
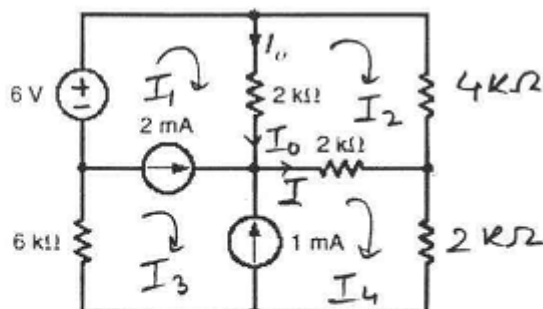


Figure P3.71

SOLUTION:



$$\text{KCL: } I = I_4 - I_2$$

$$\begin{aligned} \text{KCL: } I_1 &= I_0 + I_2 \\ I_0 &= I_1 - I_2 \end{aligned}$$

$$\begin{aligned} \text{KVL: } 4\text{k}I_2 + 2\text{k}(-I) + 2\text{k}(-I_0) &= 0 \\ 4\text{k}I_2 - 2\text{k}(I_4 - I_2) - 2\text{k}(I_1 - I_2) &= 0 \\ -2\text{k}I_1 + 8\text{k}I_2 - 2\text{k}I_4 &= 0 \end{aligned}$$

$$\begin{aligned} \text{KVL: } 6 &= 4\text{k}I_2 + 2\text{k}I_4 + 6\text{k}I_3 \\ 4\text{k}I_2 + 6\text{k}I_3 + 2\text{k}I_4 &= 6 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_3 &= 2\text{m} + I_1 \\ -I_1 + I_3 &= 2\text{m} \end{aligned}$$

$$\text{KCL: } I_4 = 1\text{m} + I_3$$

$$\begin{aligned}-2kI_1 + 8kI_2 + 0I_3 - 2kI_4 &= 0 \\ 0I_1 + 4kI_2 + 6kI_3 + 2kI_4 &= 6 \\ -I_1 + 0I_2 + I_3 + 0I_4 &= 2m \\ 0I_1 + 0I_2 - I_3 + I_4 &= 1m\end{aligned}$$

$$I_1 = -1.5 \text{ mA}$$

$$I_2 = 0 \text{ A}$$

$$I_3 = 0.5 \text{ mA}$$

$$I_4 = 1.5 \text{ mA}$$

$$I_0 = I_1 - I_2$$

$$= -1.5 - 0$$

$$I_0 = -1.5 \text{ mA}$$

3.72 Use loop analysis to find I_o in the network in Fig. P3.72.

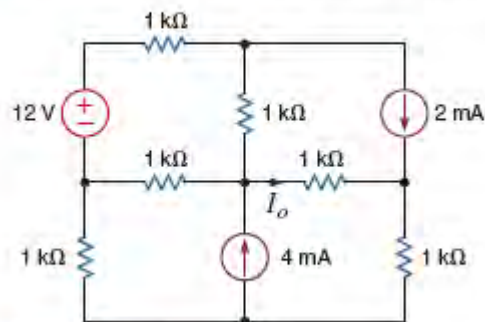
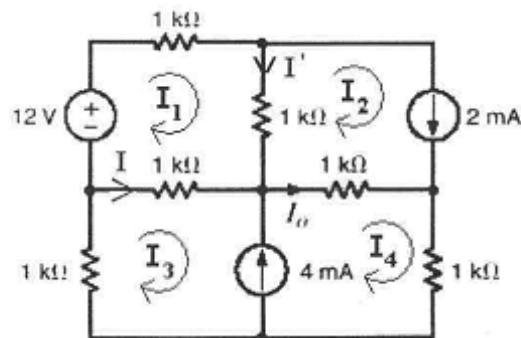


Figure P3.72

SOLUTION:



$$I_2 = 2 \text{ mA}$$

$$\text{KCL: } I_3 = I + I_1$$

$$I = I_3 - I_1$$

$$\text{KCL: } I' + I_2 = I_1$$

$$I' = I_1 - I_2 = I_1 - 2 \text{ mA}$$

$$\text{KCL: } I_o + 2 \text{ m} = I_4$$

$$I_o = I_4 - 2 \text{ m}$$

$$\text{KVL: } 12 = 1\text{k} I_1 + 1\text{k} I' + 1\text{k}(-I)$$

$$1\text{k} I_1 + 1\text{k} (I_1 - I_2) - 1\text{k} (I_3 - I_1) = 12$$

$$3\text{k} I_1 - 1\text{k} I_2 - 1\text{k} I_3 = 12$$

$$3\text{k} I_1 - 1\text{k} I_3 = 14$$

$$\text{KVL: } 1kI_3 + 1kI_1 + 1kI_0 + 1kI_4 = 0$$

$$1kI_3 + 1k(I_3 - I_1) + 1k(I_4 - 2m) + 1kI_4 = 0$$

$$\boxed{-1kI_1 + 2kI_3 + 2kI_4 = 2}$$

$$\text{KCL: } I_4 = 4m + I_3$$

$$\boxed{-I_3 + I_4 = 4m}$$

$$\text{KVL: } 12 = 1kI_1 + 1kI_2' + 1kI_0 + 1kI_4 + 1kI_3$$

$$1kI_1 + 1k(I_1 - I_2) + 1k(I_4 - 2m) + 1kI_4 + 1kI_3 = 12$$

$$2kI_1 - 1kI_2 + 1kI_3 + 2kI_4 = 14$$

$$\boxed{2kI_1 + 1kI_3 + 2kI_4 = 16}$$

$$3kI_1 - 1kI_3 + 0I_4 = 14$$

$$-1kI_1 + 2kI_3 + 2kI_4 = 2$$

$$0I_1 - I_3 + I_4 = 4m$$

$$I_1 = 4.55mA$$

$$I_3 = -0.364mA$$

$$I_4 = 3.64mA$$

$$I_0 = I_4 - 2m$$

$$I_0 = 3.64m - 2m$$

$$I_0 = 1.64mA$$

3.73 Find I_o in the circuit in Fig. P3.73 using loop analysis.

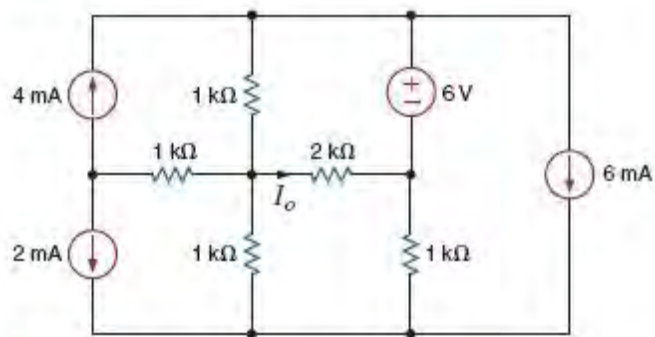
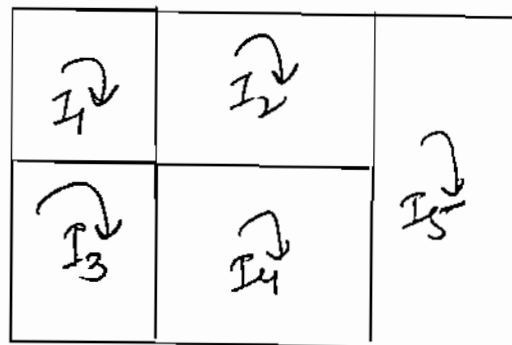


Figure P3.73

SOLUTION:



$$I_1 = \frac{4}{k}, \quad I_3 = -2/k, \quad I_5 = \frac{6}{k}$$

$$1k(I_2 - I_1) + 2k(I_2 - I_4) = -6$$

$$1k(I_4 - I_3) + 2k(I_4 - I_2) + 1k(I_4 - I_5) = 0$$

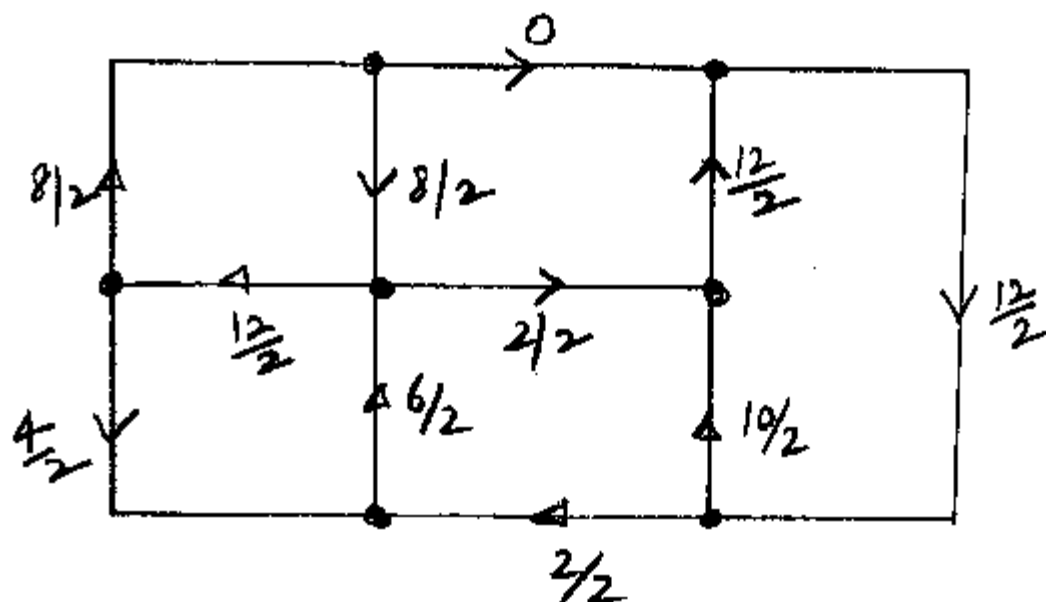
$$3kI_2 - 2kI_4 = -2$$

$$-2kI_2 + 4kI_4 = 6 - 2$$

$$\Delta = 12 - 4 = 8$$

$$\begin{bmatrix} I_2 \\ I_4 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -2 \\ 4 \end{bmatrix} = \frac{1}{8} \begin{bmatrix} -8 + 8 \\ -4 + 12 \end{bmatrix} \\ = \frac{1}{8} \begin{bmatrix} 0 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$I_0 = I_4 - I_2 = 1 - 0 \Rightarrow \boxed{I_0 = 1 \text{ mA}}$$



3.74 Find I_o in the network in Fig. P3.74 using loop analysis.

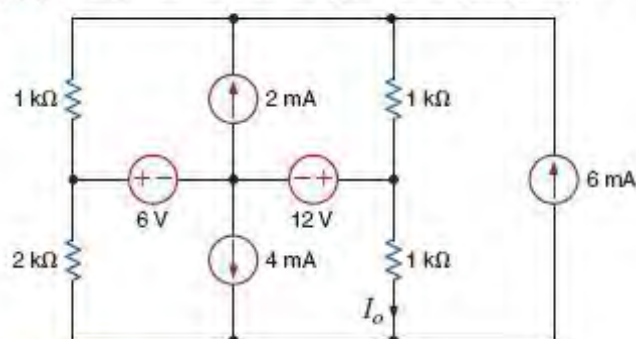
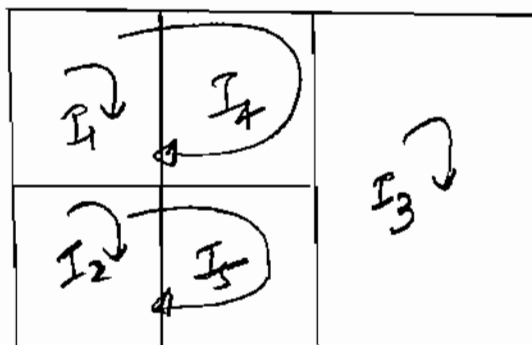


Figure P3.74

SOLUTION:



$$I_1 = \frac{-2}{k} \quad I_2 = \frac{4}{k} \quad I_3 = -\frac{6}{k}$$

$$1k(I_1 + I_2) + 1k(I_2 - I_3) = -6$$

$$2k(I_1 + I_2) + 2k(I_2 - I_3) = 6$$

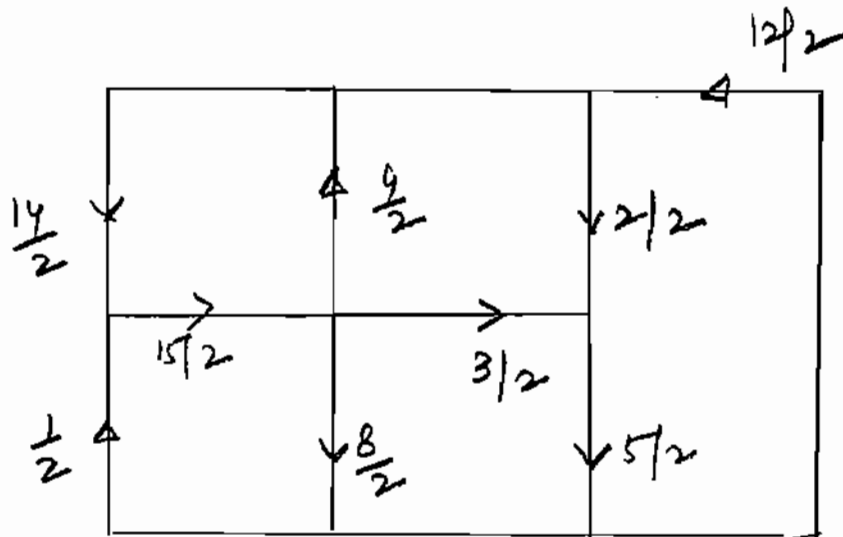
$$2kI_2 = -6 + 2 - 6$$

$$I_2 = -5/k = -\frac{10}{2}$$

$$4kI_2 = 6 - 8 - 12$$

$$I_2 = -7/2k = -7/2$$

$$I_0 = I_5 - I_3 = -\frac{7}{2k} + \frac{12}{2k} = \frac{5}{2k} A$$



3.75 Find V_o in the circuit in Fig. P3.75 using loop analysis.

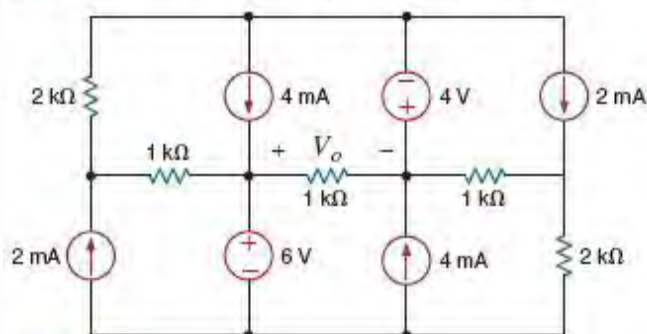
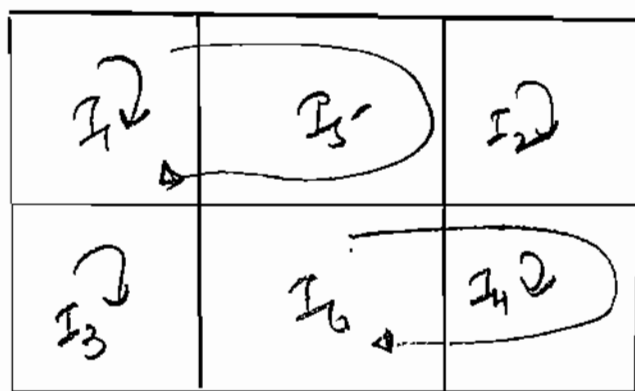


Figure P3.75

SOLUTION:



$$I_1 = \frac{4}{k}, \quad I_2 = \frac{2}{k}, \quad I_3 = \frac{2}{k}, \quad I_4 = \frac{4}{k}$$

$$2k(I_1 + I_5) + 1k(I_3 - I_6) + 1k(I_1 + I_5 - I_3) = 4$$

$$1k(I_6 - I_5) + 1k(I_4 + I_6 - I_2) + 2k(I_4 + I_6) = 6$$

$$2kI_1 + 2kI_5 + 1kI_3 - 1kI_6 + 1kI_1 + 1kI_5 - 1kI_3 = 4$$

$$1kI_6 - 1kI_5 + 1kI_4 + 1kI_6 - 1kI_2 + 2kI_4 + 2kI_6 = 6$$

$$8 + 4kI_5 - 1kI_6 + 4 - 2 = 4$$

$$4k I_6 - 1k I_5 + 12 - 2 = 6$$

$$4k I_5 - 1k I_6 = -6$$

$$-1k I_5 + 4k I_6 = -4$$

$$\Delta = (4k)(4k) - 1k^2 = 15k^2$$

$$\begin{bmatrix} I_5 \\ I_6 \end{bmatrix} = \frac{1}{15k^2} \begin{bmatrix} 4k & 1k \\ 1k & 4k \end{bmatrix} \begin{bmatrix} -6 \\ -4 \end{bmatrix}$$

$$= \frac{1}{15k^2} \begin{bmatrix} -24 & -4 \\ -6 & -16 \end{bmatrix} k$$

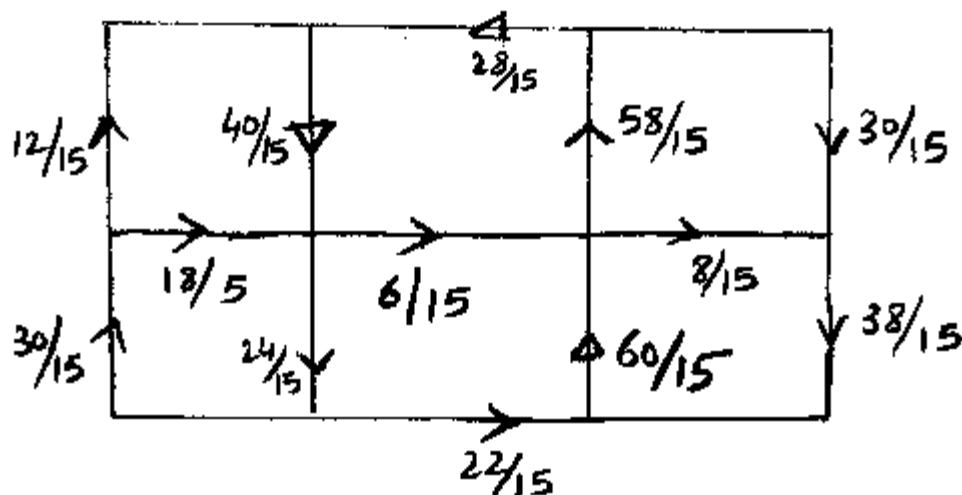
$$= \frac{1}{15k^2} \begin{bmatrix} -28k \\ -22k \end{bmatrix}$$

$$= \begin{bmatrix} -28/15k \\ -22/15k \end{bmatrix}$$

$$V_0 = 1k(I_6 - I_5)$$

$$= 1k \left(\frac{-22}{15k} + \frac{28}{15k} \right)$$

$$V_0 = \frac{6}{15} = 0.4V$$



3.76 Using loop analysis, find I_o in the circuit in Fig. P3.76.

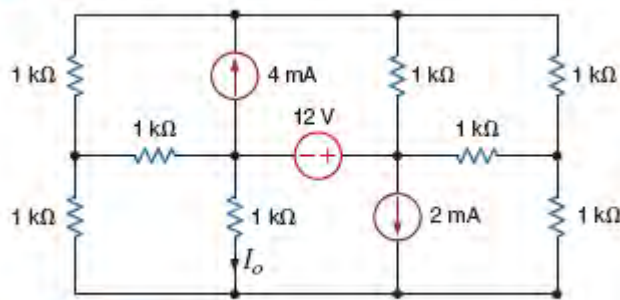
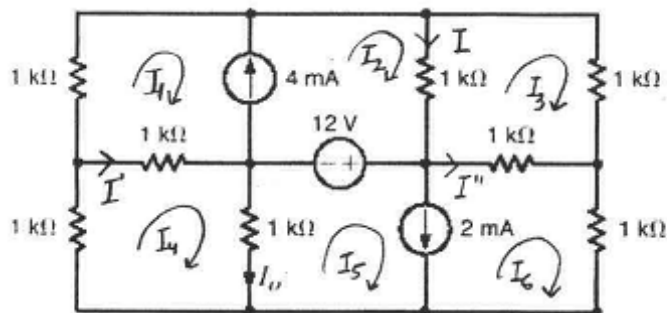


Figure P3.76

SOLUTION:



$$\begin{aligned} \text{KCL: } I_2 &= I + I_3 \\ I &= I_2 - I_3 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_3 + I'' &= I_6 \\ I'' &= I_6 - I_3 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_4 &= I_1 + I' \\ I' &= I_4 - I_1 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_o + I_5 &= I_4 \\ I_o &= I_4 - I_5 \end{aligned}$$

$$\begin{aligned} \text{KVL: } 12 &= 1kI'' + 1kI_6 + 1k(-I_o) \\ 1k(I_6 - I_3) + 1kI_6 - 1k(I_4 - I_5) &= 12 \end{aligned}$$

$$\boxed{-1kI_3 - 1kI_4 + 1kI_5 + 2kI_6 = 12}$$

$$\begin{aligned} \text{KVL: } 1kI_1' + 1kI_0 + 1kI_4 &= 0 \\ 1k(I_4 - I_1) + 1k(I_4 - I_5) + 1kI_4 &= 0 \\ \boxed{-1kI_1 + 3kI_4 - 1kI_5 = 0} \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_6 + 2m &= I_5 \\ \boxed{-I_5 + I_6 = -2m} \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_1 + 4m &= I_2 \\ \boxed{I_1 - I_2 = -4m} \end{aligned}$$

$$\begin{aligned} \text{KVL: } 1kI_1 + 1kI_3 + 1k(-I'') + 12 + 1k(-I') &= 0 \\ 1kI_1 + 1kI_3 - 1k(I_6 - I_3) - 1k(I_4 - I_1) &= -12 \\ \boxed{2kI_1 + 2kI_3 - 1kI_4 - 1kI_6 = -12} \end{aligned}$$

$$\begin{aligned} \text{KVL: } 1kI_3 + 1k(-I') + 1k(-I) &= 0 \\ 1kI_3 - 1k(I_6 - I_3) - 1k(I_2 - I_3) &= 0 \\ \boxed{-1kI_2 + 3kI_3 - 1kI_6 = 0} \end{aligned}$$

$$\begin{aligned} 0I_1 + 0I_2 - 1kI_4 + 1kI_5 + 2kI_6 &= 12 \\ -1kI_1 + 0I_2 + 0I_3 + 3kI_4 - 1kI_5 + 0I_6 &= 0 \\ 0I_1 + 0I_2 + 0I_3 + 0I_4 - I_5 + I_6 &= -2m \\ I_1 - I_2 + 0I_3 + 0I_4 + 0I_5 + 0I_6 &= -4m \\ 2kI_1 + 0I_2 + 2kI_3 - 1kI_4 + 0I_5 - 1kI_6 &= -12 \\ 0I_1 - 1kI_2 + 3kI_3 + 0I_4 + 0I_5 - 1kI_6 &= 0 \end{aligned}$$

$$I_1 = -4.93 \text{ mA}$$

$$I_2 = -0.933 \text{ mA}$$

$$I_3 = 0.933 \text{ mA}$$

$$I_4 = 0.267 \text{ mA}$$

$$I_5 = 5.73 \text{ mA}$$

$$I_6 = 3.73 \text{ mA}$$

$$I_0 = I_4 - I_5 = 0.267 \text{ mA} - 5.73 \text{ mA}$$

$$I_0 = -5.46 \text{ mA}$$

3.77 Find the mesh currents in the network in Fig. P3.77.

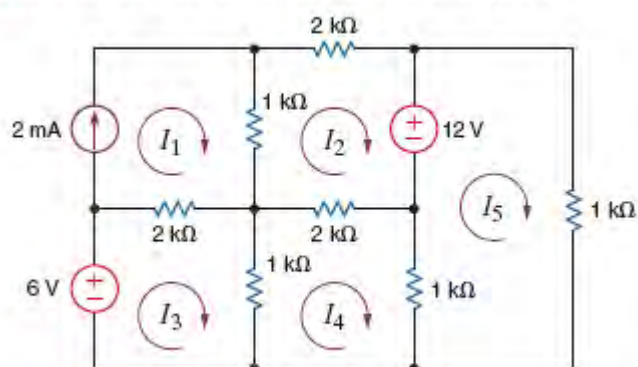
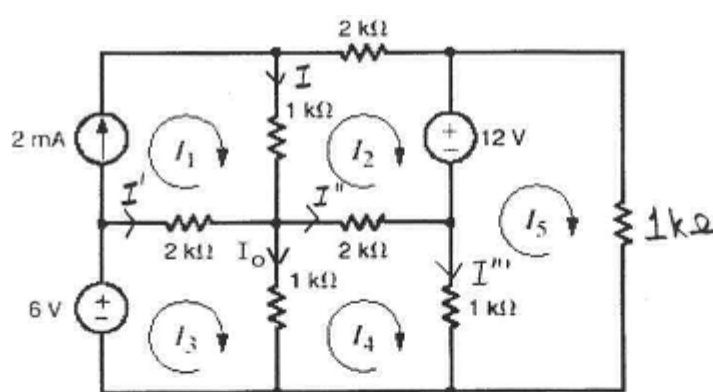


Figure P3.77

SOLUTION:



$$\begin{aligned} \text{KCL: } I_3 &= I' + I_1 \\ I' &= I_3 - I_1 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_1 &= I + I_2 \\ I &= I_1 - I_2 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I''' + I_5 &= I_4 \\ I''' &= I_4 - I_5 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_0 + I_4 &= I_3 \\ I_0 &= I_3 - I_4 \end{aligned}$$

$$\text{KCL: } I + I' = I'' + I_0$$

$$I'' = I_1 - I_2 + I_3 - I_1 - I_3 + I_4$$

$$I'' = -I_2 + I_4$$

$$\text{KVL: } 12 = 1k I_5 + 1k (-I''')$$

$$-1k(I_4 - I_5) + 1k I_5 = 12$$

$$\boxed{-1k I_4 + 2k I_5 = 12}$$

$$\text{KVL: } 2k I_2 + 12 + 2k (-I'') + 1k (-I) = 0$$

$$2k I_2 + 12 - 2k (-I_2 + I_4) - 1k (I_1 - I_2) = 0$$

$$-1k I_1 + 5k I_2 - 2k I_4 = -12$$

$$I_1 = 2\text{mA}$$

$$\boxed{5k I_2 - 2k I_4 = -10}$$

$$\text{KVL: } 2k I'' + 1k I''' + 1k (-I_0) = 0$$

$$2k (-I_2 + I_4) + 1k (I_4 - I_5) - 1k (I_3 - I_4)$$

$$\boxed{-2k I_2 - 1k I_3 + 4k I_4 - 1k I_5 = 0}$$

$$\text{KVL: } 6 = 2k I' + 1k I_0$$

$$2k (I_3 - I_1) + 1k (I_3 - I_4) = 6$$

$$\boxed{-2k I_1 + 3k I_3 - 1k I_4 = 6}$$

$$3k I_3 - 1k I_4 = 10$$

$$-1k I_4 + 2k I_5 = 12$$

$$5k I_2 - 2k I_4 = -10$$

$$-2k I_2 - 1k I_3 + 4k I_4 - 1k I_5 = 0$$

$$3k I_3 - 1k I_4 = 10$$

$$I_1 = 2\text{mA}$$

$$I_2 = -1.0986\text{mA}$$

$$I_3 = 4.0845\text{mA}$$

$$I_4 = 2.2535\text{mA}$$

$$I_5 = 7.1268\text{mA}$$

MATLAB

$$R = \begin{bmatrix} 0 & 0 & -1000 & 2000 & 5000 & 0 & -2000 & 0 \\ -2000 & -1000 & 4000 & -1000 & 0 & 3000 & -1000 & 0 \end{bmatrix};$$

$$V = [12; -10; 0; 10];$$

$$I1 = 0.002;$$

$$I = \text{inv}(R) * V;$$

$$I1$$

$$I2 = I(1)$$

$$I3 = I(2)$$

$$I4 = I(3)$$

$$I5 = I(4)$$

$$\begin{aligned} &>> \\ I_1 &= \\ &0.002 \end{aligned}$$

$$\begin{aligned} I_2 &= \\ &-0.0010986 \end{aligned}$$

$$\begin{aligned} I_3 &= \\ &0.0040845 \end{aligned}$$

$$\begin{aligned} I_4 &= \\ &0.0022535 \end{aligned}$$

$$\begin{aligned} I_5 &= \\ &0.0071268 \end{aligned}$$

>>

3.78 Use loop analysis to find V_o in the network in Fig. P3.78.

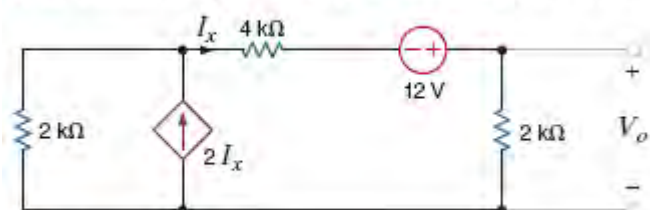
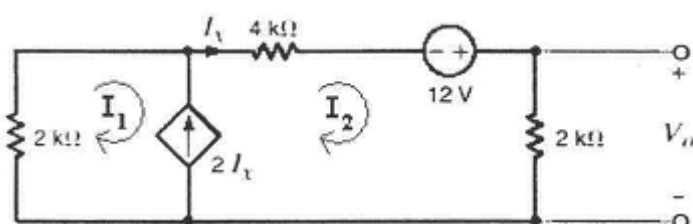


Figure P3.78

SOLUTION:



$$\text{KCL: } I_1 + 2I_x = I_x$$

$$I_1 = -I_x \text{ and } I_2 = I_x$$

$$\text{KVL: } 12 = 2kI_1 + 4kI_2 + 2kI_2$$

$$2kI_1 + 6kI_2 = 12$$

$$2k(-I_x) + 6k(I_x) = 12$$

$$4kI_x = 12$$

$$I_x = 3\text{mA}$$

$$V_o = 2kI_x = 2k(3\text{mA})$$

$$V_o = 6\text{V}$$

3.79 Find V_o in the circuit in Fig. P3.73 using nodal analysis.

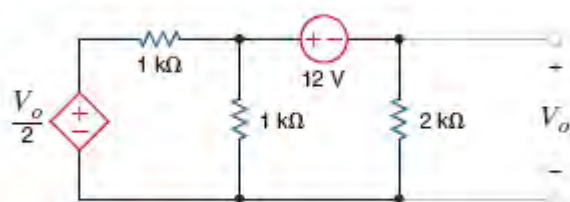
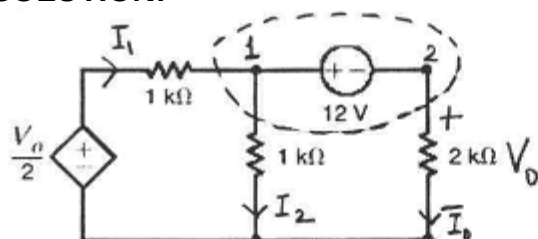


Figure P3.79

SOLUTION:



KCL at supernode:

$$I_1 = I_2 + I_o$$

$$\frac{V_o/2 - V_1}{1k} = \frac{V_1}{1k} + \frac{V_2}{2k}$$

$$V_o - 2V_1 = 2V_1 + V_2$$

$$V_2 = V_o$$

$$V_1 = 0V$$

$$V_1 - V_2 = 12$$

$$V_2 = -12V$$

$$V_o = -12V$$

3.80 Use nodal analysis to find V_o in Fig. P3.80.

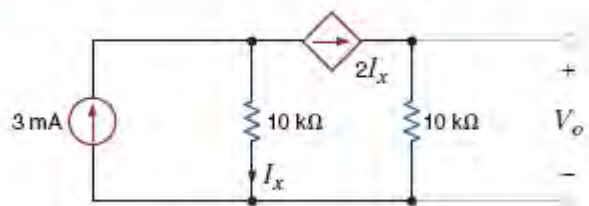
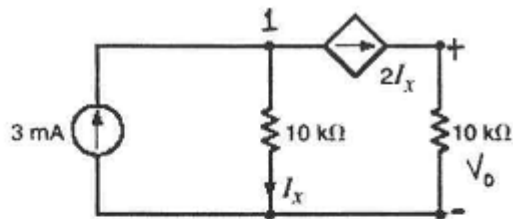


Figure P3.80

SOLUTION:



KCL at 1:

$$3\text{m} = I_x + 2I_x$$

$$3I_x = 3\text{m}$$

$$I_x = 1\text{mA}$$

$$V_o = 2I_x(10\text{k}) = 2(1\text{m})(10\text{k})$$

$$V_o = 20\text{V}$$

3.81 Find the power supplied by the 2-A current source in the network in Fig. P3.81 using loop analysis.

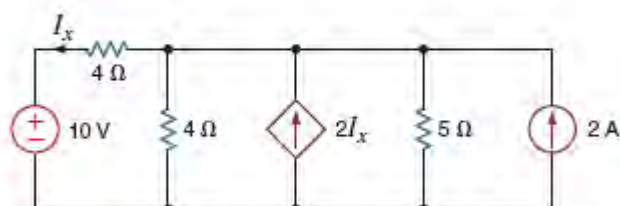
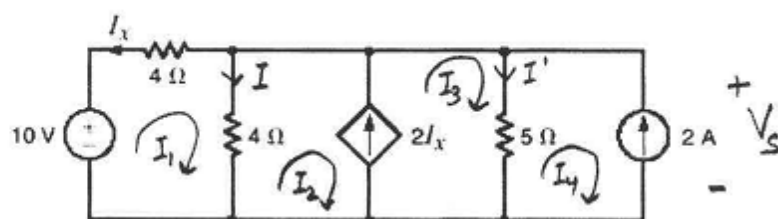


Figure P3.81

SOLUTION:



$$I_1 = -I_x$$

$$I_4 = -2A$$

$$\text{KCL: } I_1 = I + I_2$$

$$\boxed{I = I_1 - I_2}$$

$$\text{KVL: } 10 = 4I_1 + 4I$$

$$4(I_1 - I_2) + 4I_1 = 10$$

$$\boxed{8I_1 - 4I_2 = 10}$$

$$\text{KCL: } I_3 = I' + I_4$$

$$\boxed{I' = I_3 - I_4}$$

$$\text{KVL: } 10 = 4I_1 + 5I'$$

$$4I_1 + 5(I_3 - I_4) = 10$$

$$4I_1 + 5I_3 - 5(-2) = 10$$

$$4I_1 + 5I_3 = 0$$

KCL: $I_2 + 2I_x = I_3$
 $I_3 = I_2 + 2(-I_1)$

$$I_3 = -2I_1 + I_2$$

$$4I_1 + 5(-2I_1 + I_2) = 0$$

$$-6I_1 + 5I_2 = 0$$

$$8I_1 - 4I_2 = 10$$

$$-6I_1 + 5I_2 = 0$$

$$I_1 = 3.125 \text{ A}$$

$$I_2 = 3.75 \text{ A}$$

$$I_x = -I_1$$

$$I_x = -3.125 \text{ A}$$

$$V_s = I(4) = (I_1 - I_2)(4)$$

$$V_s = -2.5 \text{ V}$$

$$P_{2A} = V_s(2) = -2.5(2)$$

$$P_{2A} = -5 \text{ W} \quad \therefore \text{power supplied}$$

3.82 Find I_o in the network in Fig. P3.82.

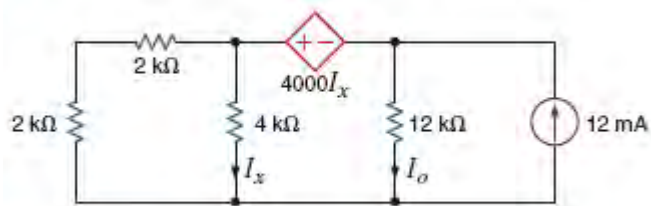
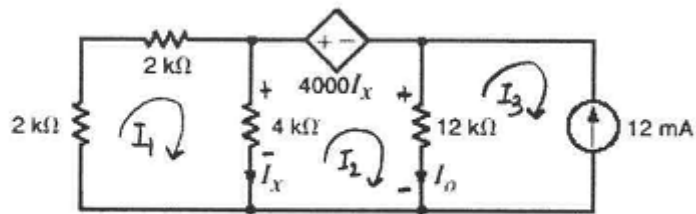


Figure P3.82

SOLUTION:



$$I_3 = -12 \text{ mA}$$

$$\text{KCL: } I_1 = I_x + I_2$$

$$\boxed{I_x = I_1 - I_2}$$

$$\text{KCL: } I_2 = I_o + I_3$$

$$\boxed{I_o = I_2 - I_3}$$

$$\text{KVL: } 4\text{k}(-I_x) + 4000 I_x + 12\text{k}(I_o) = 0$$

$$I_o = 0 \text{ A}$$

3.83 Find V_o in the circuit in Fig. P3.83 using loop analysis.

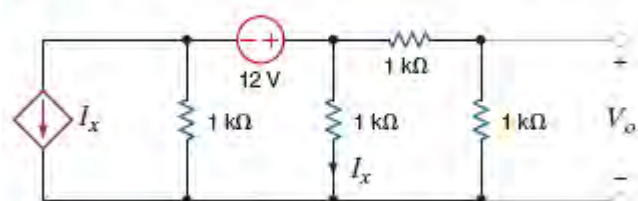
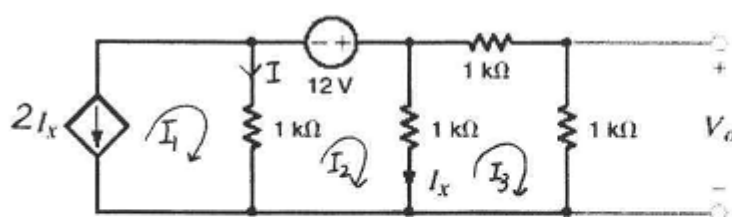


Figure P3.83

SOLUTION:



$$I_1 = -2I_x$$

$$\text{KCL: } I_1 = I + I_2$$

$$I = I_1 - I_2$$

$$\text{KCL: } I_2 = I_x + I_3$$

$$I_x = I_2 - I_3$$

$$\text{KVL: } 12 = 1kI_x + 1k(-I)$$

$$12 = 1k(I_2 - I_3) - 1k(I_1 - I_2)$$

$$-1kI_1 + 1kI_2 + 1kI_2 - 1kI_3 = 12$$

$$\boxed{-1kI_1 + 2kI_2 - 1kI_3 = 12}$$

$$\text{KVL: } 1kI_3 + 1kI_3 + 1k(-I_x) = 0$$

$$2kI_3 - 1k(I_2 - I_3) = 0$$

$$\boxed{-1kI_2 + 3kI_3 = 0}$$

$$I_1 = -2I_x = -2(I_2 - I_3)$$

$$\boxed{I_1 + 2I_2 - 2I_3 = 0}$$

$$- 1k I_1 + 2k I_2 - 1k I_3 = 12$$

$$0I_1 - 1k I_2 + 3k I_3 = 0$$

$$I_1 + 2I_2 - 2I_3 = 0$$

$$I_1 = -5.33 \text{ mA}$$

$$I_2 = 4 \text{ mA}$$

$$I_3 = 1.33 \text{ mA}$$

$$V_o = 1k(I_3)$$

$$V_o = 1k(1.33 \text{ mA})$$

$$V_o = 1.33 \text{ V}$$

3.84 Find V_o in the network in Fig. P3.84 using nodal analysis.

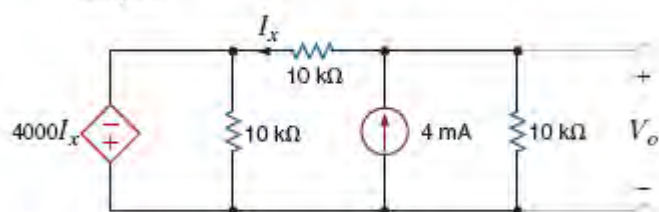
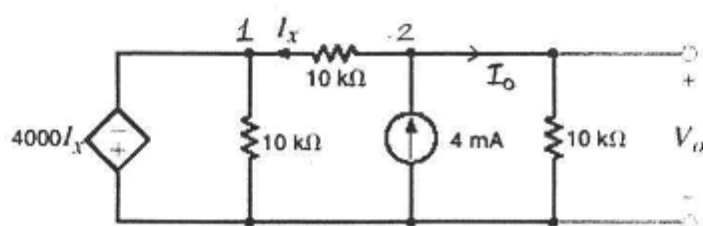


Figure P3.84

SOLUTION:



$$V_1 = -4000 I_x$$

$$I_x = \frac{V_2 - V_1}{10k}$$

$$\text{KCL at 2: } 4m = I_x + I_o$$

$$\frac{V_2 - V_1}{10k} + \frac{V_2}{10k} = 4m$$

$$V_2 - V_1 + V_2 = 40$$

$$\boxed{-V_1 + 2V_2 = 40}$$

$$V_1 = -4000 \left(\frac{V_2 - V_1}{10k} \right)$$

$$V_1 = -2/5 V_2 + 2/5 V_1$$

$$5V_1 = -2V_2 + 2V_1$$

$$\boxed{-3V_1 - 2V_2 = 0}$$

$$-V_1 + 2V_2 = 40$$

$$-3V_1 - 2V_2 = 0$$

$$V_1 = -10V$$

$$V_2 = 15V$$

$$V_0 = V_2 = 15V$$

$$V_0 = 15V$$

3.85 Find I_o in the circuit in Fig. P3.85 using loop analysis.

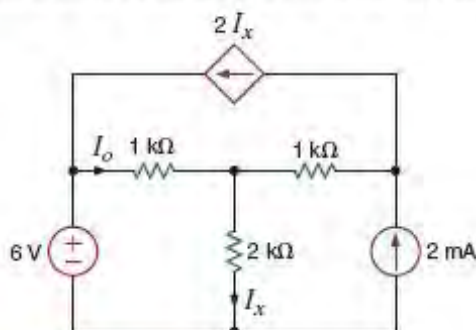


Figure P3.85

SOLUTION:

$$I_3 = -2/k$$

$$I_1 = -2I_x = -2(I_2 - I_3)$$

$$= -2I_2 + 2I_3$$

$$= -2I_2 - 4/k$$

$$1k(I_2 - I_1) + 2k(I_2 - I_3) = 6$$

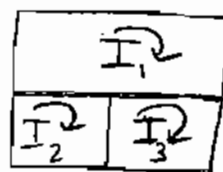
$$1kI_2 - 1k(-2I_2 - 4/k) + 2kI_2 - 2kI_3 = 6$$

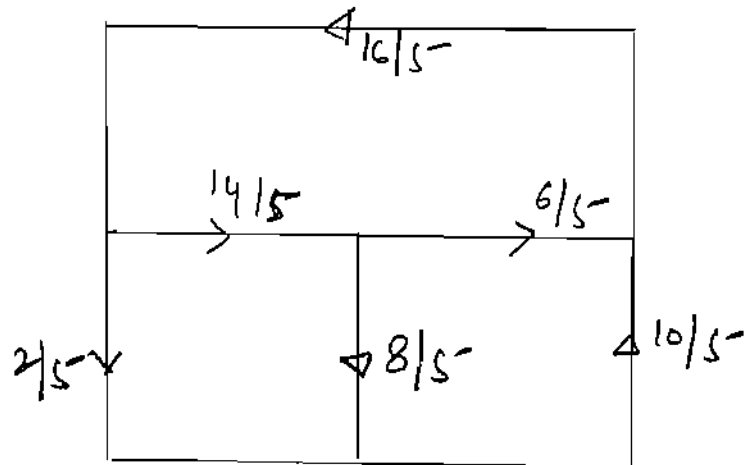
$$1kI_2 + 2kI_2 + 4 + 2kI_2 - 4 = 6$$

$$5kI_2 = -2$$

$$I_2 = -2/5k \text{ A}$$

$$I_x = I_2 - I_3 = \frac{-2}{5k} + \frac{2}{k} = \frac{-2}{5k} + \frac{10}{5k} = \frac{8}{5k}$$





$$I_0 = \frac{14}{s} \text{ mA}$$

- 3.86. Use mesh analysis to find V_o in the circuit in Fig. P3.86.

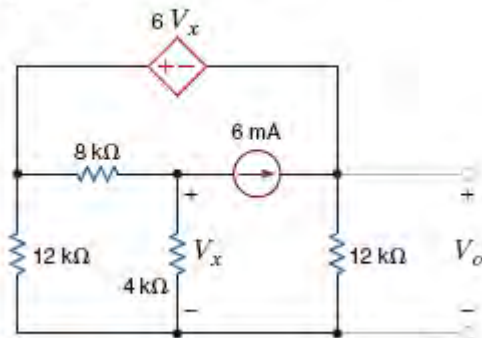
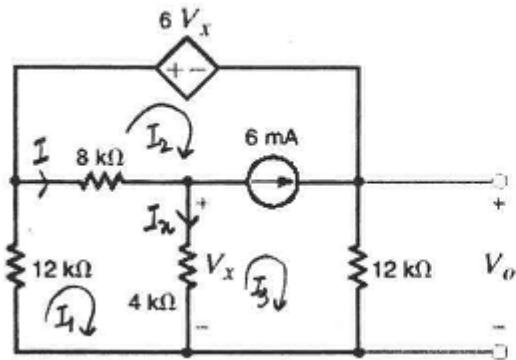


Figure P3.86

SOLUTION:



$$\begin{aligned} \text{KCL: } I_1 &= I + I_2 \\ I &= I_1 - I_2 \end{aligned}$$

$$\begin{aligned} \text{KCL: } I &= 6\text{m} + I_x \\ I_x &= I_1 - I_2 - 6\text{m} \end{aligned}$$

KVL lower left loop:

$$12\text{k} I_1 + 8\text{k} I + 4\text{k} I_x = 0$$

$$12\text{k} I_1 + 8\text{k} (I_1 - I_2) + 4\text{k} (I_1 - I_2 - 6\text{m}) = 0$$

$$\boxed{24\text{k} I_1 - 12\text{k} I_2 = 24}$$

$$\begin{aligned}\text{KVL outer loop: } 12kI_1 + 6V_x + 12kI_3 &= 0 \\ V_x &= 4kI_x = 4k(I_1 - I_2 - 6m) \\ V_x &= 4kI_1 - 4kI_2 - 24 \\ 12kI_1 + 6[4kI_1 - 4kI_2 - 24] + 12kI_3 &= 0\end{aligned}$$

$$\begin{aligned}36kI_1 - 24kI_2 + 12kI_3 &= 144 \\ \text{KCL: } I_x + I_3 &= I_1 \\ I_3 &= I_1 - [I_1 - I_2 - 6m] \\ I_3 &= I_2 + 6m\end{aligned}$$

$$\begin{aligned}36kI_1 - 24kI_2 + 12k(I_2 + 6m) &= 144 \\ \boxed{36kI_1 - 12kI_2 = 72}\end{aligned}$$

$$\begin{aligned}24kI_1 - 12kI_2 &= 24 \\ 36kI_1 - 12kI_2 &= 72\end{aligned}$$

$$\begin{aligned}I_1 &= 4mA \\ I_2 &= 6mA\end{aligned}$$

$$\begin{aligned}I_3 &= 6mA + 6mA \\ I_3 &= 12mA\end{aligned}$$

$$\begin{aligned}V_o &= 12k(I_3) = 12k(12mA) \\ V_o &= 144V\end{aligned}$$

3.87 Using mesh analysis, find V_o in the circuit in Fig. P3.87.

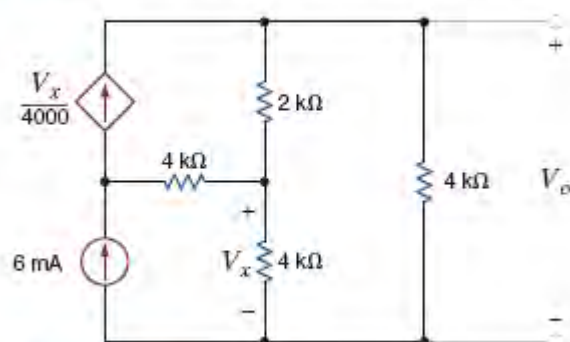
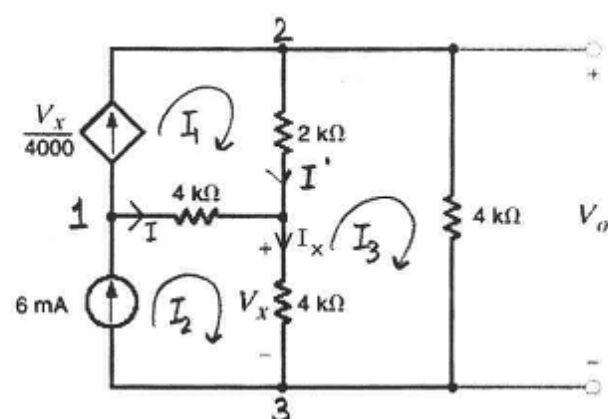


Figure P3.87

SOLUTION:



$$\text{KCL at 1: } I_2 = I + I_1$$

$$I = I_2 - I_1$$

$$\text{KCL at 2: } I_1 = I' + I_3$$

$$I' = I_1 - I_3$$

$$\text{KCL at 3: } I_x + I_3 = I_2$$

$$I_x = I_2 - I_3$$

KVL right loop:

$$2k(-I') + 4kI_3 + 4k(-I_x) = 0$$

$$-2k(I_1 - I_3) + 4kI_3 - 4k(I_2 - I_3) = 0$$

$$\boxed{-2k I_1 - 4k I_2 + 10 I_3 = 0}$$

$$I_1 = \frac{V_x}{4000}$$

$$V_x = I_x(4k) = (I_2 - I_3) 4k$$

$$I_1 = \frac{4k(I_2 - I_3)}{4000}$$

$$I_1 = I_2 - I_3$$

$$\boxed{I_1 - I_2 + I_3 = 0}$$

$$I_2 = 6mA$$

$$-2k I_1 - 4k(6m) + 10 I_3 = 0$$

$$\boxed{-2k I_1 + 10k I_3 = 24}$$

$$I_1 - I_2 + I_3 = 0$$

$$\boxed{I_1 + I_3 = 6m}$$

$$-2k I_1 + 10k I_3 = 24$$

$$I_1 + I_3 = 6m$$

$$I_1 = 3mA$$

$$I_3 = 3mA$$

$$V_o = I_3(4k) = 3m(4k)$$

$$V_o = 12V$$

3.88 Find V_o in the network in Fig. P3.88.

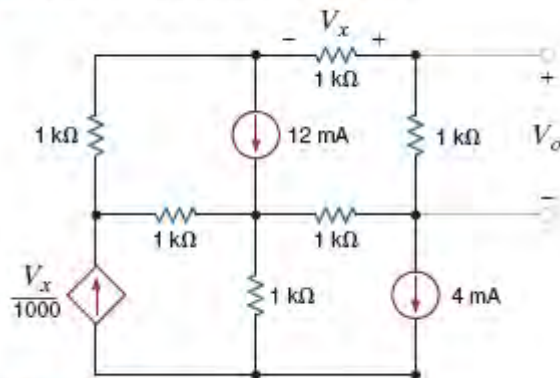
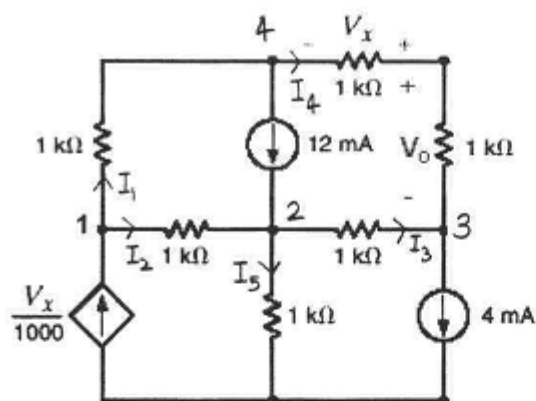


Figure P3.88

SOLUTION:



$$\begin{aligned} \text{KCL at 4: } I_1 &= 12\text{m} + I_4 \\ \frac{V_1 - V_4}{1\text{k}} &= 12\text{m} + \frac{V_4 - V_3}{2\text{k}} \end{aligned}$$

$$2V_1 - 2V_4 = 24 + V_4 - V_3$$

$$\boxed{2V_1 + V_3 - 3V_4 = 24}$$

$$\begin{aligned} \text{KCL at 3: } I_4 + I_3 &= 4\text{m} \\ \frac{V_4 - V_3}{2\text{k}} + \frac{V_2 - V_3}{1\text{k}} &= 4\text{m} \end{aligned}$$

$$V_4 - V_3 + 2V_2 - 2V_3 = 8$$

$$\boxed{2V_2 - 3V_3 + V_4 = 8}$$

$$\text{KCL at 1: } \frac{V_x}{1000} = I_1 + I_2$$

$$V_x = 1k(-I_4)$$

$$V_x = -1k I_4$$

$$\frac{-1k I_4}{1000} = \frac{V_1 - V_4}{1k} + \frac{V_1 - V_2}{1k}$$

$$-\left(\frac{V_4 - V_3}{2k}\right) = \frac{V_1 - V_4}{1k} + \frac{V_1 - V_2}{1k}$$

$$-V_4 + V_3 = 2V_1 - 2V_4 + 2V_1 - 2V_2$$

$$\boxed{4V_1 - 2V_2 - V_3 - V_4 = 0}$$

$$\text{KCL at 2: } 12\text{m} + I_2 = I_3 + I_5$$

$$12\text{m} + \frac{V_1 - V_2}{1k} = \frac{V_2 - V_3}{1k} + \frac{V_2}{1k}$$

$$12 + V_1 - V_2 = V_2 - V_3 + V_2$$

$$\boxed{-V_1 + 3V_2 - V_3 = 12}$$

$$2V_1 + 0V_2 + V_3 - 3V_4 = 24$$

$$0V_1 + 2V_2 - 3V_3 + V_4 = 8$$

$$4V_1 - 2V_2 - V_3 - V_4 = 0$$

$$-V_1 + 3V_2 - V_3 + 0V_4 = 12$$

$$V_1 = -6\text{V}$$

$$V_2 = -0.67\text{V}$$

$$V_3 = -8\text{V}$$

$$V_4 = -14.67\text{V}$$

$$I_y = \frac{V_4 - V_3}{2k} = - \frac{14.67 - (-8)}{2k}$$

$$I_y = -3.34 \text{ mA}$$

$$V_o = I_y(1k) = -3.34(1k)$$

$$V_o = -3.34 \text{ V}$$

3.89 Using loop analysis, find V_o in the circuit in Fig. P3.89.

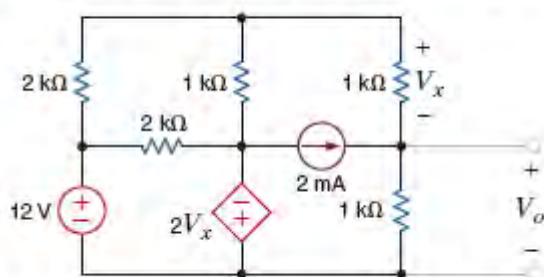
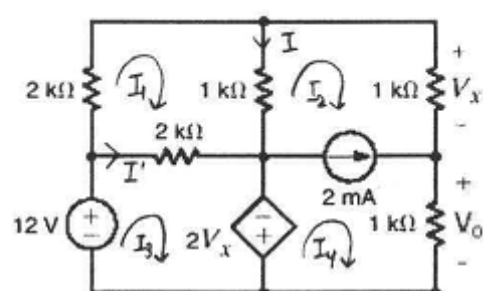


Figure P3.89

SOLUTION:



$$\text{KCL: } I_1 = I + I_2$$

$$I = I_1 - I_2$$

$$\text{KCL: } I_3 = I' + I_1$$

$$I' = I_3 - I_1$$

$$\text{KVL: } 2kI_1 + 1kI + 2k(-I') = 0$$

$$2kI_1 + 1k(I_1 - I_2) - 2k(I_3 - I_1) = 0$$

$$\boxed{5kI_1 - 1kI_2 - 2kI_3 = 0}$$

$$\text{KVL: } 2kI' = 12 + 2V_x$$

$$V_x = 1kI_2$$

$$2k(I_3 - I_1) = 12 + 2(1kI_2)$$

$$\boxed{-2kI_1 - 2kI_2 + 2kI_3 = 12}$$

$$\boxed{\text{KVL: } 2kI_1 + 1kI_2 + 1kI_4 = 12}$$

$$\text{KCL: } I_2 + 2\text{m} = I_4$$

$$\boxed{-I_2 + I_4 = 2\text{m}}$$

$$5kI_1 - 1kI_2 - 2kI_3 + 0I_4 = 0$$

$$-2kI_1 - 2kI_2 + 2kI_3 + 0I_4 = 12$$

$$2kI_1 + 1kI_2 + 0I_3 + 1kI_4 = 12$$

$$0I_1 - I_2 + 0I_3 + I_4 = 2\text{m}$$

$$I_1 = 4.5\text{mA}$$

$$I_2 = 0.5\text{mA}$$

$$I_3 = 11\text{mA}$$

$$I_4 = 2.5\text{mA}$$

$$V_o = 1kI_4 = 1k(2.5\text{m})$$

$$V_o = 2.5\text{V}$$

3.90 Using loop analysis, find V_o in the circuit in Fig. P3.90.

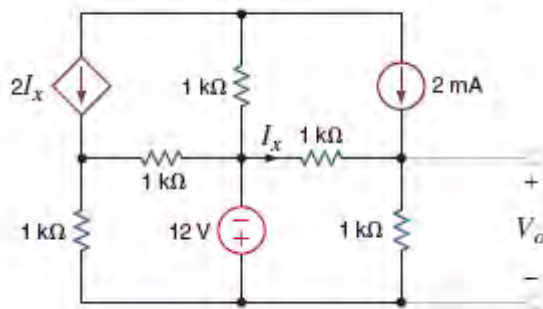
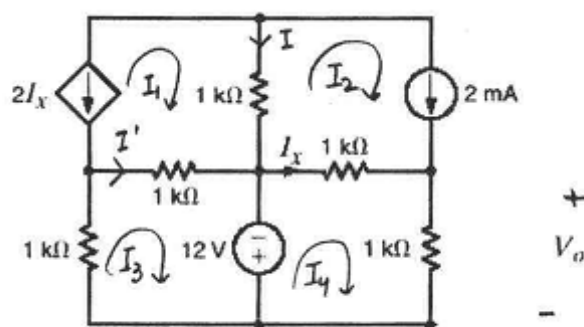


Figure P3.90

SOLUTION:



$$I_1 = -2I_x, \quad I_2 = 2\text{ mA}$$

$$\text{KCL: } I_3 = I' + I_1$$

$$I' = I_3 - I_1$$

$$\text{KCL: } I_2 + I_x = I_4$$

$$I_x = I_4 - I_2$$

$$\text{KVL: } 1\text{ k}\Omega I_3 + 1\text{ k}\Omega I' = 12$$

$$1\text{ k}\Omega (I_3 - I_1) + 1\text{ k}\Omega I_3 = 12$$

$$\boxed{-1\text{ k}\Omega I_1 + 2\text{ k}\Omega I_3 = 12}$$

$$\text{KVL: } 12 + 1\text{ k}\Omega I_x + 1\text{ k}\Omega I_4 = 0$$

$$1\text{ k}\Omega (I_4 - I_2) + 1\text{ k}\Omega I_4 = -12$$

$$\boxed{-1kI_2 + 2kI_4 = -12}$$

$$-1k(2m) + 2kI_4 = -12$$

$$2kI_4 = -12 + 2$$

$$I_4 = -5mA$$

$$V_o = 1k(I_4) = 1k(-5m)$$

$$V_o = -5V$$

3.91 Using loop analysis, find V_o in the network in Fig. P3.91.

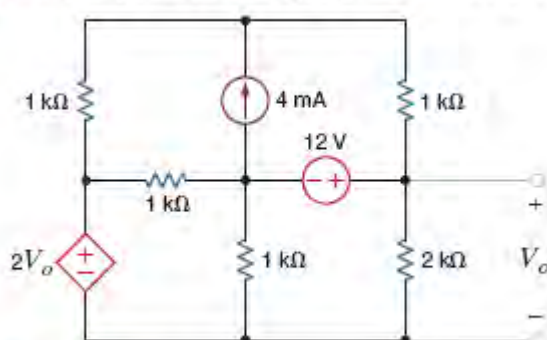
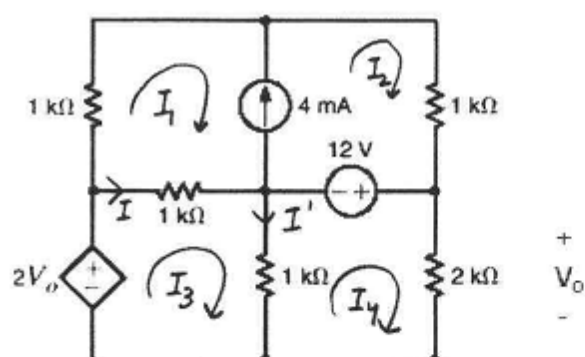


Figure P3.91

SOLUTION:



$$\text{KCL: } I_1 + 4\text{ m} = I_2$$

$$\boxed{I_1 - I_2 = -4\text{ m}}$$

$$\text{KCL: } I_3 = I + I_1$$

$$I = I_3 - I_1$$

$$\text{KCL: } I' + I_4 = I_3$$

$$I' = I_3 - I_4$$

$$\text{KVL: } 2\text{ k} I_4 + 1\text{ k}(-I') = 12$$

$$-1\text{ k}(I_3 - I_4) + 2\text{ k} I_4 = 12$$

$$\boxed{-1\text{ k} I_3 + 3\text{ k} I_4 = 12}$$

$$\text{KVL: } 1k I + 1k I' = 2V_0$$

$$V_0 = 2k I_4$$

$$1k(I_3 - I_1) + 1k(I_3 - I_4) = 2(2k I_4)$$

$$\boxed{-1k I_1 + 2k I_3 - 5k I_4 = 0}$$

$$\text{KVL: } 2V_0 = 1k I_1 + 1k I_2 + 2k I_4$$

$$V_0 = 2k I_4$$

$$\boxed{1k I_1 + 1k I_2 - 2k I_4 = 0}$$

$$I_1 - I_2 + 0I_3 + 0I_4 = -4m$$

$$0I_1 + 0I_2 - 1k I_3 + 3k I_4 = 12$$

$$-1k I_1 + 0I_2 + 2k I_3 - 5k I_4 = 0$$

$$1k I_1 + 1k I_2 + 0I_3 - 2k I_4 = 0$$

In the matrix form the equations can be represented/restructured as:

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 2 & 0 & -2 \\ 0 & 0 & 4 & -12 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ -4 \\ 44 \end{bmatrix}$$

$$V_0 = \infty$$

Since there are only 3 independent equations for 4 unknowns (1 to 4), it can't be solved for a unique solution.

This is because current equations are contradictory.
 $\therefore V_0 = \text{not defined (open circuit)}$.

3.92 Using loop analysis, find V_o in the circuit in Fig. P3.92.

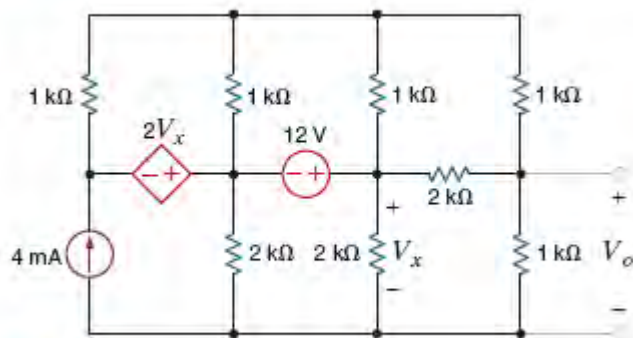
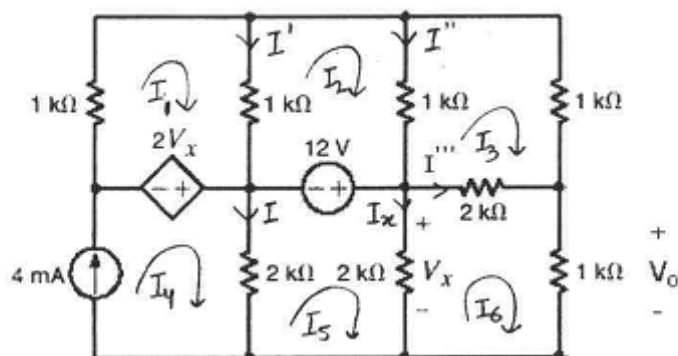


Figure P3.92

SOLUTION:



$$\text{KCL: } I_1 = I' + I_2$$

$$I' = I_1 - I_2$$

$$\text{KCL: } I_2 = I'' + I_3$$

$$I'' = I_2 - I_3$$

$$\text{KCL: } I + I_5 = I_4$$

$$I = I_4 - I_5$$

$$\text{KCL: } I_x + I_6 = I_5$$

$$I_x = I_5 - I_6$$

$$\text{KCL: } I_3 + I''' = I_6$$

$$I''' = I_6 - I_3$$

$$\text{KVL: } 2V_x + 1kI_1 + 1kI' = 0$$

$$V_x = I_x(2k) = (I_5 - I_6)2k$$

$$2(2k)(I_5 - I_6) + 1kI_1 + 1k(I_1 - I_2) = 0$$

$$4kI_5 - 4kI_6 + 2kI_1 - 1kI_2 = 0$$

$$\boxed{2kI_1 - 1kI_2 + 4kI_5 - 4kI_6 = 0}$$

$$\text{KVL: } 12 + 1k(-I') + 1kI'' = 0$$

$$-1k(I_1 - I_2) + 1k(I_2 - I_3) = -12$$

$$-1kI_1 + 2kI_2 - 1kI_3 = -12$$

$$\text{KVL: } 1k(-I'') + 1kI_3 + 2k(-I''') = 0$$

$$-1k(I_2 - I_3) + 1kI_3 - 2k(I_6 - I_3) = 0$$

$$\boxed{-1kI_2 + 4kI_3 - 2kI_6 = 0}$$

$$\text{KVL: } 12 = 2k(-I) + 2kI_x$$

$$-2k(I_4 - I_5) + 2k(I_5 - I_6) = 12$$

$$I_4 = 4\text{mA}$$

$$\boxed{4kI_5 - 2kI_6 = 20}$$

$$\text{KVL: } 2kI''' + 1kI_6 + 2k(-I_x) = 0$$

$$2k(I_6 - I_3) + 1kI_6 - 2k(I_5 - I_6) = 0$$

$$\boxed{-2kI_3 - 2kI_5 + 5kI_6 = 0}$$

$$2kI_1 - 1kI_2 + 0I_3 + 4kI_5 - 4kI_6 = 0$$

$$-1kI_1 + 2kI_2 - 1kI_3 + 0I_5 + 0I_6 = -12$$

$$0I_1 - 1kI_2 + 4kI_3 + 0I_5 - 2kI_6 = 0$$

$$0I_1 + 0I_2 + 0I_3 + 4kI_5 - 2kI_6 = 20$$

$$0I_1 + 0I_2 - 2kI_3 - 2kI_5 + 5kI_6 = 0$$

$$I_1 = -18\text{mA}$$

$$I_2 = -17\text{mA}$$

$$I_3 = -4\text{mA}$$

$$I_5 = 5.25\text{mA}$$

$$I_6 = 0.5\text{mA}$$

$$V_o = 1\text{k}(I_6) = 1\text{k}(0.5\text{mA})$$

$$V_o = 0.5\text{V}$$

3.93 Using loop analysis, find I_o in the network in Fig. P3.93.

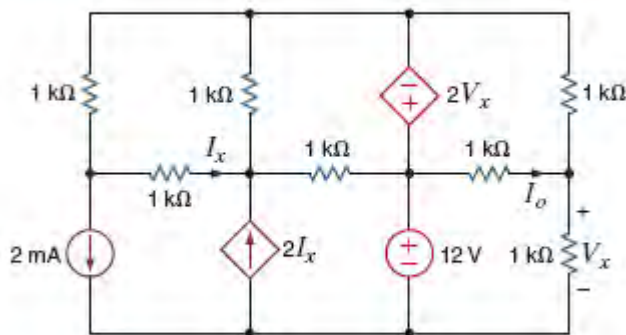
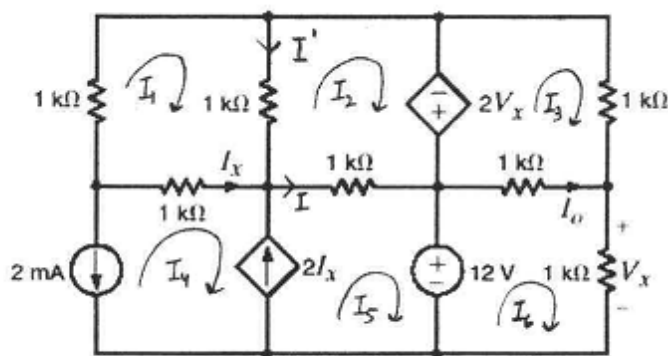


Figure P3.93

SOLUTION:



$$\text{KCL: } I_1 = I' + I_2$$

$$I' = I_1 - I_2$$

$$\text{KCL: } I_4 = I_x + I_1$$

$$I_x = I_4 - I_1$$

$$\text{KCL: } I_x + I' = I_x$$

$$\Rightarrow I_x + 2I_x + I' = I$$

$$I = 3(I_4 - I_1) + I_1 - I_2$$

$$I = -2I_1 - I_2 + 3I_4$$

$$\text{KVL: } 1kI_1 + 1kI' + 1k(-I_x) = 0$$

$$1kI_1 + 1k(I_1 - I_2) - 1k(I_4 - I_1) = 0$$

$$\boxed{3kI_1 - 1kI_2 - 1kI_4 = 0}$$

$$\text{KVL: } 1k(-I_1) + 1k(-I_2) = 2V_x$$

$$V_x = 1k I_6$$

$$-1k(I_1 - I_2) - 1k(-2I_1 - I_2 + 3I_4) = 2(1k I_6)$$

$$\boxed{1k I_1 + 2k I_2 - 3k I_4 - 2k I_6 = 0}$$

$$I_0 = I_6 - I_3$$

$$\text{KVL: } 2V_x + 1k I_3 + 1k(-I_0) = 0$$

$$2(1k I_6) + 1k I_3 - 1k(I_6 - I_3) = 0$$

$$\boxed{2k I_3 + 1k I_6 = 0}$$

$$\text{KVL: } 1k I_0 + 1k I_6 = 12$$

$$1k(I_6 - I_3) + 1k I_6 = 12$$

$$\boxed{-1k I_3 + 2k I_6 = 12}$$

$$\text{KCL: } I_5 = 2I_x + I_4$$

$$I_5 - I_4 - 2(I_4 - I_1) = 0$$

$$\boxed{2I_1 - 3I_4 + I_5 = 0}$$

$$I_4 = -2\text{mA}$$

$$\boxed{2I_1 + I_5 = -6}$$

$$\boxed{3k I_1 - 1k I_2 = -2}$$

$$\boxed{-1k I_1 + 2k I_2 - 2k I_6 = -2}$$

$$0I_1 + 0I_2 + 2k I_3 + 0I_5 + 1k I_6 = 0 \quad \text{--- (1)}$$

$$0I_1 + 0I_2 - 1k I_3 + 0I_5 + 2k I_6 = 12 \quad \text{--- (2)}$$

$$2I_1 + 0I_2 + 0I_3 + I_5 + 0I_6 = -6$$

$$3kI_1 - 1kI_2 + 0I_3 + 0I_5 + 0I_6 = -2$$

$$-1kI_1 + 2kI_2 + 0I_3 + 0I_5 - 2kI_6 = -6$$

Solving (1) and (2)

$$2I_3 = -I_6$$

$$-I_3 + 2I_6 = 12m$$

$$\Rightarrow I_6 = 4.8mA$$

$$I_3 = -2.4mA$$

$$I_0 = I_6 - I_3 = 4.8m - (-2.4m)$$

$$\boxed{I_0 = 7.2mA}$$

Note: I_1, I_2, I_5 need not be found as

$$I_0 = I_6 - I_3$$

3.94 Use loop analysis to find I_o in the circuit in Fig. P3.94.

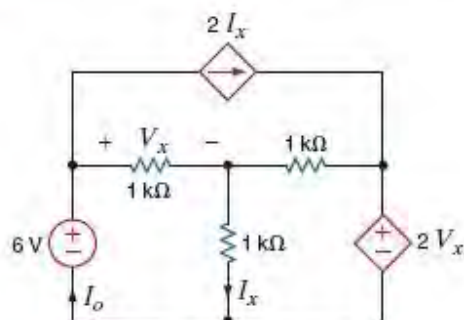
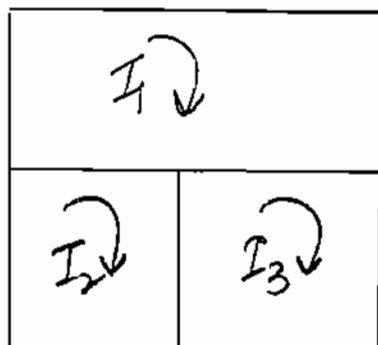


Figure P3.94

SOLUTION:



$$V_x = 1k(I_2 - I_1)$$

$$I_1 = 2I_x = 2(I_2 - I_3) \quad I_x = I_2 - I_3$$

$$1k(I_2 - I_1) + 1k(I_2 - I_3) = 6$$

$$1k(I_3 - I_2) + 1k(I_3 - I_1) = -2V_x$$

$$2kI_2 - 1kI_3 = 6 + 1kI_1$$

$$-1kI_2 + 2kI_3 = -2V_x + 1kI_1$$

$$2kI_2 - 1kI_3 = 6 + 1k(2)(I_2 - I_3)$$

$$-1kI_2 + 2kI_3 = -2k(I_2 - I_1) + 1kI_1$$

$$\cancel{2k}I_2 - 1kI_3 = 6 + \cancel{2k}I_2 - 2kI_3$$

$$-1kI_2 + 2kI_3 = -2kI_2 + 3kI_1$$

$$1kI_3 = 6$$

$$+1kI_2 + 2kI_3 = +3k(2)(I_2 - I_3)$$

$$1kI_2 + 2kI_3 = 6kI_2 - 6kI_3$$

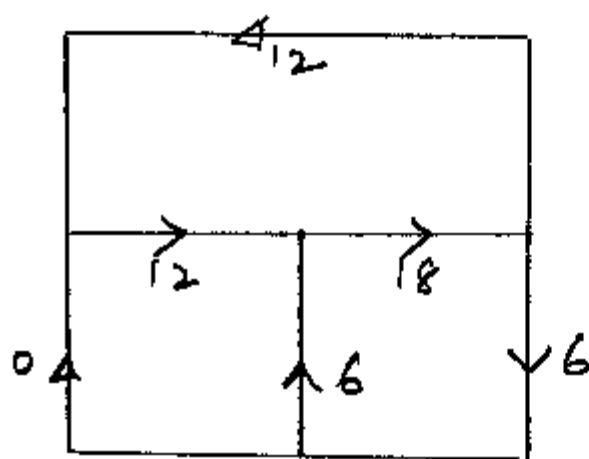
$$5kI_2 = 8kI_3 = 8k\left(\frac{6}{k}\right)$$

$$I_2 = \frac{48}{5k}, I_3 = 6mA \quad I_1 = 2\left(\frac{48}{5k} - \frac{6}{k}\right) \Rightarrow I_1 = \frac{36}{5k} = 7.2mA$$

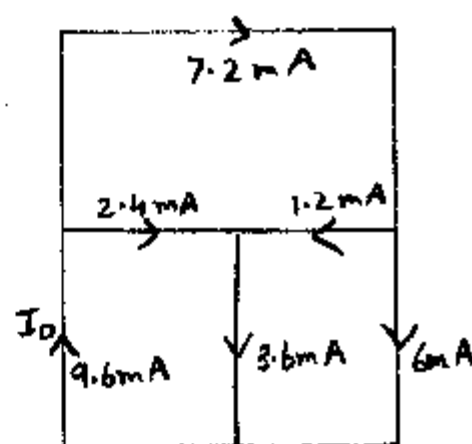
$$V_x = 1k(I_2 - I_1) = 1k(0 - I_1) = -1k(2)(0 - 6/k) = 12$$

$$V_x = 1k\left(\frac{48}{5k} - \frac{36}{5k}\right) = \frac{12}{5} = 2.4V$$

$$I_x = I_2 - I_3 = \frac{48}{5k} - \frac{6}{k} = 3.6mA$$



$$I_0 = 0$$



$$\underline{\underline{I_0 = 6 + 3.6 = 9.6mA}}$$

3.95 Find I_x in the circuit in Fig. P3.95 using loop analysis.

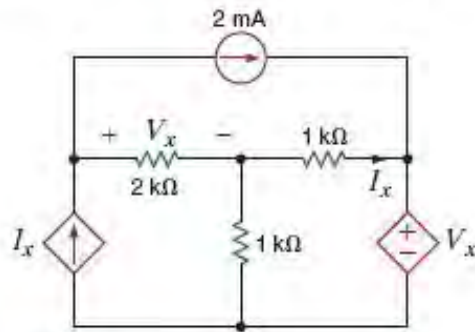
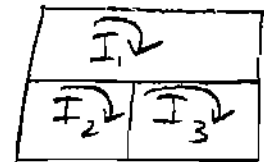


Figure P3.95

SOLUTION:

$$I_1 = \frac{2}{k} \quad I_2 = I_x = I_3 - \frac{2}{k}$$



$$1k(I_3 - I_2) + 1k(I_3 - I_1) = -2V_x = -2(2k)(I_2 - I_1)$$

$$1kI_3 - 1kI_2 + 1kI_3 - 1kI_1 = -4kI_2 + 4kI_1$$

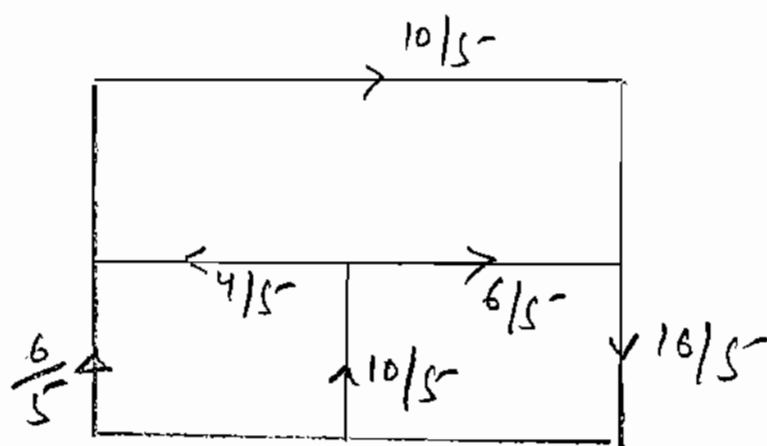
$$2kI_3 - 1k(I_3 - \frac{2}{k}) - 2 = -4k(I_3 - \frac{2}{k}) + 8$$

$$\cancel{2kI_3} - \cancel{1kI_3} + 2 - 2 = -4kI_3 + 8 + 8$$

$$5kI_3 = 16$$

$$I_3 = 16/5k \text{ A}$$

$$I_x = I_3 - 2 = \frac{16}{5} - \frac{10}{5} = \frac{6}{5k}$$



3.96 Find I_o in the circuit in Fig. P3.96 using loop analysis.

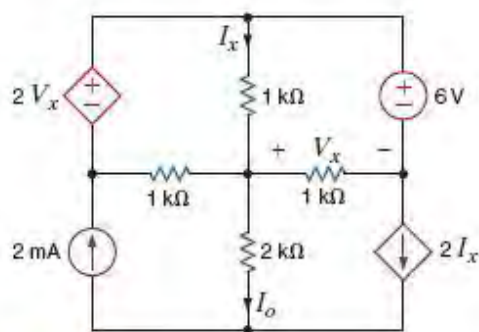
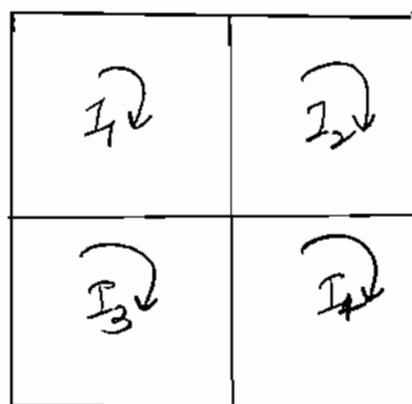


Figure P3.96

SOLUTION:



$$I_x = I_1 - I_2 \quad V_x = I_4 - I_2$$

$$1k(I_1 - I_2) + 1k(I_1 - I_3) = 2V_x$$

$$1k(I_2 - I_1) + 1k(I_2 - I_4) = -6$$

$$\cancel{1kI_1} - \cancel{1kI_2} + \cancel{1kI_1} - 1kI_3 = 2I_4k - \cancel{2I_2k}$$

$$\cancel{1kI_2} - \cancel{1kI_1} + \cancel{1kI_2} - 1kI_4 = -6$$

$$2kI_1 + 1kI_2 = 1kI_3 + 2kI_4$$

$$-1kI_1 + 2kI_2 = -6 + 1kI_4$$

$$2k I_1 + 1k I_2 = 2 + (2k)(2)(I_1 - I_2)$$

$$-1k I_1 + 2k I_2 = -6 + 1k(2)(I_1 - I_2)$$

$$2k I_1 + 1k I_2 = 2 + 4k I_1 - 4k I_2$$

$$-1k I_1 + 2k I_2 = -6 + 2k I_1 - 2k I_2$$

$$-2k I_1 + 5k I_2 = 2$$

$$-3k I_1 + 4k I_2 = -6$$

$$2k I_1 - 5k I_2 = -2$$

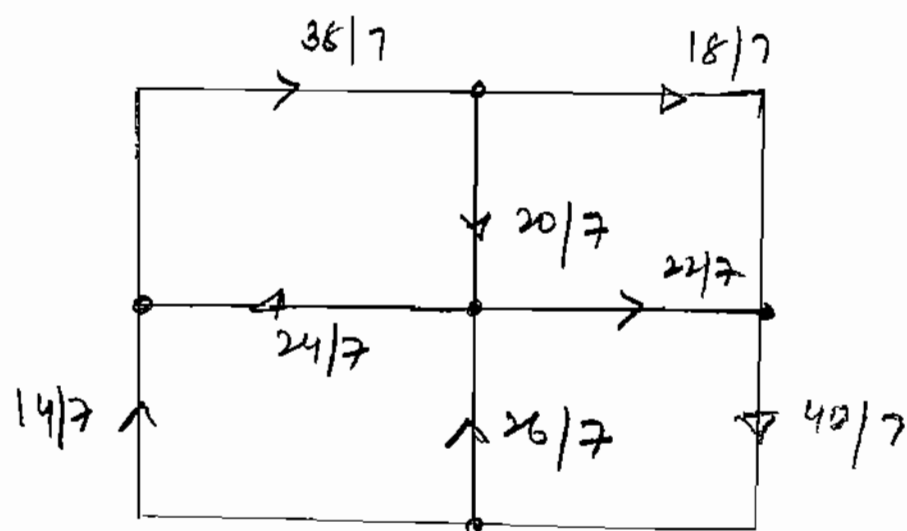
$$3k I_1 - 4k I_2 = 6$$

$$\Delta = -8k^2 + 15k^2 = 7k^2$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{7k^2} \begin{bmatrix} -4k & 5k \\ -3k & 2k \end{bmatrix} \begin{bmatrix} -2 \\ 6 \end{bmatrix}$$

$$= \frac{1}{7k^2} \begin{bmatrix} 8k + 30k \\ 6k + 12k \end{bmatrix} = \frac{1}{7k} \begin{bmatrix} 38 \\ 18 \end{bmatrix} = \begin{bmatrix} 38/7k \\ 18/7k \end{bmatrix}$$

$$I_x = I_1 - I_2 = \frac{38}{7k} - \frac{18}{7k} = \frac{20}{7k}$$



$$I_o = \frac{-26}{7k} A$$

3.97 Find V_o in the circuit in Fig. P3.97 using loop analysis.

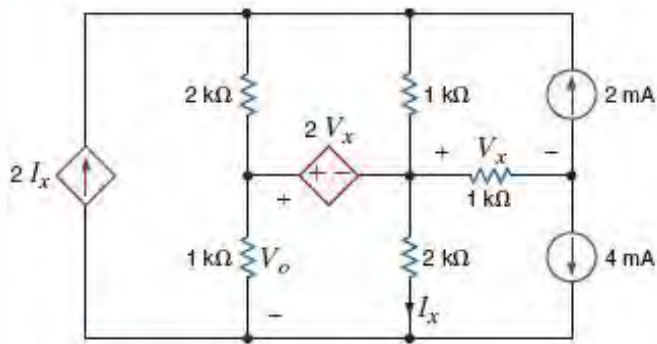
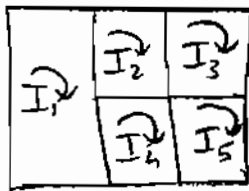


Figure P3.97

SOLUTION:



$$I_x = I_4 - I_5 \quad V_x = 1k \left(\frac{6}{k} \right) = 6V$$

$$2V_x = 12$$

$$2k(I_2 - I_1) + 1k(I_2 - I_3) = 12$$

$$1k(I_4 - I_1) + 2k(I_4 - I_5) = -12$$

$$2kI_2 - 2kI_1 + 1kI_2 - 1kI_3 = 12$$

$$1kI_4 - 1kI_1 + 2kI_4 - 2kI_5 = -12$$

$$3kI_2 - 2k(2)(I_4 - I_5) = 12 + 1kI_3$$

$$3kI_4 - 1k(2)(I_4 - I_5) = -12 + 2kI_5$$

$$3kI_2 - 4kI_4 = 12 + 1kI_3 - 4kI_5$$

$$1kI_4 = -12$$

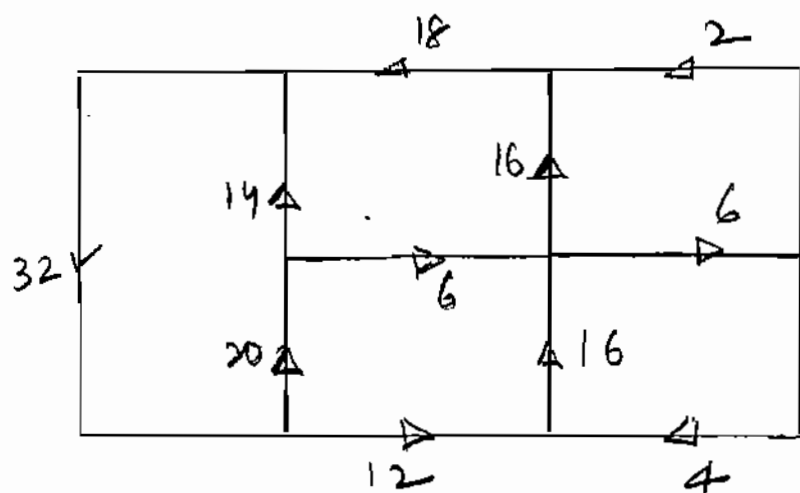
$$3kI_2 - 4kI_4 = 12 - 2 = 10$$

$$I_4 = -12/k$$

$$3k I_2 - 4k I_4 = -6$$

$$3k I_2 = -6 - 48 = -54$$

$$I_2 = -54/3k = -18/k$$



$$V_o = 1k(-20/k) = -20V$$

3.98 Solve for the mesh currents defined in the circuit in Fig. P3.98.

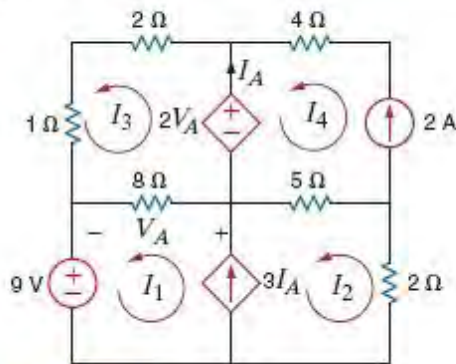


Figure P3.98

SOLUTION:

constraint equations

$$\textcircled{1} I_4 = 2A$$

$$I_1 - I_2 = 3I_A = 3(I_3 - I_4)$$

$$\textcircled{2} -I_1 + I_2 + 3I_3 - 3I_4 = 0$$

$$2I_3 + I_3 + 8(I_3 - I_1) = 2V_A = 2[8(I_1 - I_3)]$$

$$11I_3 - 8I_1 = 16I_1 - 16I_3$$

$$\textcircled{3} -24I_1 + 27I_3 = 0$$

$$5(I_2 - I_4) + 8(I_1 - I_3) + 9 + 2I_2 = 0$$

$$\textcircled{4} 8I_1 + 7I_2 - 8I_3 - 5I_4 = -9$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 \\ -1 & 1 & 3 & -3 \\ -24 & 0 & 27 & 0 \\ 8 & 7 & -8 & -5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ -9 \end{bmatrix}$$

$$I_1 = 3.8 \text{ A}$$

$$I_2 = -0.34 \text{ A}$$

$$I_3 = 3.38 \text{ A}$$

$$I_4 = 2 \text{ A}$$

3.99 Solve for the mesh currents defined in the circuit in Fig. P3.99.

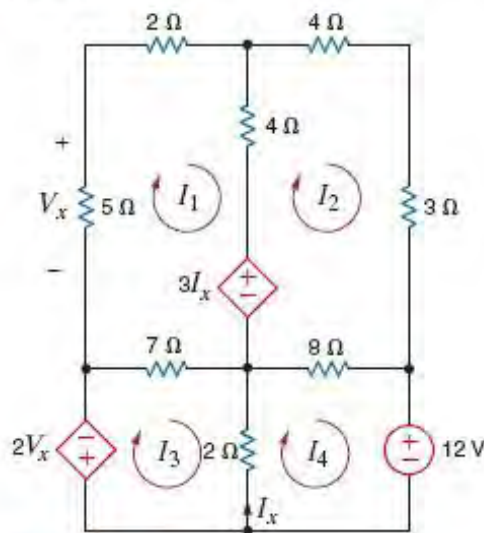


Figure P3.99

SOLUTION:

Constraint equation

$$I_2 - I_1 = 3I_x = 3(I_4 - I_3)$$

$$\textcircled{1} I_1 - I_2 - 3I_3 + 3I_4 = 0$$

$$\textcircled{2} -8I_2 - 2I_3 + 10I_4 = -12$$

$$-7I_1 + 9I_3 - 2I_4 = -2V_x = -2(-5I_1) = 10I_1$$

$$\textcircled{3} -17I_1 + 9I_3 - 2I_4 = 0$$

KVL around outside:

$$2I_1 + 7I_2 + 12 + 2V_x + 5I_1 = 0$$

$$7I_1 + 7I_2 + 2(-5I_1) = -12$$

$$\textcircled{4} -3I_1 + 7I_2 = -12$$

$$\begin{bmatrix} 1 & -1 & -3 & 3 \\ 0 & -8 & -2 & 10 \\ -17 & 0 & 9 & -2 \\ -3 & 7 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 0 \\ -12 \\ 0 \\ -12 \end{bmatrix}$$

$$I_1 = -1.37 \text{ A}$$

$$I_2 = -2.8 \text{ A}$$

$$I_3 = -3.41 \text{ A}$$

$$I_4 = -3.72 \text{ A}$$

3.100 Solve for the mesh currents defined in the circuit in Fig. P3.100.

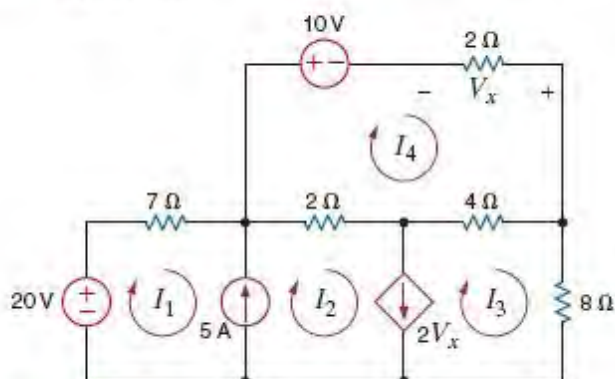


Figure P3.100

SOLUTION:

Constraint equations:

$$\textcircled{1} \quad I_2 - I_4 = 5$$

$$I_2 - I_3 = 2V_x = 2[(-2)I_4] = -4I_4$$

$$\textcircled{2} \quad I_2 - I_3 + 4I_4 = 0$$

$$\textcircled{3} \quad -2I_2 - 4I_3 + 8I_4 = -10$$

KVL around outside of circuit:

$$7I_1 + 10 + 2I_4 + 8I_3 - 20 = 0$$

$$\textcircled{4} \quad 7I_1 + 8I_3 + 2I_4 = 10$$

$$\begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 4 \\ 0 & -2 & -4 & 8 \\ 7 & 0 & 8 & 2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ -10 \\ 10 \end{bmatrix}$$

$$I_1 = -5.281 \text{ A}$$

$$I_2 = -0.238 \text{ A}$$

$$I_3 = 5.4762 \text{ A}$$

$$I_4 = 1.4286 \text{ A}$$

3.101 Using loop analysis, find I_o in the circuit in Fig. P3.101.

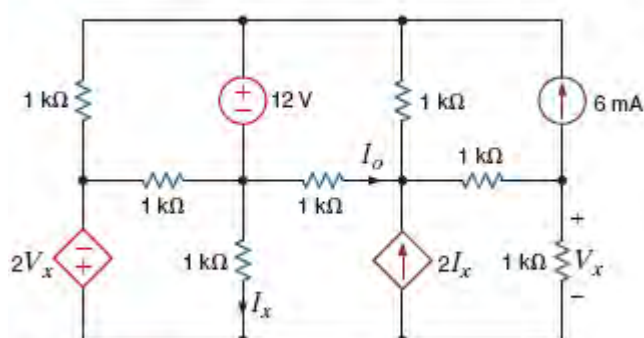
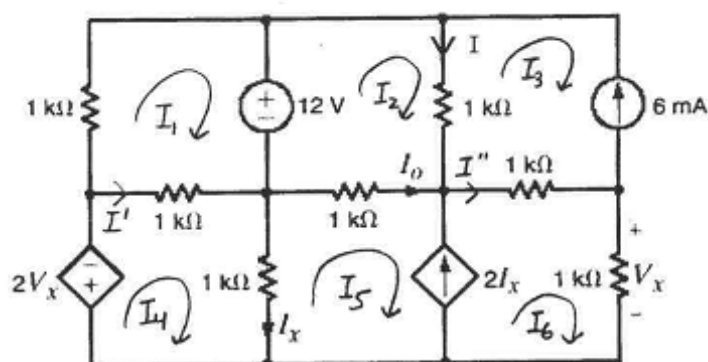


Figure P3.101

SOLUTION:



$$\text{KCL: } I_2 = I + I_3$$

$$I = I_2 - I_3$$

$$\text{KCL: } I_4 = I' + I_1$$

$$I' = I_4 - I_1$$

$$\text{KCL: } I_3 + I'' = I_6$$

$$I'' = I_6 - I_3$$

$$\text{KCL: } I_5 + I_x = I_4$$

$$I_x = I_4 - I_5$$

$$\text{KCL: } I_o + I + 2I_x = I''$$

$$I_o = I'' - I - 2I_x = I_6 - I_3 - I_2 + I_3 - 2I_x$$

$$I_o = I_6 - I_2 - 2(I_4 - I_5)$$

$$I_o = -I_2 - 2I_4 + 2I_5 + I_6$$

$$\begin{aligned} \text{KVL: } 1k I_1 + 12 + 1k(-I') &= 0 \\ 1k I_1 - 1k(I_4 - I_1) &= -12 \\ \boxed{2k I_1 - 1k I_4 &= -12} \end{aligned}$$

$$\begin{aligned} \text{KVL: } 1k I + 1k(-I_0) &= 12 \\ 1k(I_2 - I_3) - 1k(-I_2 - 2I_4 + 2I_5 + I_6) &= 12 \\ \boxed{2k I_2 - 1k I_3 + 2k I_4 - 2k I_5 - 1k I_6 &= 12} \end{aligned}$$

$$\begin{aligned} \text{KVL: } 2V_x + 1k I' + 1k I_x &= 0 \\ V_x &= 1k I_6 \\ 2(1k I_6) + 1k(I_4 - I_1) + 1k(I_4 - I_5) &= 0 \\ \boxed{-1k I_1 + 2k I_4 - 1k I_5 + 2k I_6 &= 0} \end{aligned}$$

$$\begin{aligned} \text{KCL: } I_6 &= 2I_x + I_5 \\ I_5 - I_6 + 2(I_4 - I_5) &= 0 \\ \boxed{2I_4 - I_5 - I_6 &= 0} \end{aligned}$$

$$\begin{aligned} \text{KVL: } 2V_x + 1k I' + 1k I_0 + 1k I'' + 1k I_6 &= 0 \\ 2(1k I_6) + 1k(I_4 - I_1) + 1k(-I_2 - 2I_4 + 2I_5 + I_6) + 1k(I_6 - I_3) + 1k I_6 \\ \boxed{-1k I_1 - 1k I_2 - 1k I_3 - 1k I_4 + 2k I_5 + 5k I_6 &= 0} \end{aligned}$$

$$I_3 = -6 \text{ mA}$$

$$\boxed{2k I_2 + 2k I_4 - 2k I_5 - 1k I_6 = 6}$$

$$\boxed{-1k I_1 - 1k I_2 - 1k I_4 + 2k I_5 + 5k I_6 = -6}$$

$$\begin{aligned}
 2kI_1 + 0I_2 - 1kI_4 + 0I_5 + 0I_6 &= -12 \\
 0I_1 + 2kI_2 + 2kI_4 - 2kI_5 - 1kI_6 &= 6 \\
 -1kI_1 + 0I_2 + 2kI_4 - 1kI_5 + 2kI_6 &= 0 \\
 0I_1 + 0I_2 + 2I_4 - I_5 - I_6 &= 0 \\
 -1kI_1 - 1kI_2 - 1kI_4 + 2kI_5 + 5kI_6 &= -6
 \end{aligned}$$

$$I_1 = -6.48 \text{ mA}$$

$$I_2 = 3.12 \text{ mA}$$

$$I_4 = -0.96 \text{ mA}$$

$$I_5 = 0.24 \text{ mA}$$

$$I_6 = -2.16 \text{ mA}$$

$$I_0 = -I_2 - 2I_4 + 2I_5 + I_6$$

$$I_0 = -3.12 \text{ mA} - 2(-0.96 \text{ mA}) + 2(0.24 \text{ mA}) - 2.16 \text{ mA}$$

$$I_0 = -2.88 \text{ mA}$$

3.102 Use mesh analysis to determine the power delivered by the independent 3-V source in the network in Fig. P3.102.

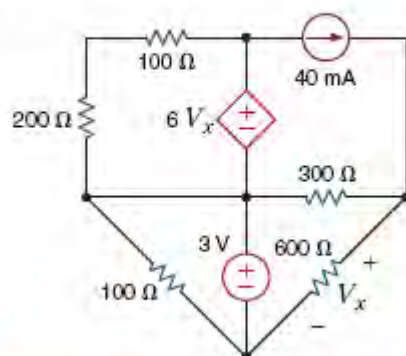
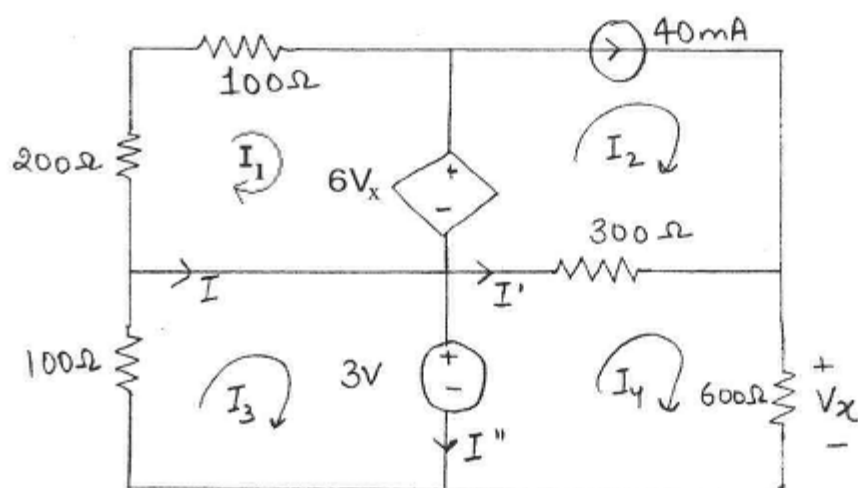


Figure P3.102

SOLUTION:



$$I_2 = 40\text{mA}$$

$$\text{KCL: } I_3 = I + I_1$$

$$I = I_3 - I_1$$

$$\text{KCL: } 40\text{mA} + I' = I_4$$

$$I' = I_4 - 40\text{mA}$$

$$\text{KVL: } 200I_1 + 100I_1 + 6V_x = 0$$

$$V_x = 600I_4$$

$$300I_1 + 3600I_4 = 0$$

$$\text{KVL: } 100I_3 + 3 = 0$$

$$I_3 = -30 \text{ mA}$$

$$\text{KVL: } 3 = 300 I' + 600 I_4$$

$$600 I_4 + 300 (I_4 - 40 \text{ m}) = 3$$

$$900 I_4 = 15$$

$$I_4 = 16.67 \text{ mA}$$

$$\text{KCL: } I_4 + I'' = I_3$$

$$I'' = I_3 - I_4$$

$$P_{3V} = I''(3) = 3(I_3 - I_4)$$

$$P_{3V} = 3(-30 \text{ m} - 16.67 \text{ m})$$

$$P_{3V} = -140.01 \text{ mW}$$

$$P_{3V} = 140.01 \text{ mW}$$

3.103 Use mesh analysis to find the power delivered by the current-control voltage source in the circuit in Fig. P3.103.

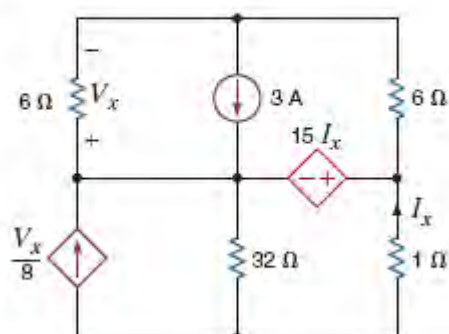
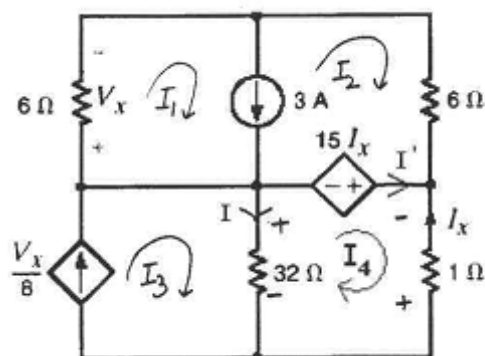


Figure P3.103

SOLUTION:



$$\text{KCL: } I + I_4 = I_3$$

$$I = I_3 - I_4$$

$$\text{KVL: } 15I_x = 1(-I_x) + 32(-I)$$

$$-16I_x - 32(I_3 - I_4) = 0$$

$$I_x = -I_4$$

$$16I_4 - 32I_3 + 32I_4 = 0$$

$$\boxed{-32I_3 + 48I_4 = 0}$$

$$\text{KCL: } I' + I_2 = I_4$$

$$I' = I_4 - I_2$$

$$\text{KCL: } I_1 = 3 + I_2$$

$$\boxed{I_1 - I_2 = 3}$$

$$\text{KVL: } 15I_x + 6I_1 + 6I_2 = 0$$

$$\boxed{6I_1 + 6I_2 - 15I_4 = 0}$$

$$I_3 = \frac{V_x}{8}$$

$$I_3 = \frac{6I_1}{8}$$

$$\boxed{6I_1 - 8I_3 = 0}$$

$$0I_1 + 0I_2 - 32I_3 + 48I_4 = 0$$

$$I_1 - I_2 + 0I_3 + 0I_4 = 3$$

$$6I_1 + 6I_2 + 0I_3 - 15I_4 = 0$$

$$6I_1 + 0I_2 - 8I_3 + 0I_4 = 0$$

$$I_1 = 4A$$

$$I_2 = 1A$$

$$I_3 = 3A$$

$$I_4 = 2A$$

$$P_{15I_x} = (15I_x)(I')$$

$$P_{15I_x} = 15(-I_4)(I_4 - I_2)$$

$$P_{15I_x} = -15(2)(2-1) = -30W$$

$$P_{15I_x} = 30W \text{ absorbed.}$$

3.104 Use both nodal and loop analyses to determine I_o in the circuit in Fig. P3.104.

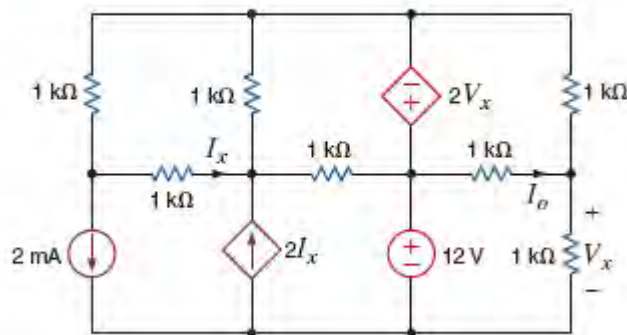
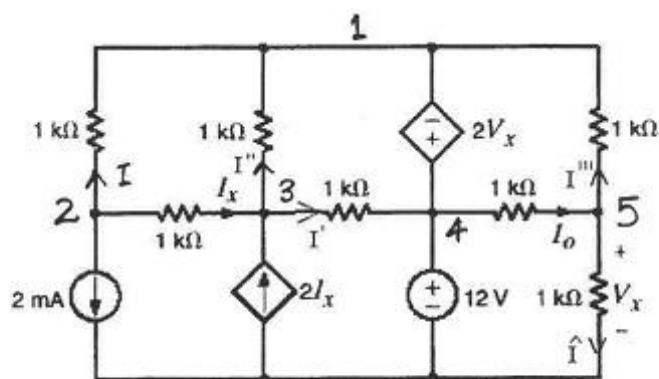


Figure P3.104

SOLUTION:



$$\text{KCL at 2: } 2\text{mA} + I + I_x = 0$$

$$\frac{V_2 - V_1}{1\text{k}} + \frac{V_2 - V_3}{1\text{k}} = -2\text{mA}$$

$$V_2 - V_1 + V_2 - V_3 = -2$$

$$\boxed{-V_1 + 2V_2 - V_3 = -2}$$

$$\text{KCL at 3: } 2I_x + I_x = I'' + I'$$

$$2 \left[\frac{V_2 - V_3}{1\text{k}} \right] + \frac{V_2 - V_3}{1\text{k}} = \frac{V_3 - V_1}{1\text{k}} + \frac{V_3 - V_4}{1\text{k}}$$

$$2V_2 - 2V_3 + V_2 - V_3 = V_3 - V_1 + V_3 - V_4$$

$$\boxed{V_1 + 3V_2 - 5V_3 + V_4 = 0}$$

$$V_4 = 12\text{V}$$

$$\boxed{V_1 + 3V_2 - 5V_3 = -12}$$

KCL at 5: $I_0 = I''' + \hat{I}$

$$\frac{V_4 - V_5}{1k} = \frac{V_5 - V_1}{1k} + \frac{V_5}{1k}$$

$$V_4 - V_5 = V_5 - V_1 + V_5$$

$$\boxed{V_1 + V_4 - 3V_5 = 0}$$

$$V_4 = 12V$$

$$\boxed{V_1 - 3V_5 = -12}$$

$$V_4 - V_1 = 2V_x$$

$$V_x = 5$$

$$\boxed{-V_1 + V_4 - 2V_5 = 0}$$

$$\boxed{-V_1 - 2V_5 = -12}$$

$$V_1 - 3V_5 = -12$$

$$-V_1 - 2V_5 = -12$$

$$V_1 = 2.4V$$

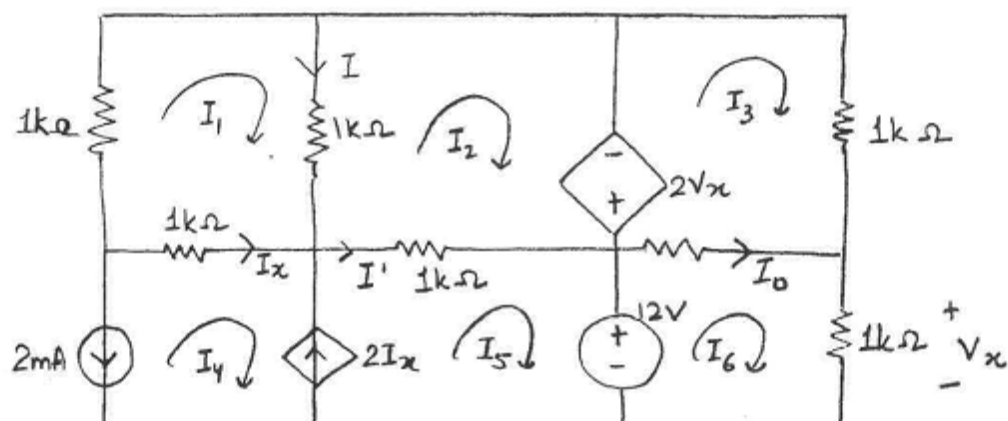
$$V_5 = 4.8V$$

$$V_4 = 3(4.8) - 2.4$$

$$V_4 = 12V$$

$$I_0 = \frac{V_4 - V_5}{1k} = \frac{12 - 4.8}{1k}$$

$$I_0 = 7.2mA$$



$$\text{KCL: } I_1 = I + I_2$$

$$I = I_1 - I_2$$

$$\text{KCL: } I_4 = I_x + I_1$$

$$I_x = I_4 - I_1$$

$$\text{KCL: } I_x + 2I_2 + I = I'$$

$$I' = I_1 - I_2 + 3[I_4 - I_1]$$

$$I' = -2I_1 - I_2 + 3I_4$$

$$\text{KCL: } I_3 + I_0 = I_6$$

$$I_0 = I_6 - I_3$$

$$\text{KVL: } 1kI_1 + 1kI + 1k(-I_x) = 0$$

$$1kI_1 + 1k(I_1 - I_2) - 1k(I_4 - I_1) = 0$$

$$\boxed{3kI_1 - 1kI_2 - 1kI_4 = 0}$$

$$\text{KVL: } 1k(-I) + 1k(-I') = 2V_x$$

$$V_x = 1kI_6$$

$$-1k(I_1 - I_2) - 1k(-2I_1 - I_2 + 3I_4) = 2(1kI_6)$$

$$\boxed{1kI_1 + 2kI_2 - 3kI_4 - 2kI_6 = 0}$$

$$\text{KVL: } 1kI_0 + 1kI_6 = 12$$

$$1k(I_6 - I_3) + 1kI_6 = 12$$

$$\boxed{-1kI_3 + 2kI_6 = 12}$$

$$\begin{aligned} \text{KCL: } I_5 &= 2I_x + I_4 \\ -I_4 + I_5 - 2[I_4 - I_1] &= 0 \\ \boxed{2I_1 - 3I_4 + I_5} &= 0 \end{aligned}$$

$$\begin{aligned} \text{KVL: } 2V_x + 1kI_3 + 1k(-I_0) &= 0 \\ 2(1kI_6) + 1kI_3 - 1k(I_6 - I_3) &= 0 \\ \boxed{2kI_3 + 1kI_6} &= 0 \end{aligned}$$

$$\begin{aligned} I_4 &= -2\text{mA} \\ \boxed{3kI_1 - 1kI_2} &= -2 \\ \boxed{1kI_1 + 2kI_2 - 2kI_6} &= -6 \\ \boxed{2I_1 + I_5} &= -6 \end{aligned}$$

$$\begin{aligned} 0I_1 + 0I_2 - 1kI_3 + 0I_5 + 2kI_6 &= 12 \\ 0I_1 + 0I_2 + 2kI_3 + 0I_5 + 1kI_6 &= 0 \\ 3kI_1 - 1kI_2 + 0I_3 + 0I_5 + 0I_6 &= -2 \\ 1kI_1 + 2kI_2 + 0I_3 + 0I_5 - 2kI_6 &= -6 \\ 2I_1 + 0I_2 + 0I_3 + I_5 + 0I_6 &= -6 \end{aligned}$$

$$\begin{aligned} I_1 &= -57.14\mu\text{A} \\ I_2 &= 1.83\text{mA} \\ I_3 &= -2.4\text{mA} \\ I_5 &= -6\text{A} \\ I_6 &= 4.8\text{mA} \end{aligned}$$

$$\begin{aligned} I_0 &= I_6 - I_3 \\ I_0 &= 4.8\text{mA} - (-2.4\text{mA}) \\ I_0 &= 7.2\text{mA} \end{aligned}$$

3.105 Use both nodal and loop analyses to find V_o in the circuit in Fig. P3.105.

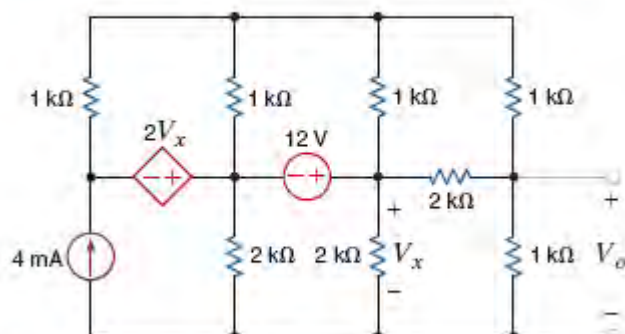
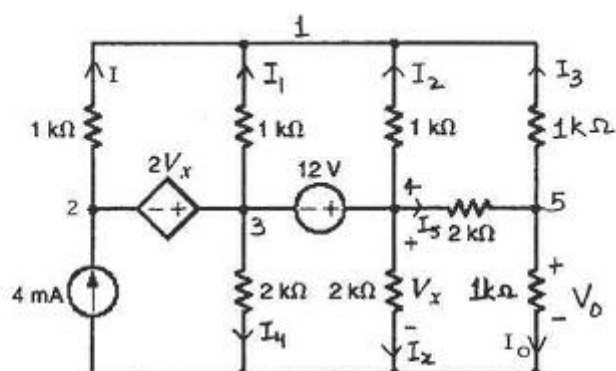


Figure P3.105

SOLUTION:



$$\text{KCL at 1: } I + I_1 + I_2 + I_3 = 0$$

$$\frac{V_2 - V_1}{1k} + \frac{V_3 - V_1}{1k} + \frac{V_4 - V_1}{1k} + \frac{V_5 - V_1}{1k} = 0$$

$$V_2 - V_1 + V_3 - V_1 + V_4 - V_1 + V_5 - V_1 = 0$$

$$\boxed{-4V_1 + V_2 + V_3 + V_4 + V_5 = 0}$$

$$\text{KCL at 5: } I_5 = I_3 + I_0$$

$$\frac{V_4 - V_5}{2k} = \frac{V_5 - V_1}{1k} + \frac{V_5}{1k}$$

$$V_4 - V_5 = 2V_5 - 2V_1 + 2V_5$$

$$\boxed{2V_1 + V_4 - 5V_5 = 0}$$

KCL at reference:

$$I_4 + I_x + I_0 = 4\text{m}$$

$$\frac{V_3}{2k} + \frac{V_4}{2k} + \frac{V_5}{1k} = 4\text{m}$$

$$\boxed{V_3 + V_4 + 2V_5 = 8}$$

$$V_4 - V_3 = 12$$

$$\boxed{-V_3 + V_4 = 12}$$

$$V_3 - V_2 = 2V_x$$

$$V_x = V_4$$

$$V_3 - V_2 = 2(V_4)$$

$$\boxed{-V_2 + V_3 - 2V_4 = 0}$$

$$-4V_1 + V_2 + V_3 + V_4 + V_5 = 0$$

$$2V_1 + 0V_2 + 0V_3 + V_4 - 5V_5 = 0$$

$$0V_1 + 0V_2 + V_3 + V_4 + 2V_5 = 8$$

$$0V_1 + 0V_2 - V_3 + V_4 + 0V_5 = 12$$

$$0V_1 - V_2 + V_3 - 2V_4 + 0V_5 = 0$$

$$V_1 = -3.5\text{V}$$

$$V_2 = -21.5\text{V}$$

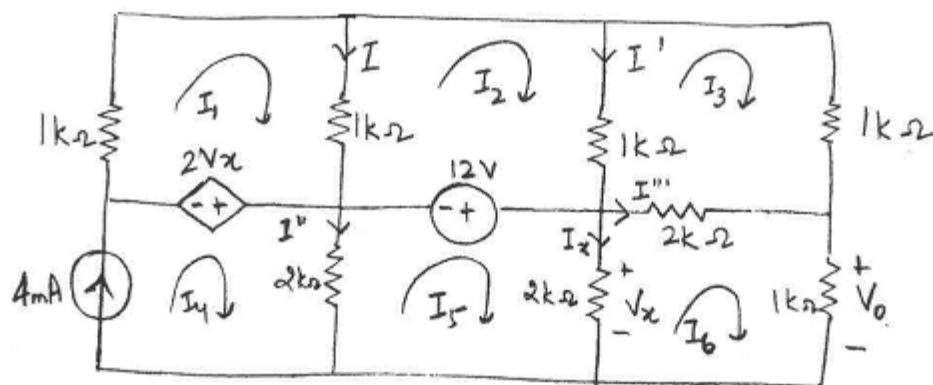
$$V_3 = -2.5\text{V}$$

$$V_4 = 9.5\text{V}$$

$$V_5 = 0.5\text{V}$$

$$V_0 = V_5 = 0.5\text{V}$$

$$V_0 = 0.5\text{V}$$



$$\text{KCL: } I_1 = I + I_2$$

$$I = I_1 - I_2$$

$$\text{KCL: } I_5 + I'' = I_4$$

$$I'' = I_4 - I_5$$

$$\text{KCL: } I_6 + I_x = I_5$$

$$I_x = I_5 - I_6$$

$$\text{KCL: } I_2 = I' + I_3$$

$$I' = I_2 - I_3$$

$$\text{KCL: } I_3 + I''' = I_6$$

$$I''' = I_6 - I_3$$

$$\text{KVL: } 2V_x + 1kI_1 + 1kI = 0$$

$$V_x = 2kI_x = 2k(I_5 - I_6)$$

$$2[2k(I_5 - I_6)] + 1kI_1 + 1k(I_1 - I_2) = 0$$

$$4kI_5 - 4kI_6 + 1kI_1 + 1kI_1 - 1kI_2 = 0$$

$$\boxed{2kI_1 - 1kI_2 + 4kI_5 - 4kI_6 = 0}$$

$$\text{KVL: } 1k(-I) + 1kI' + 12 = 0$$

$$-1k(I_1 - I_2) + 1k(I_2 - I_3) = -12$$

$$\boxed{-1kI_1 + 2kI_2 - 1kI_3 = -12}$$

$$\text{KVL: } 1k(-I') + 1kI_3 + 2k(-I''') = 0$$

$$-1k(I_2 - I_3) + 1kI_3 - 2k(I_6 - I_3) = 0$$

$$\boxed{-1kI_2 + 4kI_3 - 2kI_6 = 0}$$

$$\text{KVL: } 2k(-I'') + 2kI_x = 12$$

$$-2k(I_4 - I_5) + 2k(I_5 - I_6) = 12$$

$$\boxed{-2kI_4 + 4kI_5 - 2kI_6 = 12}$$

$$\text{KVL: } 2kI''' + 1kI_6 + 2k(-I_x) = 0$$

$$2k(I_6 - I_3) + 1kI_6 - 2k(I_5 - I_6) = 0$$

$$\boxed{-2kI_3 - 2kI_5 + 5kI_6 = 0}$$

$$I_4 = 4\text{mA}$$

$$\boxed{4kI_5 - 2kI_6 = 20}$$

$$2kI_1 - 1kI_2 + 0I_3 + 4kI_5 - 4kI_6 = 0$$

$$-1kI_1 + 2kI_2 - 1kI_3 + 0I_5 + 0I_6 = -12$$

$$0I_1 - 1kI_2 + 4kI_3 + 0I_5 - 2kI_6 = 0$$

$$0I_1 + 0I_2 + 0I_3 + 4kI_5 - 2kI_6 = 20$$

$$0I_1 + 0I_2 - 2kI_3 - 2kI_5 + 5kI_6 = 0$$

$$I_1 = -18\text{mA}$$

$$I_2 = -17\text{mA}$$

$$I_3 = -4\text{mA}$$

$$I_5 = 5.25\text{mA}$$

$$I_6 = 0.5\text{mA}$$

$$V_o = 1kI_6$$

$$V_o = 1k(0.5\text{mA})$$

$$V_o = 0.5\text{V}$$

3.106 Find I_o in the network in Fig. P3.106 using nodal analysis.

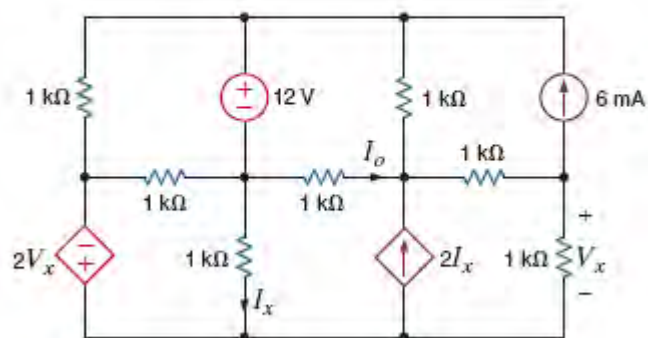
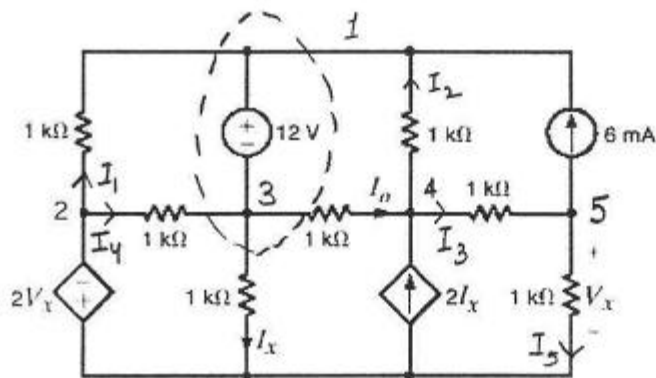


Figure P3.106

SOLUTION:



$$\text{KCL at 5: } 6\text{mA} + I_5 = I_3$$

$$6\text{mA} + \frac{V_5}{1\text{k}} = \frac{V_4 - V_5}{1\text{k}}$$

$$6 + V_5 = V_4 - V_5$$

$$\boxed{V_4 - 2V_5 = 6}$$

$$\text{KCL at 4: } I_o + 2I_x = I_2 + I_3$$

$$\frac{V_3 - V_4}{1\text{k}} + 2 \left[\frac{V_3}{1\text{k}} \right] = \frac{V_4 - V_1}{1\text{k}} + \frac{V_4 - V_5}{1\text{k}}$$

$$V_3 - V_4 + 2V_3 = V_4 - V_1 + V_4 - V_5$$

$$\boxed{V_1 + 3V_3 - 3V_4 + V_5 = 0}$$

$$V_2 = -2V_x$$

$$V_2 = -2(V_5)$$

$$\boxed{V_2 + 2V_5 = 0}$$

$$\boxed{V_1 - V_3 = 12}$$

KCL at supernode:

$$I_1 + I_4 + I_2 + 6\text{m} = I_0 + I_x$$

$$\frac{V_2 - V_1}{1\text{k}} + \frac{V_2 - V_3}{1\text{k}} + \frac{V_4 - V_1}{1\text{k}} + 6\text{m}$$

$$= \frac{V_3 - V_4}{1\text{k}} + \frac{V_3}{1\text{k}}$$

$$V_2 - V_1 + V_2 - V_3 + V_4 - V_1 + 6 = V_3 - V_4 + V_3$$

$$\boxed{-2V_1 + 2V_2 - 3V_3 + 2V_4 = -6}$$

$$0V_1 + 0V_2 + 0V_3 + V_4 - 2V_5 = 6$$

$$V_1 + 0V_2 + 3V_3 - 3V_4 + V_5 = 0$$

$$0V_1 + V_2 + 0V_3 + 0V_4 + 2V_5 = 0$$

$$V_1 + 0V_2 - V_3 + 0V_4 + 0V_5 = 12$$

$$-2V_1 + 2V_2 - 3V_3 + 2V_4 + 0V_5 = -6$$

$$V_1 = 10.8\text{V}, V_2 = 4.32\text{V}, V_3 = -1.2\text{V}$$

$$V_4 = 1.68\text{V}, V_5 = -2.16\text{V}$$

$$I_0 = \frac{V_3 - V_4}{1\text{k}} = \frac{-1.2 - 1.68}{1\text{k}}$$

$$= -2.88\text{mA}$$